ORDERED CHIRAL HOMOLOGY, THE E_3 ALGEBRA OF CHERN–SIMONS THEORY, AND THE HOLOMORPHIC REFINEMENT OF FACTORISATION HOMOLOGY

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Abstract. We define ordered chiral homology $\int_X^{\mathrm{ord}} \mathcal{A}$ for an E_1-chiral algebra on a smooth curve X as the derived pushforward of the ordered factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on the ordered Ran space $\mathrm{Ran}^{\mathrm{ord}}(X) = \coprod_n U_n(X)$: a single $\mathcal{D}$-module encoding the OPE, the KZ connection, and the factorisation product across all arities simultaneously. The Beilinson–Drinfeld Ran space $\mathrm{Ran}(X)$ is the $\Sigma_\bullet$-coinvariant quotient of $\mathrm{Ran}^{\mathrm{ord}}(X)$, and the averaging map $\mathrm{av}\colon R\Gamma(\mathrm{Ran}^{\mathrm{ord}}(X), \mathcal{F}^{\mathrm{ord}}) \to R\Gamma(\mathrm{Ran}(X), \mathcal{F}^{\Sigma})$ recovers the Beilinson–Drinfeld chiral homology; for genuinely E_1-chiral algebras the descent is lossy, carrying GRT_1-dependent data from the Drinfeld associator. For $V_k(\mathfrak{g})$, we construct a filtered E_3-chiral algebra $\mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})$ in $\mathcal{D}$-modules on $\mathrm{Ran}(X)$: its P_3 bracket is $\{X, Y\} = h_{\mathrm{KZ}} \cdot (X, Y)$, where (X, Y) is the Killing form and $h_{\mathrm{KZ}} = 1/(k + h^\vee)$ is the KZ coupling. The E_3-deformation space is $\hbar H^3(\mathfrak{g})[[\hbar]]$; formal disk restriction recovers the Costello–Francis–Gwilliam perturbative Chern–Simons E_3-algebra, and the Etingof–Kazhdan classification of quantum vertex R-matrices gives a sixth identification, all in natural bijection. Five computations span the $G/L/C/M$ classification: Heisenberg ($P(t) = 1+t+t^2$, regular KZ), $V_k(\mathfrak{sl}_2)$ (KZ conformal blocks with all E_3 operations exhibited), $\beta\gamma$ (nilpotent R-matrix, class C), Virasoro (irregular KZ connection, Stokes data, class M), and the Yangian $Y_\hbar(\mathfrak{sl}_2)$ (the first example where ordered chiral homology is strictly richer than symmetric). The Etingof–Kazhdan quantum vertex algebra is exhibited with all five braided VOA axioms verified, and $\mathcal{W}_{1+\infty}$ is realised as a class M chiral quantum group via the cohomological Hall algebra. The chiral quantum group equivalence theorem identifies three structures: vertex R-matrix, chiral A_∞-structure, and chiral coproduct, forming an equivalence triangle mediated by the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ and its deconcatenation coproduct.

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1. Introduction

Beilinson and Drinfeld's chiral homology [4] is the fundamental invariant of a chiral algebra on an algebraic curve. For a chiral algebra $\mathcal{A}$ on a smooth proper curve X, the chiral homology $\int_X \mathcal{A}$ computes the space of conformal blocks: the derived global sections of a factorisation $\mathcal{D}$-module on the Ran space $\mathrm{Ran}(X)$. The construction requires that $\mathcal{A}$ be a *commutative* factorisation algebra: the factorisation product, which merges local observables at

disjoint collections of points, must be symmetric under permutation of the inputs.

Every vertex algebra in the standard landscape (Heisenberg, affine Kac–Moody, Virasoro, $\mathcal{W}$-algebras, $\beta\gamma$ systems, lattice algebras) satisfies this commutativity hypothesis. The OPE $Y(a,z)Y(b,w)$ and $Y(b,w)Y(a,z)$ agree (after clearing poles) by the locality axiom, and the factorisation product descends to the symmetric quotient $X^{(n)} = X^n/\Sigma_n$.

The hypothesis fails for the class of algebras that the first volume of this series [23] identifies as the *genuinely* E_1*-chiral* objects: the Yangians $Y_\hbar(\mathfrak{g})$, the Etingof–Kazhdan quantum vertex algebras [13], and more broadly any chiral algebra whose operator product is governed by a non-trivial vertex R-matrix $S(z)$ satisfying the quantum Yang–Baxter equation. For these algebras, the factorisation product is *braided*, not commutative: swapping two groups of inputs introduces a twist by $S(z)$.

The deficiency is not in the algebras but in the available integration theory. There is no $\mathcal{D}$-module obstruction to working on ordered spaces: $\mathcal{D}$-modules on X^n, on the ordered Fulton–MacPherson compactification $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(X)$, and on Mok's log-smooth compactifications are perfectly standard objects, studied since Kashiwara and Malgrange. The correct foundational object is the *ordered Ran space* $\mathrm{Ran}^{\mathrm{ord}}(X) = \coprod_n U_n(X)$, the moduli of ordered tuples of distinct points. The Beilinson–Drinfeld Ran space is the $\Sigma_\bullet$-coinvariant quotient $\mathrm{Ran}(X) = \mathrm{Ran}^{\mathrm{ord}}(X)/\Sigma_\bullet$, and the commutative factorisation structure on $\mathrm{Ran}(X)$ is the equivariant shadow of the ordered factorisation structure on $\mathrm{Ran}^{\mathrm{ord}}(X)$ that exists for any chiral algebra, commutative or not.

This paper constructs the ordered factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on $\mathrm{Ran}^{\mathrm{ord}}(X)$, defines ordered chiral homology as the derived pushforward of this single object, and establishes the basic properties of the resulting invariant.

1.1. **Main results.** Three results emerge from the ordered theory. The first constructs the $\mathcal{D}$-module and defines ordered chiral homology; the second separates it from topological Hochschild; the third identifies the E_3 structure that only the ordered picture sees.

Theorem 1.1 (Ordered chiral homology)**.** *Let $\mathcal{A}$ be a chiral algebra on a smooth proper curve X of genus g, and let $r(z)$ denote the classical r-matrix (the arity-2 projection of the Maurer–Cartan element $\Theta_{\mathcal{A}}$ in the E_1 convolution algebra).*

(i) (Ordered factorisation $\mathcal{D}$-module on $\mathrm{Ran}^{\mathrm{ord}}(X)$.) *The OPE of $\mathcal{A}$, extended meromorphically across the diagonals, defines a factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on the ordered Ran space $\mathrm{Ran}^{\mathrm{ord}}(X) = \coprod_n U_n(X)$. Its restriction to the n-th stratum is the holonomic $\mathcal{D}$-module*

$$\mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big|_{U_n} = \mathcal{A}^{\boxtimes n}\big|_{U_n}, \qquad \mathcal{F}_n^{\mathrm{ord}}(\mathcal{A}) := j_* j^*(\mathcal{A}^{\boxtimes n}) \in \mathcal{D}\text{-mod}(X^n), \tag{1.1}$$

where $j\colon U_n = X^n \setminus \bigcup_{i<j} \Delta_{ij} \hookrightarrow X^n$ is the inclusion of the ordered configuration space. No Σ_n-equivariance is imposed. The factorisation product, the KZ connection, and the chiral differential are encoded in the single object $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on $\mathrm{Ran}^{\mathrm{ord}}(X)$.

(ii) (Ordered chiral homology.) *The* ordered chiral homology *of $\mathcal{A}$ over X is the derived pushforward*

$$\int_X^{\mathrm{ord}} \mathcal{A} \;:=\; R\Gamma_{\mathrm{dR}}\big(\mathrm{Ran}^{\mathrm{ord}}(X),\ \mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big). \tag{1.2}$$

The arity-by-arity decomposition $\bigoplus_{n\geq 0} R\Gamma_{\mathrm{dR}}(X^n,\, \mathcal{F}_n^{\mathrm{ord}}(\mathcal{A}))$ computes this pushforward stratum by stratum.

(iii) (Symmetric descent.) *The quotient $q\colon \mathrm{Ran}^{\mathrm{ord}}(X) \to \mathrm{Ran}(X)$ induces the averaging map*

$$\int_X^{\mathrm{ord}} \mathcal{A} = R\Gamma\big(\mathrm{Ran}^{\mathrm{ord}}(X),\, \mathcal{F}^{\mathrm{ord}}\big) \xrightarrow{\mathrm{av}} R\Gamma\big(\mathrm{Ran}(X),\, \mathcal{F}^{\Sigma}\big) = \int_X \mathcal{A}, \tag{1.3}$$

which is a quasi-isomorphism for E_∞-chiral $\mathcal{A}$. For E_1-chiral $\mathcal{A}$, the descent is lossy: the kernel of $\mathrm{av}\colon R\Gamma_{\mathrm{dR}}(X^n, \mathcal{F}_n^{\mathrm{ord}}) \twoheadrightarrow R\Gamma_{\mathrm{dR}}(X^n, \mathcal{F}_n^{\mathrm{ord}})_{\Sigma_n}$ is non-trivial at $n \geq 3$, carrying GRT_1-dependent data from the Drinfeld associator.

Theorem 1.2 (Topological Hochschild vs. ordered chiral homology of $D^\times$)**.** *Let $\mathcal{A}$ be a chiral algebra on a curve X, and let $D^\times = \mathrm{Spec}\ \mathbb{C}((z))$ be the punctured formal disk.*

(i) (Topological Hochschild.) *Viewing $\mathcal{A}$ as an E_1-topological algebra by forgetting the holomorphic structure, the factorisation homology $\int_{S^1} \mathcal{A}$ is the Hochschild homology $\mathrm{HH}_*(\mathcal{A})$, computed by the cyclic bar complex with the Hochschild differential using the associative product $m_2\colon \mathcal{A}\otimes\mathcal{A} \to \mathcal{A}$ and the higher A_∞ operations m_k.*

(ii) (Ordered chiral homology of $D^\times$.) *The ordered chiral chain complex at arity n on $D^\times$ is*

$$\mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A}) = \Big(\Omega^*_{\log}\big(\overline{\mathrm{FM}}_n^{\mathrm{ord}}(D^\times)\big) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n},\ \ d_{\mathrm{total}}\Big) \tag{1.4}$$

where $\Omega^*_{\log}$ denotes logarithmic differential forms on the Fulton–MacPherson compactification, s^{-1} is the desuspension ($|s^{-1}v| = |v| - 1$), $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal, and

$$d_{\mathrm{total}} = d_{\mathrm{dR}} + d_{\mathrm{bar}} + \nabla_{\mathrm{KZ}} \tag{1.5}$$

with d_{dR} the de Rham differential, d_{bar} the bar differential encoding the OPE, and

$$\nabla_{\mathrm{KZ}} = \sum_{i<j} r_{ij}(z_i - z_j)\, dz_{ij}, \qquad dz_{ij} = d(z_i - z_j), \tag{1.6}$$

the Knizhnik–Zamolodchikov connection. The Arnold forms $\omega_{ij} = d\log(z_i - z_j)$ are generators of the cohomology algebra $H^*(\mathrm{Conf}_n(\mathbb{C}))$

and appear as coefficients in the bar differential d_{bar}; they are not *the connection* 1*-form of* ∇_{KZ}.

(iii) (Formality bridge.) *For* E_∞*-chiral* $\mathcal{A}$*, the formality bridge* (*Theorem* 5.4) *gives a quasi-isomorphism* $\mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A}) \xrightarrow{\sim} \mathrm{HH}_*(\mathcal{A})$*: ordered and symmetric chiral homologies agree as chain complexes, though the transferred* A_∞*-structure may carry non-trivial higher operations for class* $L/C/M$*. For genuinely* E_1*-chiral* $\mathcal{A}$ (*where* $S(z) \neq \mathrm{id}$)*, this quasi-isomorphism does not exist.*

Theorem 1.3 (The E_3-algebra and Chern–Simons)**.** *Let* $\mathfrak{g}$ *be a simple finite-dimensional Lie algebra and* $V_k(\mathfrak{g})$ *the affine Kac–Moody vertex algebra at level* k*.*

(i) (The E_3 structure.) *The curve geometry provides the* E_2 *structure on* $\overline{B}^\Sigma(V_k(\mathfrak{g}))$ (*Warning* 2.5)*. By the Higher Deligne Conjecture* (*Lurie* [18]*, Francis* [15])*, the Hochschild cohomology of this* E_2*-coalgebra*

$$Z_{\mathrm{ch}}^{\mathrm{der}}(V_k(\mathfrak{g})) = \mathrm{HH}^*\big(\overline{B}^\Sigma(V_k(\mathfrak{g})),\ \overline{B}^\Sigma(V_k(\mathfrak{g}))\big) \tag{1.7}$$

carries a natural E_3*-algebra structure* ($\mathsf{E}_2 + 1 = \mathsf{E}_3$)*.*

(ii) (Deformation classification.) *The space of* E_3*-deformations of* $Z_{\mathrm{ch}}^{\mathrm{der}}(V_k(\mathfrak{g}))$ *is non-canonically isomorphic to* $\hbar H^3(\mathfrak{g})[[\hbar]]$*. For simple* $\mathfrak{g}$*,* $H^3(\mathfrak{g})$ *is one-dimensional, so there is a single deformation parameter at each order in* $\hbar$*.*

(iii) (Comparison.) *The same* $\hbar H^3(\mathfrak{g})[[\hbar]]$ *classifies, by Costello–Francis–Gwilliam* [9]*: perturbative Chern–Simons quantisations, filtered* E_3*-algebras deforming* $C^*(\mathfrak{g})$*, braided monoidal deformations of* $\mathrm{Rep}_{\mathrm{fin}}(\mathfrak{g})$*, and quasi-triangular quasi-Hopf deformations of* $U(\mathfrak{g})$*; and by Etingof–Kazhdan* [12, 13]*: quantum vertex* R*-matrices on* $V(\mathfrak{g}, K)$*, forming a* GRT_1*-torsor. All six classification spaces are in natural bijection.*

(iv) (Topological enhancement.) *By Khan–Zeng* [20]*, the Sugawara Virasoro element at non-critical level upgrades the holomorphic-topological symmetry of the* 3*d gauge theory to fully topological symmetry. Khan–Zeng* [20] *prove this at the classical* (*PVA*) *level; the quantum extension holds because the Sugawara element exists as an exact operator in* $V_k(\mathfrak{g})$ *at every non-critical level, satisfying the full Virasoro commutation relations without quantum corrections to the BRST-exactness argument* (*see Proposition 6.20*)*. At the critical level* $k = -h^\vee$*, the Sugawara element is undefined and the topological enhancement does not apply; the topological nature of the perturbative* E_3 *at* $k = -h^\vee + \hbar\lambda_1 + \cdots$ *holds order-by-order in* $\hbar$ (*since the Sugawara element exists at any nonzero departure from critical level*)*.*

1.2. **Architecture of the paper.** The logical sequence is forced by the construction. One needs the $\mathsf{E}_1/\mathsf{E}_\infty$ hierarchy and the ordered Ran space $\mathrm{Ran}^{\mathrm{ord}}(X)$ (§2) before building the ordered factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$

on $\mathrm{Ran}^{\mathrm{ord}}(X)$ (§3), this single $\mathcal{D}$-module before defining ordered chiral homology as its derived pushforward and establishing symmetric descent to the Beilinson–Drinfeld theory (§4), the definition before separating it from topological Hochschild through four levels of formality (§5), and the formality analysis before identifying the E_3 structure via Chern–Simons theory (§6). The examples in §7 cover all four shadow classes $G/L/C/M$. Section 8 places the construction against Ben-Zvi–Brochier–Jordan and Latyntsev. Section 9 closes with the chiral quantum group equivalence: vertex R-matrices, chiral A_∞-structures, and chiral coproducts are three projections of the ordered bar complex.

2. Background: chiral algebras and the $\mathsf{E}_1/\mathsf{E}_\infty$ hierarchy

The ordered theory rests on a single structural observation: every chiral algebra has two independent invariants (chirality hierarchy and topological ladder), and the standard Beilinson–Drinfeld theory sees only their E_∞-equivariant shadow. This section recalls the two hierarchies and the bridge theorem that identifies E_1-chiral algebras with E_2-topological data.

2.1. Two orthogonal axes.

A chiral algebra carries two independent pieces of structure: how commutative its OPE is (the chirality hierarchy, $\mathsf{E}_\infty \subset \mathcal{P}_\infty \subset \mathsf{E}_1$) and how many real dimensions its configuration spaces see (the topological ladder, $\mathsf{E}_1 \subset \mathsf{E}_2 \subset \cdots \subset \mathsf{E}_\infty$). Conflating the two is the source of most errors in the subject; separating them is the starting point of the ordered theory.

Definition 2.1 (Chirality hierarchy)**.** For chiral algebras on a fixed algebraic curve X, the *chirality hierarchy* E_∞-chiral $\subset \mathcal{P}_\infty$-chiral $\subset \mathsf{E}_1$-chiral records the commutativity of the OPE:

(i) An E_∞-*chiral algebra* has symmetric OPE: the vertex operators $Y(a,z)$ and $Y(b,w)$ commute after clearing poles. Every standard vertex algebra (Heisenberg, affine Kac–Moody, Virasoro, $\mathcal{W}$-algebras, lattice algebras) is E_∞-chiral. The symmetry means the factorisation product descends to $X^{(n)} = X^n/\Sigma_n$.

(ii) An E_1-*chiral algebra* has braided OPE: the vertex operators satisfy S-locality

$$(z-w)^N Y(a,z)Y(b,w) = (z-w)^N Y(b,w)Y(a,z) \cdot S(z-w) \tag{2.1}$$

where $S(z)$ is a vertex R-matrix satisfying the quantum Yang–Baxter equation, unitarity, the shift condition, and the hexagon axiom (Etingof–Kazhdan [13]). The factorisation product lives on ordered configurations X^n, not on $X^{(n)}$.

Definition 2.2 (The ordered Ran space)**.** For a smooth algebraic curve X, the *ordered Ran space* is

$$\mathrm{Ran}^{\mathrm{ord}}(X) = \coprod_{n\geq 1} U_n(X), \qquad U_n(X) = X^n \setminus \bigcup_{i<j} \Delta_{ij},$$

the moduli of ordered finite subsets of X: a point of $\mathrm{Ran}^{\mathrm{ord}}(X)$ is an ordered tuple $(x_1, \ldots, x_n)$ of distinct points of X. The *ordered factorisation structure* on $\mathrm{Ran}^{\mathrm{ord}}(X)$ encodes the merging of ordered subsets: for $(x_1, \ldots, x_k)$ and $(x_{k+1}, \ldots, x_n)$ with disjoint underlying sets, the concatenation $(x_1, \ldots, x_n)$ defines a map

$$\mathrm{union}_{k,n-k} \colon \mathrm{Ran}^{\mathrm{ord}}(X)_k \times_{\mathrm{disj}} \mathrm{Ran}^{\mathrm{ord}}(X)_{n-k} \to \mathrm{Ran}^{\mathrm{ord}}(X)_n.$$

Definition 2.3 (The Ran space as coinvariant quotient)**.** The Beilinson–Drinfeld *Ran space* $\mathrm{Ran}(X) = \coprod_{n\geq 1} X^{(n)}$, where $X^{(n)} = X^n/\Sigma_n$ is the n-th symmetric product, is the $\Sigma_\bullet$-coinvariant quotient:

$$\mathrm{Ran}(X) = \mathrm{Ran}^{\mathrm{ord}}(X)/\Sigma_\bullet.$$

The factorisation structure on $\mathrm{Ran}(X)$ is the $\Sigma_\bullet$-equivariant shadow of the ordered factorisation structure on $\mathrm{Ran}^{\mathrm{ord}}(X)$: for disjoint finite subsets $I, J \subset X$, the merging map $\mathrm{Ran}(X)_I \times \mathrm{Ran}(X)_J \to \mathrm{Ran}(X)_{I\cup J}$ is the descent of $\mathrm{union}_{k,n-k}$ through the symmetric group quotient.

Remark 2.4 (Chiral algebras, E_2, and the role of the curve)*.* An E_∞-chiral algebra $\mathcal{A}$ on a complex curve X is a factorisation algebra on $\mathrm{Ran}(X)$; for an E_1-chiral algebra, the factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ lives on the ordered Ran space $\mathrm{Ran}^{\mathrm{ord}}(X)$ (Definition 2.2), and descends to $\mathrm{Ran}(X)$ only when the braiding is trivial. In either case, $\mathcal{A}$ is *not* an E_2-algebra in the operadic sense. The E_2 structure lives on the symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$, provided by the curve geometry (Warning 2.5). The holomorphic data (the spectral parameter z, the r-matrix $r(z)$, the KZ connection) refines this topological E_2 structure; forgetting the complex structure of X recovers the bare E_2-coalgebra.

Warning 2.5 (Bar-cobar gives E_1; the curve gives E_2)**.** The abstract bar-cobar adjunction for E_1-algebras is:

$$\mathrm{Bar}\colon \mathsf{E}_1\text{-Alg} \rightleftarrows \mathsf{E}_1\text{-CoAlg}\colon \mathrm{Cobar}.$$

The output is an E_1-coalgebra with the *deconcatenation* coproduct: $\Delta(a_1 \otimes \cdots \otimes a_n) = \sum_{k=0}^{n}(a_1 \otimes \cdots \otimes a_k) \otimes (a_{k+1} \otimes \cdots \otimes a_n)$. This is E_1 (it respects the linear ordering of tensor factors).

The symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ of a chiral algebra on a curve X carries the *coshuffle* coproduct: $\Delta(a_1 \cdots a_n) = \sum_{I\sqcup J=\{1,\ldots,n\}} a_I \otimes a_J$ (sum over all unordered bipartitions). This is $\mathsf{E}_2/\mathsf{E}_\infty$: it uses the Σ_n-symmetry from unordered configuration spaces $\mathrm{Conf}_k(X)$ of the curve, whose real $2k$-dimensional topology carries the $\mathrm{FM}_2 \simeq \mathsf{E}_2$ operad.

The passage from $\overline{B}^{\mathrm{ord}}$ (E_1, deconcatenation) to $\overline{B}^{\Sigma}$ (E_2, coshuffle) is the Σ_n-coinvariant quotient. This quotient *introduces* the curve geometry: it replaces ordered configurations with unordered ones, and the deconcatenation with the coshuffle.

Without the curve, bar-cobar gives E_1, and Higher Deligne gives the centre E_2 ($\mathsf{E}_1 + 1$; this is the classical Deligne conjecture for associative algebras).

With the curve, the bar is E_2, and Higher Deligne gives the centre E_3 (E_2+1). The extra operadic level is the curve.

Definition 2.6 (Topological ladder)**.** The *topological ladder* $\mathsf{E}_1 \subset \mathsf{E}_2 \subset \mathsf{E}_3 \subset \cdots \subset \mathsf{E}_\infty$ is indexed by real manifold dimension:

(i) $n = 1$: operations on $\mathrm{Conf}_k(\mathbb{R})$. E_1-algebras $=$ A_∞-algebras $=$ homotopy associative algebras.
(ii) $n = 2$: operations on $\mathrm{Conf}_k(\mathbb{R}^2) \cong \mathrm{Conf}_k(\mathbb{C})$. E_2-algebras govern braided monoidal categories.
(iii) $n = 3$: operations on $\mathrm{Conf}_k(\mathbb{R}^3)$. E_3-algebras arise from perturbative Chern–Simons theory (Costello–Francis–Gwilliam [9]).

Theorem 2.7 (Bridge; [23, Theorem 1.1(ii)])**.** *The chiral bar-cobar adjunction on a complex curve X is the holomorphic refinement of the $n = 2$ chapter of E_n Koszul duality. The symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ of a chiral algebra $\mathcal{A}$ on X is an E_2-coalgebra (from the coshuffle coproduct on $\mathrm{Conf}_k(X) \cong \mathrm{Conf}_k(\mathbb{R}^2)$), not merely an E_1-coalgebra (which is what the abstract bar-cobar adjunction would give). For an E_1-chiral algebra, the braid-group action on $\mathrm{Conf}_k(\mathbb{C})$ is non-trivial (the R-matrix braiding); for an E_∞-chiral algebra, it factors through the symmetric group. In both cases, the E_2 structure on $\overline{B}^{\Sigma}$ comes from the curve, not from bar-cobar.*

Remark 2.8 (The bridge at the level of operads). The real Fulton–MacPherson compactification $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{R})$ is isomorphic to the Stasheff associahedron K_n (Sinha [29]). The inclusion $\mathbb{R} \hookrightarrow \mathbb{C}$ induces an embedding

$$K_n = \overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{R}) \hookrightarrow \overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C}) \tag{2.2}$$

of the A_∞ operad into the ordered Fulton–MacPherson operad. An E_1-chiral algebra structure is an operadic map $A_\infty \to \mathrm{End}_{\mathcal{A}}^{\mathrm{ch,ord}}$ factoring through (2.2). The image of K_n in $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$ is real-codimension n: the A_∞ operations are a thin slice of the full chiral operations.

3. $\mathcal{D}$-modules on ordered configuration spaces

The integration theory for E_∞-chiral algebras lives on symmetric products $X^{(n)} = X^n/\Sigma_n$, but the E_1 theory requires ordered configurations X^n. What are the $\mathcal{D}$-modules on X^n that encode the OPE, and what happens to them at the diagonals? The answer is the ordered factorisation $\mathcal{D}$-module: a holonomic $\mathcal{D}$-module on X^n whose singularities along the diagonals are precisely the OPE poles.

3.1. **Ordered configuration spaces and their compactifications.** The configuration space $U_n(X) = X^n \setminus \bigcup \Delta_{ij}$ and its Fulton–MacPherson compactification are the geometric substrate for ordered chiral homology. The compactification resolves the diagonals by blowup, producing boundary strata indexed by ordered collision patterns: rooted planar trees with n labelled leaves.

Definition 3.1 (Ordered configuration space)**.** Let X be a smooth algebraic curve over $\mathbb{C}$. For $n \geq 0$, the *ordered configuration space* is the smooth quasi-projective variety

$$U_n = U_n(X) := X^n \setminus \bigcup_{1 \leq i < j \leq n} \Delta_{ij} \tag{3.1}$$

where $\Delta_{ij} = \{(x_1, \ldots, x_n) \in X^n : x_i = x_j\}$ is the (i,j)-diagonal. For $n = 0$, $U_0 = \{\varnothing\} = \operatorname{Spec} \mathbb{C}$. For $n = 1$, $U_1 = X$.

Definition 3.2 (Ordered FM compactification)**.** The *ordered Fulton–MacPherson compactification* $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(X)$ is the real oriented blowup of X^n along all diagonals Δ_{ij} $(i < j)$, iterated in order of increasing codimension. It is a smooth manifold with corners whose interior is $U_n(X)$ and whose boundary strata are indexed by *ordered collision patterns*: rooted planar trees T with n labelled leaves, where each internal vertex records a group of points colliding simultaneously.

For each ordered collision pattern T with internal vertices $v_1, \ldots, v_r$ of valences $k_1, \ldots, k_r$, the corresponding boundary stratum is

$$\partial_T \overline{\mathrm{FM}}_n^{\mathrm{ord}}(X) \cong \prod_{i=1}^{r} \overline{\mathrm{FM}}_{k_i}^{\mathrm{ord}}(T_{x_i}X) \ \times \ \overline{\mathrm{FM}}_{n'}^{\mathrm{ord}}(X) \tag{3.2}$$

where $T_{x_i}X \cong \mathbb{C}$ is the tangent space at the collision point x_i, and $n' = n - \sum_i (k_i - 1)$ is the number of surviving points after collisions.

Remark 3.3 ($\mathcal{D}$-modules on ordered spaces). The category $\mathcal{D}\text{-mod}(X^n)$ of $\mathcal{D}$-modules on X^n is the standard abelian category of quasi-coherent sheaves with flat connection (Kashiwara [19], Hotta–Takeuchi–Tanisaki [17]). The diagonals $\Delta_{ij} \subset X^n$ are smooth divisors; the $*$-extension j_* and the !-extension $j_!$ along $j \colon U_n \hookrightarrow X^n$ are the standard meromorphic and distributional extensions. The classical Riemann–Hilbert correspondence applies to regular holonomic $\mathcal{D}$-modules on X^n; for irregular holonomic $\mathcal{D}$-modules, one needs the irregular Riemann–Hilbert correspondence of D'Agnolo–Kashiwara [11]. Both cases arise in the chiral setting (see Proposition 3.6 and Warning 3.7).

3.2. **The ordered factorisation $\mathcal{D}$-module.** Given the configuration spaces and their compactifications, one extends the external product $\mathcal{A}^{\boxtimes n}$ meromorphically across the diagonals. The resulting $*$-extension $j_* j^*(\mathcal{A}^{\boxtimes n})$ is the ordered factorisation $\mathcal{D}$-module: a holonomic $\mathcal{D}$-module on X^n whose singularity type along each diagonal Δ_{ij} is dictated by the OPE pole order of $\mathcal{A}$.

Construction 3.4 (Ordered factorisation $\mathcal{D}$-module)**.** Let $\mathcal{A}$ be a chiral algebra on X. The *external product* $\mathcal{A}^{\boxtimes n}$ is the $\mathcal{D}$-module on X^n defined by

$$\mathcal{A}^{\boxtimes n} := \mathcal{A} \boxtimes \mathcal{A} \boxtimes \cdots \boxtimes \mathcal{A} = \mathrm{pr}_1^* \mathcal{A} \otimes_{\mathcal{O}_{X^n}} \mathrm{pr}_2^* \mathcal{A} \otimes_{\mathcal{O}_{X^n}} \cdots \otimes_{\mathcal{O}_{X^n}} \mathrm{pr}_n^* \mathcal{A} \tag{3.3}$$

where $\mathrm{pr}_i \colon X^n \to X$ is the i-th projection.

The *ordered factorisation $\mathcal{D}$-module* at arity n is the $*$-extension:

$$\mathcal{F}_n^{\mathrm{ord}}(\mathcal{A}) := j_* j^*(\mathcal{A}^{\boxtimes n}) \in \mathcal{D}\text{-mod}(X^n). \tag{3.4}$$

Concretely, sections of $\mathcal{F}_n^{\mathrm{ord}}$ over an open $V \subset X^n$ are sections of $\mathcal{A}^{\boxtimes n}$ over $V \cap U_n$ that extend meromorphically across the diagonals. The poles along Δ_{ij} are the OPE poles: if the OPE $Y(a,z)b = \sum_{k \geq -N} a_{(k)} b\, z^{-k-1}$ has poles of order at most N for all $a, b \in \mathcal{A}$, then sections of $\mathcal{F}_n^{\mathrm{ord}}$ have poles of order at most N along each Δ_{ij}.

Definition 3.5 (Ordered factorisation $\mathcal{D}$-module on $\mathrm{Ran}^{\mathrm{ord}}(X)$)**.** For a chiral algebra $\mathcal{A}$ on X, the *ordered factorisation $\mathcal{D}$-module* is the $\mathcal{D}$-module on $\mathrm{Ran}^{\mathrm{ord}}(X)$ (Definition 2.2) whose restriction to the n-th stratum $U_n(X)$ is

$$\mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big|_{U_n} = \mathcal{A}^{\boxtimes n}\big|_{U_n}$$

with the factorisation product: for the merging map $\mathrm{union}_{k,n-k}$, the factorisation isomorphism

$$\mathrm{union}^*_{k,n-k}\mathcal{F}^{\mathrm{ord}}\big|_{U_n} \;\simeq\; \mathcal{F}^{\mathrm{ord}}\big|_{U_k} \boxtimes \mathcal{F}^{\mathrm{ord}}\big|_{U_{n-k}}$$

is the OPE (the meromorphic extension across the diagonals). The KZ connection endows $\mathcal{F}^{\mathrm{ord}}$ with a flat connection on each stratum, compatible with the factorisation maps. The arity-by-arity $\mathcal{D}$-modules $\mathcal{F}_n^{\mathrm{ord}}(\mathcal{A})$ of Construction 3.4 are the restrictions of this single object to the strata of $\mathrm{Ran}^{\mathrm{ord}}(X)$.

Proposition 3.6 (Properties of the ordered factorisation $\mathcal{D}$-module)**.** *Let $\mathcal{A}$ be a chiral algebra on a smooth proper curve X.*

(i) (Holonomicity and singularity type.) *$\mathcal{F}_n^{\mathrm{ord}}(\mathcal{A})$ is a holonomic $\mathcal{D}$-module on X^n. The singularity type along $\bigcup \Delta_{ij}$ depends on the pole order of the r-matrix:*
- Class G (Heisenberg): $r(z) = k/z$, *pole order* 1. *Regular singularities.*
- Class L (affine KM): $r(z) = k\Omega/z$ (*trace-form*), *pole order* 1. *Regular singularities.*
- Class C ($\beta\gamma$): *r is a constant nilpotent endomorphism* (*d-log absorbs the simple OPE pole*). *Regular singularities.*
- Class M (Virasoro, $\mathcal{W}_N$): $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$, *pole order* 3. Irregular *singularities of Poincaré rank* 2.

(ii) (The KZ connection.) *The flat connection on $\mathcal{F}_n^{\mathrm{ord}}(\mathcal{A})$ restricts on U_n to the standard product connection on $\mathcal{A}^{\boxtimes n}|_{U_n}$ plus the Knizhnik–Zamolodchikov correction:*

$$\nabla_n = \nabla_{\mathcal{A}}^{\boxtimes n} + \sum_{i<j} r_{ij}(z_i - z_j)\, dz_{ij} \tag{3.5}$$

where $r_{ij}(z) \in \mathrm{End}(\mathcal{A} \otimes \mathcal{A})((z))$ is the classical r-matrix (*arity*-2 *projection of $\Theta_{\mathcal{A}}$*) *acting on the (i,j) tensor factors, and $dz_{ij} = d(z_i - z_j)$ is the connection* 1*-form.*

For affine Kac–Moody at level k in the trace-form convention, this reads

$$\nabla_n^{\mathrm{KM}} = \nabla^{\boxtimes n} + \sum_{i<j} \frac{k\,\Omega_{ij}}{z_i - z_j}\, d(z_i - z_j) = \nabla^{\boxtimes n} + \sum_{i<j} \frac{\Omega_{ij}}{(k+h^\vee)(z_i - z_j)}\, d(z_i - z_j), \tag{3.6}$$

which has a simple pole *along Δ_{ij}: the connection is Fuchsian.*

(iii) (Flatness.) $\nabla_n^2 = 0$ *on U_n, following from the classical Yang–Baxter equation for $r(z)$.*

(iv) (Equivariant descent.) *For E_∞-chiral $\mathcal{A}$ (where $S(z) = \mathrm{id}$ up to Koszul signs): $\mathcal{F}_n^{\mathrm{ord}}$ is Σ_n-equivariant, and $(\mathcal{F}_n^{\mathrm{ord}})^{\Sigma_n} \simeq \mathcal{F}_n^{\Sigma}$ is the Beilinson–Drinfeld factorisation $\mathcal{D}$-module on $X^{(n)}$.*

(v) (Non-equivariance for E_1.) *For E_1-chiral $\mathcal{A}$: $\mathcal{F}_n^{\mathrm{ord}}$ is* not *Σ_n-equivariant. The braiding $S(z)$ on the (i,j) factor introduces monodromy as the i-th and j-th points circle each other.*

Proof. Part (i): holonomicity follows from j_* preserving the holonomic property (Kashiwara's theorem [19]). The external product $\mathcal{A}^{\boxtimes n}$ is holonomic on X^n (being an external tensor of holonomic $\mathcal{D}$-modules on curves). The singularity type is determined by the pole structure of the connection. For class G and L, the r-matrix $r(z)$ has a simple pole: writing the connection locally as $d + r(z)\,dz$ with $r(z) = A/z + O(1)$, the singularity is regular (Fuchsian). For class C, the d-log absorption of the OPE simple pole produces a connection with at worst logarithmic singularity: regular. For class M, the Virasoro r-matrix has a cubic pole $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$; the connection $d + r(z)\,dz$ has a pole of order 3 in dz, giving an irregular singularity of Poincaré rank $3-1=2$. Holonomic $\mathcal{D}$-modules on proper varieties have finite-dimensional de Rham cohomology regardless of the regularity of singularities (Mebkhout [27]).

Part (ii): the KZ connection arises from the Maurer–Cartan element. In the convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathsf{E}_1} = \bigoplus_n \mathrm{End}((s^{-1}\bar{\mathcal{A}})^{\otimes n})$, the MC element $\Theta_{\mathcal{A}}$ at arity 2 gives $r(z) = \Theta_{\mathcal{A}}|_{\text{arity } 2} \in \mathrm{End}(\mathcal{A}\otimes\mathcal{A})((z))$. The connection 1-form is $r_{ij}(z_i - z_j)\,dz_{ij}$ where $dz_{ij} = d(z_i - z_j)$. Note: the Arnold forms $\omega_{ij} = d\log(z_i - z_j) = dz_{ij}/(z_i - z_j)$ are generators of $H^*(\mathrm{Conf}_n(\mathbb{C}))$ and appear as coefficients in the bar differential d_{bar}; they are distinct from the connection 1-form.

Part (iii): the flatness condition $\nabla_n^2 = 0$ splits into three contributions: $(\nabla^{\boxtimes n})^2 = 0$ (individual connections); the cross-term $[\nabla^{\boxtimes n}, \sum r_{ij}\,dz_{ij}] = 0$ (from the shift condition $[\partial_z, r(z)] = -r'(z)$ paired with $d(dz_{ij}) = 0$); and $(\sum r_{ij}\,dz_{ij})^2 = 0$, which expands as $\sum_{i<j<k}([r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}])\,dz_{ij}\wedge dz_{jk} = 0$, where the coefficient vanishes by the classical Yang–Baxter equation.

Part (iv): for E_∞-chiral $\mathcal{A}$, the locality axiom gives $S(z) = \mathrm{id}$ (up to signs from the Koszul convention), so the KZ monodromy is trivial and $\mathcal{F}_n^{\mathrm{ord}}$ descends to $X^{(n)}$.

Part (v): for E_1-chiral $\mathcal{A}$, the KZ monodromy around Δ_{ij} is $\exp(2\pi i \cdot r_{ij})$, which acts by the quantum R-matrix S_{ij} (Drinfeld–Kohno theorem). Since $S \neq \mathrm{id}$, $\mathcal{F}_n^{\mathrm{ord}}$ does not descend. □

Warning 3.7 (Regularity dichotomy: gauge theory vs. gravity)**.** The singularity type of the factorisation $\mathcal{D}$-module encodes a fundamental physical distinction.

- Regular singularities (class $G/L/C$): the connection is Fuchsian; the classical Riemann–Hilbert correspondence of Deligne applies. The 3d theory is a gauge theory (Chern–Simons for class L; abelian for class G; free matter for class C).
- Irregular singularities (class M): the connection has a higher-order pole; the classical Riemann–Hilbert correspondence does not apply, and one needs the irregular Riemann–Hilbert correspondence of D'Agnolo–Kashiwara [11]. The 3d theory involves gravity (holomorphic-topological gravity in the Beltrami parametrisation for Virasoro). The irregular singularity carries Stokes data and wall-crossing phenomena (for the Virasoro algebra, at $c = -22/5$ the Stokes multipliers exhibit a wall-crossing).

The regularity type depends on the pole order of $r(z)$, which is the OPE pole order minus one (d-log absorption): class $G/L/C$ have r-matrix pole order ≤ 1, class M has pole order ≥ 2.

4. Ordered chiral homology

The ordered factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ is a holonomic $\mathcal{D}$-module with flat connection. Its de Rham cohomology is a finite-dimensional graded vector space at each arity. Assembling these across arities, with the chiral differential from FM boundary strata, produces the ordered chiral homology of $\mathcal{A}$ over X.

4.1. **Definition.** The de Rham cohomology of a holonomic $\mathcal{D}$-module on a proper variety is finite-dimensional (Mebkhout). Applied to $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on $\mathrm{Ran}^{\mathrm{ord}}(X)$, this gives a graded vector space at each arity; the cross-arity chiral differential assembles these into a chain complex.

Definition 4.1 (Ordered chiral homology via $\mathrm{Ran}^{\mathrm{ord}}(X)$)**.** Let $\mathcal{A}$ be a chiral algebra on a smooth proper curve X, and let $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ be the ordered factorisation $\mathcal{D}$-module on $\mathrm{Ran}^{\mathrm{ord}}(X)$ (Definition 3.5). The *ordered chiral homology* is the derived pushforward to the point:

$$\int_X^{\mathrm{ord}} \mathcal{A} \;:=\; R\Gamma_{\mathrm{dR}}\big(\mathrm{Ran}^{\mathrm{ord}}(X),\ \mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big) \;=\; \pi_*\mathcal{F}^{\mathrm{ord}}(\mathcal{A}) \tag{4.1}$$

where $\pi\colon \mathrm{Ran}^{\mathrm{ord}}(X) \to \mathrm{pt}$ is the projection. The arity-by-arity decomposition

$$\int_X^{\mathrm{ord}} \mathcal{A} \;=\; \Big(\bigoplus_{n\geq 0} R\Gamma_{\mathrm{dR}}\big(X^n,\ \mathcal{F}_n^{\mathrm{ord}}(\mathcal{A})\big),\ \ d_{\mathrm{ch}}\Big) \tag{4.2}$$

is the computation of this pushforward stratum by stratum, where $\mathrm{dR}(\mathcal{F}) = \mathcal{F} \otimes_{\mathcal{D}} \Omega^{\bullet}_{X^n}$ is the de Rham complex of the $\mathcal{D}$-module $\mathcal{F}$ and d_{ch} is the *chiral differential*: the alternating sum of the restriction maps to the codimension-1 boundary strata of $\overline{\mathrm{FM}}^{\mathrm{ord}}_n(X)$, corresponding to pairwise collisions of adjacent points. At each arity, the *ordered chiral homology at arity n* is

$$\int_{X,n}^{\mathrm{ord}} \mathcal{A} \;:=\; R\Gamma_{\mathrm{dR}}\big(X^n,\ \mathcal{F}^{\mathrm{ord}}_n(\mathcal{A})\big). \tag{4.3}$$

4.2. **Symmetric descent.** The quotient map $q\colon \mathrm{Ran}^{\mathrm{ord}}(X) \to \mathrm{Ran}(X)$ (Definition 2.3) induces the natural map from ordered to symmetric chiral homology.

Proposition 4.2 (Symmetric descent recovers Beilinson–Drinfeld)**.** *Let $\mathcal{A}$ be an E_∞-chiral algebra on a smooth proper curve X. The quotient $q\colon \mathrm{Ran}^{\mathrm{ord}}(X) \to \mathrm{Ran}(X)$ induces the averaging map*

$$\int_X^{\mathrm{ord}} \mathcal{A} = R\Gamma\big(\mathrm{Ran}^{\mathrm{ord}}(X),\, \mathcal{F}^{\mathrm{ord}}\big) \xrightarrow{\mathrm{av}} R\Gamma\big(\mathrm{Ran}(X),\, \mathcal{F}^{\Sigma}\big) = \int_X \mathcal{A}, \tag{4.4}$$

which is a quasi-isomorphism. At each arity, the Σ_n-coinvariants of the ordered chiral homology recover the Beilinson–Drinfeld chiral homology:

$$\Big(\int_{X,n}^{\mathrm{ord}} \mathcal{A}\Big)_{\Sigma_n} \;\simeq\; R\Gamma_{\mathrm{dR}}\big(X^{(n)},\ \mathcal{F}^{\Sigma}_n(\mathcal{A})\big) \;=:\; \int_{X,n} \mathcal{A}. \tag{4.5}$$

Proof. For E_∞-chiral $\mathcal{A}$, $\mathcal{F}^{\mathrm{ord}}_n$ is Σ_n-equivariant (Proposition 3.6(iv)). The equivariant descent for $\mathcal{D}$-modules works through the quotient stack $[X^n/\Sigma_n]$: the category of Σ_n-equivariant $\mathcal{D}$-modules on X^n is equivalent to the category of $\mathcal{D}$-modules on $[X^n/\Sigma_n]$. Since Σ_n is a finite group acting in characteristic zero, Maschke's theorem ensures that representations of Σ_n are semisimple; the quotient stack $[X^n/\Sigma_n]$ and the coarse moduli space $X^{(n)} = X^n/\Sigma_n$ have equivalent $\mathcal{D}$-module categories. The quotient $\pi\colon X^n \to X^{(n)}$ is ramified along the diagonals, but equivariant descent is not affected by ramification in characteristic zero.

The de Rham functor commutes with Σ_n-coinvariants (which are exact for the symmetric group acting on vector spaces over $\mathbb{C}$ of characteristic zero), giving: $R\Gamma_{\mathrm{dR}}(X^n, \mathcal{F}^{\mathrm{ord}}_n)_{\Sigma_n} \simeq R\Gamma_{\mathrm{dR}}(X^{(n)}, (\mathcal{F}^{\mathrm{ord}}_n)^{\Sigma_n}) \simeq R\Gamma_{\mathrm{dR}}(X^{(n)}, \mathcal{F}^{\Sigma}_n)$. The right side is the Beilinson–Drinfeld chiral homology at arity n.

The cross-arity chiral differential d_{ch} is Σ-equivariant for E_∞-chiral algebras because the factorisation maps (collision/OPE) are symmetric: the OPE $Y(a,z)b$ and $Y(b,z)a$ agree up to signs. This ensures that the total complex, not just the fixed-arity pieces, descends to the symmetric quotient. □

Proposition 4.3 (Lossy descent for E_1-chiral algebras)**.** *For an E_1-chiral algebra $\mathcal{A}$, the natural map*

$$\mathrm{av}_n\colon \int_{X,n}^{\mathrm{ord}} \mathcal{A} \;\longrightarrow\; \Big(\int_{X,n}^{\mathrm{ord}} \mathcal{A}\Big)_{\Sigma_n} \tag{4.6}$$

is surjective but not injective for $n \geq 3$. The kernel $\ker(\mathrm{av}_n)$ *carries ordered data invisible to the Beilinson–Drinfeld theory:*

(i) *At $n = 2$: the full spectral r-matrix $r(z)$ versus the scalar $\kappa = \mathrm{av}(r(z))$.*
(ii) *At $n = 3$: the Drinfeld associator Φ appears in* $\ker(\mathrm{av}_3)$*. The space of sections of* av_3 *compatible with the Maurer–Cartan equation is a* GRT_1*-torsor.*
(iii) *At $n \geq 4$: the kernel carries higher-arity ordered shadow data from the E_1 obstruction tower.*

Proof. By [23, the non-splitting proposition], the short exact sequence $0 \to \ker(\mathrm{av}) \to \mathfrak{g}_{\mathcal{A}}^{\mathsf{E}_1} \to \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} \to 0$ does not split as dg Lie algebras. The cross-arity obstruction (the bar differential is Σ_{n-2}-equivariant but not Σ_n-equivariant) forces non-trivial elements of $\ker(\mathrm{av}_n)$ at $n \geq 3$. The GRT_1-torsor structure at $n = 3$ follows from Drinfeld's theorem on associators combined with the Etingof–Kazhdan quantisation theorem. □

4.3. **Genus $g \geq 1$ curvature.** At genus zero the bar differential squares to zero and ordered chiral homology is an ordinary chain complex. At genus $g \geq 1$, this fails: the fiberwise bar differential acquires curvature proportional to the Koszul invariant $\kappa(\mathcal{A})$ times the Hodge class $\omega_g = c_1(\lambda)$ on $\overline{\mathcal{M}}_g$. The curvature is a *base* class (on moduli), not a *fiber* form (on the curve).

Remark 4.4 (Curvature at higher genus). At genus $g \geq 1$, the fiberwise bar differential acquires curvature:

$$d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g \tag{4.7}$$

where $\kappa(\mathcal{A})$ is the Koszul invariant of $\mathcal{A}$ and $\omega_g = c_1(\lambda)$ is the first Chern class of the Hodge line bundle on $\overline{\mathcal{M}}_g$; ω_g is a class on the *moduli space* $\overline{\mathcal{M}}_g$, not a form on the curve.

Three points require emphasis.

(a) The $\mathcal{D}$-module connection at *fixed* arity n remains flat: $\nabla_n^2 = 0$ on U_n. The curvature (4.7) appears in the *cross-arity* chiral differential d_{ch}, which connects different arities.
(b) The curvature $\kappa(\mathcal{A}) \cdot \omega_g$ is a *scalar* (it acts on all tensor factors uniformly), so it is Σ_n-invariant: it survives descent to the symmetric quotient unchanged.
(c) The period-corrected differential $D^{(g)}$ of [23, §II.3] restores $d^2 = 0$ at the total level by incorporating the Hodge correction.

5. Topological Hochschild versus ordered chiral homology of $D^\times$

Two chain complexes compute invariants of a chiral algebra on a punctured disk: the cyclic bar complex (topological Hochschild, from the circle $S^1 = \partial D$) and the ordered chiral chain complex (from the punctured disk $D^\times$ itself, with its complex structure). For E_∞-chiral algebras, operad formality identifies

the two. For E_1-chiral algebras, they diverge: the ordered complex carries holomorphic data from the spectral parameter that the cyclic bar complex cannot see.

5.1. **Two computations.** Let $\mathcal{A}$ be a chiral algebra on a curve X. Fix a point $p \in X$ and let $D = \operatorname{Spec} \mathbb{C}[[z]]$ be the formal disk at p, with $D^\times = \operatorname{Spec} \mathbb{C}((z))$ the punctured formal disk. The two chain complexes are the following.

Definition 5.1 (Topological Hochschild)**.** The *topological Hochschild homology* of $\mathcal{A}$ is the factorisation homology of $\mathcal{A}$, viewed as an E_1-topological algebra, over the circle $S^1 = \partial D$:

$$\mathrm{HH}_*^{\mathrm{top}}(\mathcal{A}) := \int_{S^1} \mathcal{A}. \tag{5.1}$$

This is computed by the cyclic bar complex

$$B_n^{\mathrm{cyc}}(\mathcal{A}) = \mathcal{A}^{\otimes n} \Big/ \big(a_1 \otimes \cdots \otimes a_n \sim (-1)^{|a_n|(|a_1|+\cdots+|a_{n-1}|)} a_n \otimes a_1 \otimes \cdots \otimes a_{n-1}\big) \tag{5.2}$$

with the Hochschild differential $d_{\mathrm{HH}}(a_1, \ldots, a_n) = \sum_{i=1}^{n-1} \pm(a_1, \ldots, m_2(a_i, a_{i+1}), \ldots, a_n) + \pm(m_2(a_n, a_1), a_2, \ldots, a_{n-1})$ and, for algebras with non-trivial A_∞-structure, the higher terms involving m_k for $k \geq 3$.

The input data: the associative product $m_2 \colon \mathcal{A} \otimes \mathcal{A} \to \mathcal{A}$ and the higher A_∞ operations m_k. No dependence on z; no meromorphic forms; no Arnold relations; no configuration-space geometry.

Definition 5.2 (Ordered chiral homology of $D^\times$)**.** The *ordered chiral chain complex on $D^\times$ at arity n* is:

$$\mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A}) := \Big(\Omega^*_{\log}\big(\overline{\mathrm{FM}}_n^{\mathrm{ord}}(D^\times)\big) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n},\ \ d_{\mathrm{dR}} + d_{\mathrm{bar}} + \nabla_{\mathrm{KZ}}\Big). \tag{5.3}$$

The ingredients:

(a) $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(D^\times)$: the ordered FM compactification of configurations of n distinct points in $D^\times$. This is a smooth manifold with corners.

(b) $\Omega^*_{\log}$: logarithmic differential forms, with logarithmic singularities along the boundary divisors of the FM compactification. The Arnold forms $\omega_{ij} = d\log(z_i - z_j)$ are among the generators; these are coefficients in d_{bar}, not in ∇_{KZ}.

(c) d_{bar}: the bar differential encoding the full OPE. At the codimension-1 boundary stratum where $z_i \to z_j$ $(i < j)$, the bar differential extracts the residue:

$$d_{\mathrm{bar}}\big(a_1 \otimes \cdots \otimes a_n\big)\big|_{\Delta_{ij}} = \sum_k \pm\, a_1 \otimes \cdots \otimes a_{i\,(k)}\, a_j \otimes \cdots \otimes a_n \ \otimes\ \omega_{ij}^{(k)} \tag{5.4}$$

where $a_{(k)}b$ are the OPE modes and $\omega_{ij}^{(k)}$ are the corresponding forms on the boundary stratum.

(d) ∇_{KZ}: the KZ connection

$$\nabla_{\mathrm{KZ}} = \sum_{i<j} r_{ij}(z_i - z_j)\, d(z_i - z_j), \tag{5.5}$$

with connection 1-form $r(z)\,dz$, not $r(z)\,d\log z$.

The *ordered chiral homology of* $D^\times$ is the derived pushforward of $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ over $\mathrm{Ran}^{\mathrm{ord}}(D^\times)$; the arity-by-arity computation gives:

$$\int_{D^\times}^{\mathrm{ord}} \mathcal{A} = R\Gamma_{\mathrm{dR}}\big(\mathrm{Ran}^{\mathrm{ord}}(D^\times),\ \mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big) = \bigoplus_{n\geq 0} H^*\big(\mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A})\big). \tag{5.6}$$

The input data: the full OPE $Y(a,z)b = \sum_k a_{(k)}b\, z^{-k-1}$, the Arnold forms on $\mathrm{Conf}_n^{\mathrm{ord}}(D^\times)$, the FM compactification, and the r-matrix. The output depends on the complex structure of $D^\times$ (the coordinate z).

5.2. **Four levels of formality.** Operad formality (Kontsevich–Tamarkin) guarantees that ordered and symmetric give the same answer for E_∞-chiral algebras. But this does not guarantee simplicity: the transferred A_∞-structure may carry non-trivial higher operations. Four formality conditions, each strictly stronger than the last, control the relationship between ordered chiral homology and topological Hochschild.

Definition 5.3 (Formality hierarchy)**.** (a) (Operad formality.) The chain operad $C_*(\overline{\mathrm{FM}}_n(\mathbb{C}))$ is formal as an operad: it is quasi-isomorphic to its cohomology $H^*(\overline{\mathrm{FM}}_n(\mathbb{C}))$ via a quasi-isomorphism depending on a Drinfeld associator Φ (Kontsevich–Tamarkin). This is *unconditional*: it depends only on the operad, not on any algebra.

(b) (Algebra formality.) The transferred A_∞-structure on $H^*(\mathcal{A})$ has $m_k^{\mathrm{tr}} = 0$ for $k \geq 3$. This depends on the *specific algebra*: it holds for class G (Heisenberg), and fails for class $L/C/M$.

(c) (Convolution algebra formality.) The modular L_∞ brackets $\ell_k = 0$ for $k \geq 3$ in $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$. This holds *only* for class G ([23, prop:sc-formal-iff-class-g]).

(d) (SC-formality.) The Swiss-cheese type operations $m_k^{\mathrm{SC}} = 0$ for $k \geq 3$. This is equivalent to class G.

Operad formality (a) is the strongest general result; it guarantees that ordered and symmetric give the *same answer as chain complexes* for E_∞-chiral algebras. It does *not* guarantee that the answer is simple: the transferred A_∞-structure may carry non-trivial higher operations (levels (b)–(d) fail).

Theorem 5.4 (Formality bridge for E_∞-chiral algebras)**.** *Let $\mathcal{A}$ be an E_∞-chiral algebra and Φ a Drinfeld associator. The Kontsevich–Tamarkin operad formality quasi-isomorphism induces a quasi-isomorphism of chain complexes:*

$$\mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A}) \xrightarrow[\Phi]{\sim} B_n^{\mathrm{cyc}}(\mathcal{A}). \tag{5.7}$$

In particular, $\int_{D^\times}^{\mathrm{ord}} \mathcal{A} \simeq \mathrm{HH}_*^{\mathrm{top}}(\mathcal{A})$ as graded vector spaces.

The transferred A_∞-structure on the cohomological model $H^*(\mathcal{A})$ may still carry non-trivial higher operations m_k^{tr} for $k \geq 3$. For class L (affine KM), class C ($\beta\gamma$), and class M (Virasoro): these transferred operations are non-trivial. What operad formality guarantees is that the ordered and symmetric complexes compute the same cohomology.

Proof. The proof proceeds in six steps: the KT formality theorem, the identification of the two chain complexes, the transfer of the bar differential, the vanishing of higher OPE modes, the transferred A_∞-structure, and the failure for genuinely E_1-chiral algebras.

Step 1: Kontsevich–Tamarkin operad formality. The chain operad $C_*(\overline{\mathrm{FM}}_n(\mathbb{C}))$, with structure maps given by the operad composition on Fulton–MacPherson compactifications, is formal: there exists a quasi-isomorphism of operads

$$\varphi_\Phi \colon C_*(\overline{\mathrm{FM}}_n(\mathbb{C})) \xrightarrow{\sim} H^*(\overline{\mathrm{FM}}_n(\mathbb{C})) \tag{5.8}$$

depending on the choice of a Drinfeld associator $\Phi \in \exp(\hat{\mathfrak{t}}_3)$ (Kontsevich [6], Tamarkin [7]). The target $H^*(\overline{\mathrm{FM}}_n(\mathbb{C}))$ is the Arnold algebra: it is generated by classes $\omega_{ij} = [d\log(z_i - z_j)]$ of degree 1 ($1 \leq i < j \leq n$), subject to the Arnold relation $\omega_{ij}\omega_{jk} + \omega_{jk}\omega_{ki} + \omega_{ki}\omega_{ij} = 0$ for all triples $i < j < k$ (Arnold [1]). The quasi-isomorphism φ_Φ is Σ_n-equivariant and respects the operad composition maps; these properties are inherited from the Σ_n-equivariance of the FM operad structure.

Step 2: Identification of the two chain complexes. The ordered chiral chain complex at arity n (Definition 5.2) is the tensor product

$$\mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A}) = \Omega^*_{\log}\big(\overline{\mathrm{FM}}_n^{\mathrm{ord}}(D^\times)\big) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n}$$

with the total differential $d_{\mathrm{dR}} + d_{\mathrm{bar}} + \nabla_{\mathrm{KZ}}$. The de Rham factor carries the full algebra of logarithmic differential forms on the FM compactification. The cyclic bar complex $B_n^{\mathrm{cyc}}(\mathcal{A})$ uses only the cohomology $H^*(\overline{\mathrm{FM}}_n^{\mathrm{ord}})$, which is the Arnold algebra. The quasi-isomorphism (5.8) induces a chain map from the de Rham model to the Arnold model:

$$\varphi_\Phi \otimes \mathrm{id} \colon \Omega^*_{\log}(\overline{\mathrm{FM}}_n^{\mathrm{ord}}) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n} \longrightarrow H^*(\overline{\mathrm{FM}}_n^{\mathrm{ord}}) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n}. \tag{5.9}$$

Since φ_Φ is a quasi-isomorphism of differential graded vector spaces, so is $\varphi_\Phi \otimes \mathrm{id}$ (by the Künneth theorem; the algebra factor $(s^{-1}\bar{\mathcal{A}})^{\otimes n}$ is degreewise finite-dimensional).

Step 3: Transfer of the bar differential. The bar differential d_{bar} (equation (5.4)) acts at the codimension-1 boundary stratum Δ_{ij} where $z_i \to z_j$, extracting the full OPE:

$$d_{\mathrm{bar}}\big(a_1 \otimes \cdots \otimes a_n\big)\big|_{\Delta_{ij}} = \sum_{k \geq 0} \pm\, a_1 \otimes \cdots \otimes a_{i\,(k)}\, a_j \otimes \cdots \otimes a_n \;\otimes\; \omega_{ij}^{(k)}$$

where $\omega_{ij}^{(k)}$ is the differential form on the boundary stratum that pairs with OPE mode $a_{(k)}b$. For $\mathcal{A}$ an E_∞-chiral algebra, the $\mathcal{D}$-module underlying the OPE is Σ_n-equivariant: the braiding is trivial ($S(z) = \mathrm{id}$), so d_{bar} commutes

with the Σ_n-action on $\overline{\mathrm{FM}}_n^{\mathrm{ord}}$. Since φ_Φ is a Σ_n-equivariant operad map, the transferred differential $d_{\mathrm{bar}}^{\mathrm{tr}} = \pi \circ d_{\mathrm{bar}} \circ \iota$, where $\iota\colon H^* \to \Omega^*_{\log}$ is the inclusion of cohomology representatives and $\pi\colon \Omega^*_{\log} \to H^*$ is the projection from the homotopy transfer data of φ_Φ, is well defined on the target complex.

Step 4: Vanishing of higher OPE modes. The OPE $Y(a,z)b = \sum_{k\geq 0} a_{(k)}b\, z^{-k-1}$ pairs each mode with a differential form on the FM compactification. The leading mode $a_{(0)}b = m_2(a,b)$ pairs with the 1-form $\omega_{ij} = d\log(z_i - z_j)$: this is the Arnold generator in degree 1, and it represents a non-trivial cohomology class in $H^1(\overline{\mathrm{FM}}_n^{\mathrm{ord}})$.

The higher modes $a_{(k)}b$ for $k \geq 1$ pair with forms $\omega_{ij}^{(k)}$ supported on the boundary strata of the FM compactification. These forms arise from the Laurent expansion of the OPE along the exceptional divisor of the blowup $\overline{\mathrm{FM}}_n \to \mathbb{C}^n$: the mode $a_{(k)}b$ multiplies the residue form $z_{ij}^{-k-1}\, dz_{ij}$ by a regular form on the remaining coordinates, where $z_{ij} = z_i - z_j$ is the local coordinate on the boundary stratum. For $k \geq 1$, the pole order exceeds 1, so the corresponding form $\omega_{ij}^{(k)}$ involves derivatives of the Green kernel along the exceptional fiber.

Under the projection $\pi\colon \Omega^*_{\log}(\overline{\mathrm{FM}}_n^{\mathrm{ord}}) \to H^*(\overline{\mathrm{FM}}_n^{\mathrm{ord}})$, these boundary forms vanish:

$$\pi(\omega_{ij}^{(k)}) = 0 \in H^*(\overline{\mathrm{FM}}_n^{\mathrm{ord}}) \quad \text{for all } k \geq 1. \tag{5.10}$$

The argument proceeds by degree analysis in the Arnold algebra. The cohomology $H^*(\mathrm{Conf}_n(\mathbb{C}))$ is generated in degree 1 by the classes $[\omega_{ij}]$; every class of degree ≥ 2 is decomposable, i.e. a product of degree-1 generators subject to the Arnold relation. The forms $\omega_{ij}^{(k)}$ for $k \geq 1$ are localised on the codimension-1 boundary stratum Δ_{ij} and carry fiber degree ≥ 1 along the exceptional divisor, in addition to the base degree from ω_{ij} itself. They are indecomposable in the fiber direction: each involves the local coordinate of a single boundary stratum and cannot be expressed as a product of Arnold generators from distinct boundary strata. Since the Arnold algebra has no indecomposable classes in degree ≥ 2, these forms must be exact in $\Omega^*_{\log}(\overline{\mathrm{FM}}_n^{\mathrm{ord}})$. Explicitly, $\omega_{ij}^{(k)}$ for $k \geq 1$ is d_{dR} of a form of lower degree supported on Δ_{ij}, so it lies in the kernel of π.

The transferred bar differential on the Arnold model therefore reduces to the contribution from $k = 0$ alone:

$$d_{\mathrm{bar}}^{\mathrm{tr}}\big(a_1 \otimes \cdots \otimes a_n\big) = \sum_{i<j} \pm\, a_1 \otimes \cdots \otimes m_2(a_i, a_j) \otimes \cdots \otimes a_n \ \otimes\ [\omega_{ij}].$$

This is precisely the Hochschild differential d_{HH} on the cyclic bar complex $B_n^{\mathrm{cyc}}(\mathcal{A})$.

Step 5: Transferred A_∞-structure and independence of the chain-level comparison. The homotopy transfer theorem (Kadeishvili [5]) applied to the quasi-isomorphism φ_Φ produces a transferred A_∞-structure $\{m_k^{\mathrm{tr}}\}_{k\geq 2}$ on $H^*(\mathcal{A})$: $m_2^{\mathrm{tr}} = m_2$, and for $k \geq 3$ the operations m_k^{tr} are given by sums

over planar rooted trees with k leaves, each internal edge decorated by the contracting homotopy $h\colon \Omega^*_{\log} \to \Omega^{*-1}_{\log}$ (satisfying $\mathrm{id} - \iota\pi = [d_{\mathrm{dR}}, h]$) and each vertex by the original multiplication.

These transferred operations encode the interaction between the OPE pole structure and the configuration-space geometry: they are non-trivial precisely when algebra-level formality (Definition 5.3(b)) fails. This occurs for class L (affine Kac–Moody), class C ($\beta\gamma$), and class M (Virasoro, $\mathcal{W}_N$).

The chain-level quasi-isomorphism $\mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A}) \xrightarrow{\sim} B_n^{\mathrm{cyc}}(\mathcal{A})$ does *not* depend on these higher operations. The map $\varphi_\Phi \otimes \mathrm{id}$ is a quasi-isomorphism of chain complexes by Step 2, and the transferred bar differential equals d_{HH} by Step 4. The two complexes compute the same cohomology, establishing (5.7), regardless of whether the A_∞-structure is formal. What m_k^{tr} for $k \geq 3$ controls is the *multiplicative* structure on the cohomology: the A_∞-algebra refinement of $H^*(\mathcal{A})$ as an algebra over the operad $H^*(\overline{\mathrm{FM}})$.

Step 6: Why the argument fails for genuinely E_1-chiral algebras. For a genuinely E_1-chiral algebra, the R-matrix braiding $S(z) \neq \mathrm{id}$ breaks the Σ_n-equivariance of the bar differential: d_{bar} no longer commutes with the Σ_n-action, because the OPE $Y(a,z)b$ and $Y(b,-z)a$ differ by $S(z) \neq \mathrm{id}$. The $\mathcal{D}$-module underlying the OPE does not descend to the symmetric quotient $\mathrm{Conf}_n(\mathbb{C})/\Sigma_n$, so the transfer through φ_Φ cannot reduce the ordered complex to the cyclic bar complex. The ordered chiral complex retains holomorphic data from the spectral-parameter dependence of $S(z)$ that has no counterpart in B_n^{cyc} (Theorem 5.5). □

Theorem 5.5 (Formality failure for genuinely E_1-chiral algebras)**.** *Let $\mathcal{A}$ be a genuinely E_1-chiral algebra (tier (c): the R-matrix $S(z)$ is not derived from any E_∞-chiral OPE; $S(z) \neq$ id). The chain-level formality quasi-isomorphism (5.7) does* not *exist.*

(i) (Operad formality holds.) *The Kontsevich–Tamarkin formality of $C_*(\overline{\mathrm{FM}}_n(\mathbb{C}))$ as an operad is unconditional: it depends only on the operad structure, not on the specific algebra.*

(ii) (Equivariance fails.) *The $\mathcal{D}$-module $\mathcal{F}_n^{\mathrm{ord}}$ is not Σ_n-equivariant (because $S(z) \neq$ id), so the ordered complex does* not *descend to the symmetric quotient. The ordered chiral homology is strictly richer than topological Hochschild: the ordered complex carries holomorphic data from the spectral parameter dependence of $S(z)$ that has no counterpart in the cyclic bar complex.*

(iii) (Dimension count.) *By Arnold's theorem, the cohomology algebra $H^*(\mathrm{Conf}_n(\mathbb{C}))$ has Poincaré polynomial $\prod_{j=0}^{n-1}(1+jt)$, with total dimension $n!$. The ordered chiral chain complex at arity n therefore has*

$$\dim \mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A}) = (\dim \mathcal{A})^n \cdot n!, \tag{5.11}$$

while the cyclic bar complex has $\dim B_n^{\mathrm{cyc}} = (\dim \mathcal{A})^n$. *The ratio* $n!$ *measures the additional holomorphic data carried by the ordered chiral complex beyond the topological Hochschild complex.*

Proof of the dimension count. The Poincaré polynomial of $H^*(\mathrm{Conf}_n(\mathbb{C}))$ is $P_n(t) = \prod_{j=0}^{n-1}(1+jt)$ (Arnold [1]). Setting $t = 1$ gives $P_n(1) = \prod_{j=0}^{n-1}(1+j) = n!$. $P_2(1) = 1 \cdot 2 = 2$; at $n = 3$, $P_3(1) = 1 \cdot 2 \cdot 3 = 6$; at $n = 4$, $P_4(1) = 1 \cdot 2 \cdot 3 \cdot 4 = 24$. The ordered chiral complex is $\Omega^*_{\log}(\overline{\mathrm{FM}}_n^{\mathrm{ord}}) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n}$; at the cohomological level, the $\Omega^*_{\log}$ factor contributes the Arnold algebra, whose total dimension is $n!$. □

6. The E_3 structure and Chern–Simons theory

The symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ of a chiral algebra on a curve X is an E_2-coalgebra (from the coshuffle coproduct on $\mathrm{Conf}_k(X) \cong \mathrm{Conf}_k(\mathbb{R}^2)$). The Higher Deligne Conjecture promotes the Hochschild cohomology of an E_2-algebra to E_3. Applied to $\overline{B}^{\Sigma}(\mathcal{A})$, this gives the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ a natural E_3-algebra structure: E_2 from the curve, $+1$ from the centre construction, $= \mathsf{E}_3$. The comparison with perturbative Chern–Simons theory (Costello–Francis–Gwilliam) identifies this E_3 structure with the bulk observables of the 3d gauge theory.

6.1. The derived chiral centre.

The ordered and symmetric chiral Hochschild cochain complexes differ by the presence or absence of Σ_k-coinvariants. The derived chiral centre is the cohomology of the symmetric version; its ordered refinement carries the full E_1 data.

Definition 6.1 (Symmetric and ordered chiral Hochschild complexes)**.** Let $\mathcal{A}$ be a chiral algebra on a curve X.

(i) The *symmetric chiral Hochschild cochain complex* is:

$$\mathrm{HH}^{*,\Sigma}_{\mathrm{ch}}(\mathcal{A},\mathcal{A}) = \bigoplus_{k\geq 0} \Omega^*(\overline{\mathrm{FM}}_k(\mathbb{C})) \otimes_{\Sigma_k} \mathrm{Hom}(\mathcal{A}^{\otimes k}, \mathcal{A}), \tag{6.1}$$

with the chiral Hochschild differential. The derived chiral centre is $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) := \mathrm{HH}^{*,\Sigma}_{\mathrm{ch}}(\mathcal{A},\mathcal{A})$.

(ii) The *ordered chiral Hochschild cochain complex* is:

$$\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(\mathcal{A},\mathcal{A}) = \bigoplus_{k\geq 0} \Omega^*(\overline{\mathrm{FM}}_k^{\mathrm{ord}}(\mathbb{C})) \otimes \mathrm{Hom}(\mathcal{A}^{\otimes k}, \mathcal{A}), \tag{6.2}$$

without Σ_k-coinvariants, with the ordered chiral Hochschild differential.

Proposition 6.2 (The E_3 structure on the derived centre)**.** *Let* $\mathcal{A}$ *be a chiral algebra on a smooth algebraic curve* X.

(i) (Source of the E_2 structure.) *The curve geometry provides the* E_2-*coalgebra structure on* $\overline{B}^{\Sigma}(\mathcal{A})$ (*Warning* 2.5).

(ii) (Higher Deligne gives E_3.) *By the Higher Deligne Conjecture (Lurie [18], Francis [15]), the Hochschild cohomology of an E_2-algebra carries a natural E_3-algebra structure. Applied to $\overline{B}^{\Sigma}(\mathcal{A})$:*

$$Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = \mathrm{HH}^*(\overline{B}^{\Sigma}(\mathcal{A}), \overline{B}^{\Sigma}(\mathcal{A})) \quad \textit{is an } \mathsf{E}_3\textit{-algebra.} \tag{6.3}$$

The dimension count: E_2 (curve, $\dim_{\mathbb{R}} X = 2$) $+1$ (centre construction) $= \mathsf{E}_3$.

(iii) *The E_3 structure exists for* any *chiral algebra $\mathcal{A}$ on a curve (E_∞-chiral or E_1-chiral). It does not require a Virasoro element, conformal symmetry, or any additional hypothesis beyond the curve geometry.*

Warning 6.3 (The bar complex itself is not E_3)**.** The symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ does *not* carry an E_3-algebra structure. The obstruction is topological: E_3 requires $\mathrm{Conf}_k(\mathbb{R}^3)$, but a curve provides only $\mathrm{Conf}_k(\mathbb{R}^2)$. For $\widehat{\mathfrak{sl}}_2$, the obstruction space $\mathrm{HH}^3_{\mathsf{E}_2}(\overline{B}^{\Sigma}, \overline{B}^{\Sigma})$ is one-dimensional and the obstruction class is non-zero. The E_3-algebra is the *derived centre*, not the bar complex.

6.2. **The CFG comparison.** Costello, Francis, and Gwilliam construct a filtered E_3-algebra $\mathcal{A}^\lambda$ from BV quantisation of Chern–Simons theory on $\mathbb{R}^3$. The derived chiral centre of $V_k(\mathfrak{g})$ and the CFG algebra both deform $C^*(\mathfrak{g})$ over $\hbar H^3(\mathfrak{g})[[\hbar]]$. The question is whether these deformation families are isomorphic as E_3-algebras.

Theorem 6.4 (Costello–Francis–Gwilliam [9])**.** *Let $\mathfrak{g}$ be a simple finite-dimensional Lie algebra. BV quantisation of Chern–Simons theory on $\mathbb{R}^3$ with gauge algebra $\mathfrak{g}$ and invariant pairing λ yields a filtered E_3-algebra $\mathcal{A}^\lambda$ on the Chevalley–Eilenberg cochains $C^*(\mathfrak{g})$.*

(i) *The classical observables are $C^*(\mathfrak{g}) = \mathrm{Sym}(\mathfrak{g}^\vee[-1])$ with the CE differential: a commutative (E_∞) algebra.*

(ii) *The quantum observables $\mathcal{A}^\lambda$ are a filtered E_3-algebra deforming $C^*(\mathfrak{g})$, with deformation space $\hbar H^3(\mathfrak{g})[[\hbar]]$.*

(iii) *For any E_3-algebra $\mathcal{A}$, $\mathrm{Mod}_{\mathcal{A}}$ is E_2-monoidal (braided monoidal). Filtered Koszul duality gives $\mathrm{Perf}_{C^*(\mathfrak{g})} \simeq \mathrm{Rep}_{\mathrm{fin}}(\mathfrak{g})^{\mathrm{dg}}$, so deforming $C^*(\mathfrak{g})$ as E_3 deforms $\mathrm{Rep}_{\mathrm{fin}}(\mathfrak{g})$ as braided monoidal: this recovers the Drinfeld–Jimbo quantum group.*

(iv) (Theorem 1.4 of [9].) *Canonical bijection between: perturbative CS quantisations, filtered E_3-algebras deforming $C^*(\mathfrak{g})$, braided monoidal deformations of $\mathrm{Rep}_{\mathrm{fin}}(\mathfrak{g})$, and quasi-triangular quasi-Hopf algebras deforming $U(\mathfrak{g})$. All non-canonically isomorphic to $\hbar H^3(\mathfrak{g})[[\hbar]]$.*

Conjecture 6.5 (Deformation families of E_3-algebras)**.** *For the affine Kac–Moody vertex algebra $V_k(\mathfrak{g})$ at generic level k, the E_3-algebra $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ of Proposition 6.2 and the CFG E_3-algebra $\mathcal{A}^\lambda$ are related as follows.*

(i) (Matching deformation spaces.) *Both $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ and $\mathcal{A}^\lambda$ are parametrised by $\hbar H^3(\mathfrak{g})[[\hbar]]$: for the derived chiral centre, the deformation parameter is the departure from critical level $k = -h^\vee + \hbar\lambda_1 + \hbar^2\lambda_2 + \cdots$;*

for the CFG algebra, it is the Chern–Simons coupling. This matching of deformation spaces *is a necessary condition for isomorphism.*

(ii) (Base point mismatch.) *At the critical level $k = -h^\vee$, the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(V_{-h^\vee}(\mathfrak{g}))$ is related to the Feigin–Frenkel centre* $\mathrm{Fun}(\mathrm{Op}_{\mathfrak{g}^\vee}(D))$ (*opers on the formal disk*), *not directly to $C^*(\mathfrak{g})$. The comparison requires a localisation or restriction functor, and it is at this base point that the identification is most delicate.*

(iii) (Formal deformation families.) *We conjecture that the formal deformation* families *are isomorphic:*

$$\{Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))\}_{k\in -h^\vee+\hbar\mathbb{C}[[\hbar]]} \;\simeq\; \{\mathcal{A}^\lambda\}_{\lambda\in\hbar H^3(\mathfrak{g})[[\hbar]]} \tag{6.4}$$

as families of E_3-algebras over $\hbar H^3(\mathfrak{g})[[\hbar]]$, rather than as individual fibers.

(iv) (Sugawara constraint.) *The topological enhancement* (*Theorem 1.3(iv)*) *holds at generic k but fails at $k = -h^\vee$, where the Sugawara element is undefined. The perturbative identification at $k = -h^\vee + \hbar\lambda_1 + \cdots$ holds order-by-order in $\hbar$, since the Sugawara element exists at any nonzero $\hbar$.*

(v) (What is proved.) *The deformation* spaces *agree* (*both $\hbar H^3(\mathfrak{g})[[\hbar]]$*), *which is a necessary condition but not sufficient for isomorphism. The comparison at generic k is accessible through the algebraic string dictionary of* [23].

6.3. **Explicit E_3 operations for $V_k(\mathfrak{sl}_2)$.** The abstract E_3 structure of Proposition 6.2 acquires concrete content for $\mathfrak{g} = \mathfrak{sl}_2$: the five-dimensional derived chiral centre $\mathbb{C} \oplus \mathfrak{sl}_2[-1] \oplus \mathbb{C}[-2]$ carries an explicit cup product, Gerstenhaber bracket, P_3 bracket, and BV operator, all computable from the OPE.

The underlying graded vector space. By Proposition 6.2 and the chiral Hochschild computation of Example 7.2, the derived chiral centre of $V_k(\mathfrak{sl}_2)$ at generic level $k \neq -2$ is the graded vector space

$$Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{sl}_2)) \;=\; \underbrace{\mathbb{C}}_{=\mathrm{HH}^0} \oplus \underbrace{\mathfrak{sl}_2[-1]}_{=\mathrm{HH}^1} \oplus \underbrace{\mathbb{C}[-2]}_{=\mathrm{HH}^2}, \tag{6.5}$$

with total dimension $1 + 3 + 1 = 5$. Write $\mathbf{1}$ for the generator of $\mathrm{HH}^0 = \mathbb{C}$ (the vacuum), $\{e, f, h\}$ for the standard basis of $\mathrm{HH}^1 = \mathfrak{sl}_2$ (outer derivations: infinitesimal level-preserving deformations parametrised by $\mathfrak{g}$ itself), and η for the generator of $\mathrm{HH}^2 = \mathbb{C}$ (the central charge deformation cocycle). The $\mathfrak{sl}_2$-module structure is: HH^0 trivial, HH^1 adjoint, HH^2 trivial.

Conventions. The Killing form normalisation is $(e, f) = (f, e) = 1$, $(h, h) = 2$, all other pairings zero. Structure constants: $[e, f] = h$, $[h, e] = 2e$, $[h, f] = -2f$. The inverse Casimir is $\Omega = e \otimes f + f \otimes e + \frac{1}{2}\, h \otimes h$, and $h^\vee = 2$. The KZ coupling constant is $h_{\mathrm{KZ}} = 1/(k + h^\vee) = 1/(k + 2)$.

Remark 6.6 (Notation: h_{KZ} vs $\hbar$). Two closely related parameters appear throughout this paper. The *KZ coupling* $h_{\mathrm{KZ}} = 1/(k+h^\vee)$ is the normalisation constant in the KZ connection $\nabla = d + h_{\mathrm{KZ}} \sum \Omega_{ij}\, dz_{ij}$; it vanishes as $k \to \infty$ (weak-coupling limit) and diverges at the critical level $k = -h^\vee$. The *departure from critical level* $\hbar = k + h^\vee$ is the complementary parameter: $\hbar = 0$ at critical level, $\hbar \to \infty$ at weak coupling, and $h_{\mathrm{KZ}} = 1/\hbar$. In the $\mathfrak{sl}_2$ computation below, $h^\vee = 2$, so $h_{\mathrm{KZ}} = 1/(k+2)$ and $\hbar = k+2$. We write h_{KZ} for the P_3 bracket normalisation (where the KZ coupling is the natural parameter) and $\hbar$ for deformation-theoretic contexts (where the departure from critical level controls the formal neighbourhood).

An E_3-algebra carries three levels of operations:

- Level 0 (E_∞): a graded-commutative product $\mu\colon \mathrm{HH}^p \otimes \mathrm{HH}^q \to \mathrm{HH}^{p+q}$.
- Level 1 (E_2 beyond E_1): the Gerstenhaber bracket $[-,-]\colon \mathrm{HH}^p \otimes \mathrm{HH}^q \to \mathrm{HH}^{p+q-1}$ of degree -1, derived from the fundamental class of $S^1 = \mathrm{Conf}_2(\mathbb{R}^2)/\mathbb{R}^+$.
- Level 2 (E_3 beyond E_2): the P_3 bracket $\{-,-\}\colon \mathrm{HH}^p \otimes \mathrm{HH}^q \to \mathrm{HH}^{p+q-2}$ of degree -2, derived from the fundamental class of $S^2 = \mathrm{Conf}_2(\mathbb{R}^3)/\mathbb{R}^+$.

We compute each in turn.

Proposition 6.7 (Explicit E_3 operations on $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{sl}_2))$)**.** *Let $k \neq -2$. The E_3-algebra structure on $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{sl}_2))$ is determined by the following data.*

(I) Cup product *(E_∞ level).*

(a) $\mathbf{1}$ *is the unit:* $\mathbf{1}\cdot f = f \cdot \mathbf{1} = f$ *for all* f.
(b) $\mu(X,Y) = 0$ *for all* $X, Y \in \mathrm{HH}^1 = \mathfrak{sl}_2$.
(c) $\mu(X,\eta) = 0$ *for all* $X \in \mathrm{HH}^1$, *and* $\eta\cdot\eta = 0$.

(II) Gerstenhaber bracket *(E_2 level, degree -1).*

$$[\mathbf{1},-] = 0, \qquad [X,Y] = 0 \quad \textit{for all } X,Y \in \mathrm{HH}^1 = \mathfrak{sl}_2,$$
$$[X,\eta] = 0 \quad \textit{for all } X \in \mathrm{HH}^1, \qquad [\eta,\eta] = 0.$$

(III) The P_3 bracket *(E_3 level, degree -2).*

$$\{X,Y\} = h_{\mathrm{KZ}} \cdot (X,Y) = \frac{1}{k+h^\vee}\,(X,Y) \quad \textit{for } X,Y \in \mathrm{HH}^1, \tag{6.6}$$
$$\{X,\eta\} = 0 \quad \textit{for } X \in \mathrm{HH}^1, \tag{6.7}$$
$$\{\mathbf{1},-\} = 0, \tag{6.8}$$
$$\{\eta,\eta\} = 0, \tag{6.9}$$

where (X,Y) denotes the normalised Killing form and $h_{\mathrm{KZ}} = 1/(k+h^\vee) = 1/(k+2)$ is the KZ coupling constant (Remark 6.6).

Proof. We establish each level by $\mathfrak{sl}_2$-equivariance, degree counting, and the BV relation.

Cup product. The cup product $\mu\colon \mathrm{HH}^p \otimes \mathrm{HH}^q \to \mathrm{HH}^{p+q}$ is graded-commutative: $\mu(a,b) = (-1)^{pq}\,\mu(b,a)$. For $X, Y \in \mathrm{HH}^1$, this gives $\mu(X,Y) = -\mu(Y,X)$: the restriction to $\mathrm{HH}^1 \otimes \mathrm{HH}^1 \to \mathrm{HH}^2 = \mathbb{C}$ is an $\mathfrak{sl}_2$-invariant antisymmetric bilinear form on the adjoint representation valued in the trivial representation. Since $\Lambda^2(\mathfrak{sl}_2)^{\mathfrak{sl}_2} \cong \mathrm{Hom}_{\mathfrak{sl}_2}(\Lambda^2(\mathrm{ad}), \mathbb{C}) = 0$ (the exterior square of the adjoint decomposes as $\Lambda^2(\mathrm{ad}) \cong \mathrm{ad}$, which has no trivial summand), the cup product $\mu(X,Y) = 0$ for all $X, Y \in \mathrm{HH}^1$.

The products $\mu(\mathrm{HH}^1, \mathrm{HH}^2) \subset \mathrm{HH}^3 = 0$ and $\mu(\mathrm{HH}^2, \mathrm{HH}^2) \subset \mathrm{HH}^4 = 0$ vanish for degree reasons.

BV operator. The framed E_2 structure on $\overline{B}^{\Sigma}(V_k(\mathfrak{sl}_2))$ (from genus-1 sewing) equips the derived centre with a BV operator $\Delta\colon \mathrm{HH}^n \to \mathrm{HH}^{n-1}$ satisfying $\Delta^2 = 0$ and the BV relation

$$[f,g] = \Delta(f \cdot g) - (\Delta f) \cdot g - (-1)^{|f|}\, f \cdot (\Delta g). \tag{6.10}$$

The component $\Delta\colon \mathrm{HH}^1 \to \mathrm{HH}^0$ is an $\mathfrak{sl}_2$-equivariant linear map from the adjoint representation to the trivial representation. Since the adjoint is irreducible and non-trivial, $\Delta|_{\mathrm{HH}^1} = 0$.

The component $\Delta\colon \mathrm{HH}^2 \to \mathrm{HH}^1$ is an $\mathfrak{sl}_2$-equivariant linear map from the trivial to the adjoint. Again, $\Delta|_{\mathrm{HH}^2} = 0$.

All other components vanish trivially ($\Delta|_{\mathrm{HH}^0}$ lands in $\mathrm{HH}^{-1} = 0$; $\Delta|_{\mathrm{HH}^n} = 0$ for $n \geq 3$). Therefore $\Delta = 0$ on the entire derived centre.

Gerstenhaber bracket. With $\Delta = 0$ and $\mu|_{\mathrm{HH}^1 \otimes \mathrm{HH}^1} = 0$, the BV relation (6.10) gives: for $X, Y \in \mathrm{HH}^1$,

$$[X,Y] = \Delta(\mu(X,Y)) - (\Delta X) \cdot Y + X \cdot (\Delta Y) = \Delta(0) - 0 + 0 = 0.$$

For $X \in \mathrm{HH}^1$, $\eta \in \mathrm{HH}^2$: $[X,\eta] = \Delta(\mu(X,\eta)) - (\Delta X) \cdot \eta - (-1)^1\, X \cdot (\Delta\eta) = 0 - 0 + 0 = 0$ (since $\mu(X,\eta) \in \mathrm{HH}^3 = 0$, $\Delta X = 0$, $\Delta\eta = 0$). Similarly $[\eta,\eta] \in \mathrm{HH}^3 = 0$.

The vanishing of the Gerstenhaber bracket on the derived centre is a structural consequence of the $\mathfrak{sl}_2$-equivariance constraints: the adjoint representation provides no equivariant maps to or from the trivial representation. This is specific to the *derived centre* $\mathrm{HH}^*(\overline{B}^{\Sigma}, \overline{B}^{\Sigma})$; the Gerstenhaber bracket on the full Hochschild cochain complex $C^*(\overline{B}^{\Sigma}, \overline{B}^{\Sigma})$ is non-trivial (it carries the commutator of derivations at the cochain level).

P_3 *bracket.* The degree-(-2) bracket $\{-,-\}\colon \mathrm{HH}^p \otimes \mathrm{HH}^q \to \mathrm{HH}^{p+q-2}$ satisfies the graded symmetry

$$\{a,b\} = -(-1)^{(|a|-2)(|b|-2)}\,\{b,a\} \tag{6.11}$$

of a degree-(-2) Lie bracket, together with the Leibniz rule $\{a, b \cdot c\} = \{a,b\} \cdot c + (-1)^{(|a|-2)|b|}\, b \cdot \{a,c\}$.

Step 1: $\{\mathbf{1}, -\} = 0$. The Leibniz rule applied to $\mathbf{1} = \mathbf{1} \cdot \mathbf{1}$ gives $\{\mathbf{1}, f\} = 2\,\{\mathbf{1}, f\}$ for all f; hence $\{\mathbf{1}, f\} = 0$.

Step 2: $\{X,Y\} = h_{\mathrm{KZ}}(X,Y)$. For $X, Y \in \mathrm{HH}^1$, $\{X,Y\} \in \mathrm{HH}^0 = \mathbb{C}$. The symmetry (6.11) at $|X| = |Y| = 1$ gives $\{X,Y\} = -(-1)^{(-1)(-1)}\,\{Y,X\} =$

$-(-1)^1\{Y,X\} = \{Y,X\}$: the bracket is *symmetric* on $\mathrm{HH}^1 \otimes \mathrm{HH}^1$. By $\mathfrak{sl}_2$-equivariance, $\{X,Y\}$ is an invariant symmetric bilinear form on the adjoint representation valued in $\mathbb{C}$. The space of such forms is one-dimensional, spanned by the Killing form (X,Y). Hence $\{X,Y\} = \alpha \cdot (X,Y)$ for a scalar α depending on k.

The scalar α is determined by the r-matrix residue, independently of any comparison with the CFG construction. The P_3 bracket on the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g})) = \mathrm{HH}^*(\overline{B}^{\Sigma}(V_k(\mathfrak{g})), \overline{B}^{\Sigma}(V_k(\mathfrak{g})))$ is the E_2-Hochschild cohomology operation derived from the fundamental class of $S^2 \subset \mathrm{Conf}_2(\mathbb{R}^3)$. At arity 2, this operation is the residue of the classical r-matrix paired with the fundamental class of $S^1 \subset \mathrm{Conf}_2(\mathbb{C})$. In the KZ normalisation (see (7.9) and the conventions of §6.10):

$$r(z) = \frac{\Omega}{(k+h^\vee)\,z}, \qquad \mathrm{Res}_{z=0}\big[r(z)\big] = \frac{\Omega}{k+h^\vee}. \tag{6.12}$$

The residue acts on $X \otimes Y \in \mathfrak{g} \otimes \mathfrak{g}$ by contraction with the inverse Casimir: $\Omega(X \otimes Y) = (X,Y)$ (the normalised Killing form). The P_3 bracket on $\mathrm{HH}^1 \cong \mathfrak{g}$ is therefore

$$\{X,Y\} = \mathrm{Res}_{z=0}\big[r(z)(X \otimes Y)\big] = \frac{(X,Y)}{k+h^\vee} = h_{\mathrm{KZ}} \cdot (X,Y), \tag{6.13}$$

giving $\alpha = h_{\mathrm{KZ}} = 1/(k+h^\vee)$. For $\mathfrak{sl}_2$, $h^\vee = 2$ and $\alpha = 1/(k+2)$.

Verification: at $k \to \infty$, $h_{\mathrm{KZ}} \to 0$ and the bracket vanishes (the weak-coupling limit is the abelian E_3). At the critical level $k = -h^\vee$, h_{KZ} diverges (the bracket degenerates, reflecting the Feigin–Frenkel singularity). Both limits are consistent with the behaviour of the KZ connection.

Step 3: $\{X,\eta\} = 0$. The bracket $\{X,\eta\} \in \mathrm{HH}^1 = \mathfrak{sl}_2$ for $X \in \mathrm{HH}^1$, $\eta \in \mathrm{HH}^2$ is an $\mathfrak{sl}_2$-equivariant map $\mathrm{ad} \otimes \mathbb{C} \to \mathrm{ad}$, hence $\{X,\eta\} = \beta\, X$ for some scalar β. The P_3 Jacobi identity applied to $X, Y \in \mathrm{HH}^1$ and $\eta \in \mathrm{HH}^2$, combined with $\{X,Y\} = h_{\mathrm{KZ}}\,(X,Y) \in \mathrm{HH}^0$, gives $\{\{X,\eta\},\eta\} = 0$ (since $\{\{X,\eta\},\eta\} \in \mathrm{HH}^{-1} = 0$) and $\{\eta,\{X,\eta\}\} \in \mathrm{HH}^{-1} = 0$, while $\{X,\{\eta,\eta\}\} = \{X,0\} = 0$. The Jacobi identity is thus satisfied for all β; the constraint comes instead from the self-consistency of the E_3 deformation. At the cohomological level, one applies the iterated Jacobi identity to $X, Y \in \mathrm{HH}^1$ and η:

$$\begin{aligned}
\{\{X,Y\},\eta\} &= \{X,\{Y,\eta\}\} - (-1)^{(|X|-2)(|Y|-2)}\,\{Y,\{X,\eta\}\} \\
&= \{X,\beta Y\} - (-1)^1\,\{Y,\beta X\} \\
&= \beta\, h_{\mathrm{KZ}}\,(X,Y) + \beta\, h_{\mathrm{KZ}}\,(Y,X) \\
&= 2\beta\, h_{\mathrm{KZ}}\,(X,Y).
\end{aligned}$$

The left side: $\{\{X,Y\},\eta\} = \{h_{\mathrm{KZ}}(X,Y)\,\mathbf{1},\eta\} = h_{\mathrm{KZ}}(X,Y)\,\{\mathbf{1},\eta\} = 0$. Hence $2\beta\, h_{\mathrm{KZ}}\,(X,Y) = 0$ for all X, Y. Since $(e,f) = 1 \neq 0$ and $h_{\mathrm{KZ}} \neq 0$ at generic k, we conclude $\beta = 0$.

Step 4: $\{\eta,\eta\} = 0$. The bracket $\{\eta,\eta\} \in \mathrm{HH}^2 = \mathbb{C}$ and the symmetry (6.11) at $|\eta| = 2$ gives $\{\eta,\eta\} = -(-1)^{0\cdot 0}\{\eta,\eta\} = -\{\eta,\eta\}$, hence $\{\eta,\eta\} = 0$. □

Remark 6.8 (The Killing form as the E_3 shadow of the Cartan 3-form). The bracket (6.6) has a transparent topological origin. The E_3-deformation space is $\hbar\, H^3(\mathfrak{sl}_2)[[\hbar]] \cong \hbar\,\mathbb{C}[[\hbar]]$, one-dimensional at each $\hbar$-order (Theorem 6.4(iv)), where $\hbar = k + h^\vee$ is the departure from critical level. The generator of $H^3(\mathfrak{sl}_2)$ is the Cartan 3-form

$$\omega_3(X,Y,Z) = (X,[Y,Z]), \tag{6.14}$$

the unique (up to scalar) $\mathfrak{sl}_2$-invariant element of $\Lambda^3(\mathfrak{sl}_2^*)$. The degree-$(-2)$ bracket $\{X,Y\}$ is the contraction of ω_3 with the KZ coupling $h_{\mathrm{KZ}} = 1/\hbar$:

$$\{X,Y\} = h_{\mathrm{KZ}} \cdot \omega_3(X,Y,-)\big|_{\mathrm{HH}^2\to\mathrm{HH}^0} = h_{\mathrm{KZ}} \cdot (X,Y).$$

The last equality uses the fact that the contraction $\omega_3(X,Y,-) \in \mathfrak{sl}_2^*$ is paired with the identity map $\mathfrak{sl}_2 \to \mathfrak{sl}_2$ to produce a scalar, and the Killing form absorbs the structure constants: $\omega_3(X,Y,Z) = (X,[Y,Z])$ contracts to (X,Y) after tracing over Z with the Casimir. The P_3 bracket on $Z^{\mathrm{der}}_{\mathrm{ch}}$ is the binary shadow of the ternary Cartan form; the normalisation $h_{\mathrm{KZ}} = 1/(k+h^\vee)$ is fixed by the r-matrix residue (equation (6.12)), not by comparison with CFG.

6.4. **Quantum corrections from the vertex R-matrix: E_3 operations for the EK quantum VOA.** The classical $V_k(\mathfrak{sl}_2)$ has trivial braiding $S(z) = \mathrm{id}$; the Etingof–Kazhdan quantum VOA V_{EK} of Example 9.9 has $S(z) = \mathrm{id} + h_{\mathrm{KZ}}\,\Omega/z + O(z^{-2})$. Does the non-trivial R-matrix change the E_3-algebra structure on the derived chiral centre? On cohomology, no: $\mathfrak{sl}_2$-equivariance forces the P_3 bracket to be proportional to the Killing form with the same coefficient $h_{\mathrm{KZ}} = 1/(k+2)$. On cochains, yes: the quantum corrections encode the braided monoidal structure of $\mathrm{Rep}(Y_\hbar(\mathfrak{sl}_2))$.

The underlying graded vector space. The EK quantum VOA is a flat formal deformation of $V_k(\mathfrak{sl}_2)$ over $\mathbb{C}[[h_{\mathrm{KZ}}]]$, with $h_{\mathrm{KZ}} = 1/(k+2)$. Since the chiral Hochschild cohomology is computed by a bounded complex of finite-dimensional $\mathfrak{sl}_2$-modules, the deformation does not change the underlying graded vector space:

$$Z^{\mathrm{der}}_{\mathrm{ch}}(V_{\mathrm{EK}}) \;=\; \mathbb{C}[[h_{\mathrm{KZ}}]] \;\oplus\; \mathfrak{sl}_2[[h_{\mathrm{KZ}}]][-1] \;\oplus\; \mathbb{C}[[h_{\mathrm{KZ}}]][-2], \tag{6.15}$$

with total rank 5 over $\mathbb{C}[[h_{\mathrm{KZ}}]]$. The generators $\mathbf{1} \in \mathrm{HH}^0$, $\{e,f,h\} \in \mathrm{HH}^1$, and $\eta \in \mathrm{HH}^2$ are defined as in Proposition 6.7, now taken over the base ring $\mathbb{C}[[h_{\mathrm{KZ}}]]$.

Equivariance under the quantum R-matrix. The Yang R-matrix $R(u) = u\,I + \hbar\,P$ on $V \otimes V$ is $\mathfrak{sl}_2$-equivariant: $[R(u),\, \rho(X)\otimes I + I \otimes \rho(X)] = 0$ for all $X \in \mathfrak{sl}_2$. The full vertex R-matrix $S(z)$ on $V_{\mathrm{EK}}^{\otimes 2}$ (equation (9.29)) is likewise $\mathfrak{sl}_2$-equivariant, since it is constructed from $\mathcal{R}(z)$, the spectral

universal R-matrix of the Yangian $Y_\hbar(\mathfrak{sl}_2)$, which commutes with the $\mathfrak{sl}_2$-action by the defining property of the Drinfeld coproduct.

This equivariance is the key structural constraint: all three levels of the E_3 operations on $Z^{\mathrm{der}}_{\mathrm{ch}}(V_{\mathrm{EK}})$ are $\mathfrak{sl}_2$-equivariant maps between representations. The equivariance arguments of Proposition 6.7 therefore apply verbatim to the quantum case.

Proposition 6.9 (E_3 operations on $Z^{\mathrm{der}}_{\mathrm{ch}}(V_{\mathrm{EK}})$)**.** *Let V_{EK} be the Etingof–Kazhdan quantum vertex algebra for $\mathfrak{sl}_2$ at KZ coupling $h_{\mathrm{KZ}} = 1/(k+2) \neq 0$ (Example 9.9). The E_3-algebra structure on the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(V_{\mathrm{EK}})$ coincides with the classical E_3 structure of Proposition 6.7: the cup product and Gerstenhaber bracket vanish, and the P_3 bracket is*

$$\{X, Y\}_q \;=\; h_{\mathrm{KZ}} \cdot (X, Y) \qquad \text{for } X, Y \in \mathrm{HH}^1 = \mathfrak{sl}_2, \tag{6.16}$$

with all other brackets zero. In particular, the vertex R-matrix $S(z)$ does not renormalise the P_3 bracket on cohomology.

Proof. The argument has four ingredients: equivariance, one-dimensionality of the obstruction space, exactness of the rational classical r-matrix, and a GRT_1 consistency check.

Step 1: cup product and BV operator. The cup product $\mu\colon \mathrm{HH}^p \otimes \mathrm{HH}^q \to \mathrm{HH}^{p+q}$ is a graded-commutative product on $\mathfrak{sl}_2$-modules over $\mathbb{C}[[h_{\mathrm{KZ}}]]$. For $X, Y \in \mathrm{HH}^1 = \mathfrak{sl}_2[[h_{\mathrm{KZ}}]]$, the product $\mu(X, Y) \in \mathrm{HH}^2 = \mathbb{C}[[h_{\mathrm{KZ}}]]$ is an $\mathfrak{sl}_2$-equivariant antisymmetric bilinear form on the adjoint representation: $\Lambda^2(\mathrm{ad})^{\mathfrak{sl}_2} = 0$, so $\mu(X, Y) = 0$ over $\mathbb{C}[[h_{\mathrm{KZ}}]]$.

The BV operator $\Delta\colon \mathrm{HH}^n \to \mathrm{HH}^{n-1}$ is $\mathfrak{sl}_2$-equivariant. The components $\Delta|_{\mathrm{HH}^1}\colon \mathrm{ad} \to \mathbb{C}$ and $\Delta|_{\mathrm{HH}^2}\colon \mathbb{C} \to \mathrm{ad}$ vanish by Schur's lemma. Hence $\Delta = 0$ on the derived centre, and the BV relation gives $[X, Y] = 0$ for all $X, Y \in \mathrm{HH}^1$, just as in the classical case.

Step 2: the P_3 bracket is proportional to the Killing form. For $X, Y \in \mathrm{HH}^1$, the bracket $\{X, Y\}_q \in \mathrm{HH}^0 = \mathbb{C}[[h_{\mathrm{KZ}}]]$ is an $\mathfrak{sl}_2$-equivariant symmetric bilinear form on the adjoint representation: $\mathrm{Sym}^2(\mathrm{ad})^{\mathfrak{sl}_2} \cong \mathbb{C}$. Hence $\{X, Y\}_q = f(h_{\mathrm{KZ}}) \cdot (X, Y)$ for some $f(h_{\mathrm{KZ}}) \in \mathbb{C}[[h_{\mathrm{KZ}}]]$.

The remaining brackets ($\{X, \eta\}_q$, $\{\eta, \eta\}_q$, $\{\mathbf{1}, -\}_q$) vanish by the same equivariance and Jacobi arguments as in Proposition 6.7 (the argument for $\beta = 0$ uses $h_{\mathrm{KZ}} \neq 0$, $(e, f) = 1 \neq 0$, and $\{\mathbf{1}, -\} = 0$; all remain valid over $\mathbb{C}[[h_{\mathrm{KZ}}]]$).

Step 3: the scalar $f(h_{\mathrm{KZ}}) = h_{\mathrm{KZ}}$ (from exactness of the rational r-matrix). The P_3 bracket on cohomology is computed by the d-log residue of the classical r-matrix (Construction 6.13, equation (6.25)): for $X, Y \in \mathrm{HH}^1 \cong \mathfrak{g}$, the bracket $\{X, Y\}_q$ is determined by the coefficient of the simple pole in $r(z)$.

For the EK quantum vertex algebra, the spectral universal R-matrix $\mathcal{R}(z)$ of $Y_{h_{\mathrm{KZ}}}(\mathfrak{sl}_2)$ evaluates on the fundamental representation to the Yang R-matrix

$$R(z) = z\, I + h_{\mathrm{KZ}}\, P \;\in\; \mathrm{End}(V \otimes V)[z], \tag{6.17}$$

which is *polynomial of degree one* in the spectral parameter z (equation (9.31)). The classical r-matrix extracted from $R(z)$ is

$$r(z) = \frac{h_{\mathrm{KZ}}\,\Omega}{z}, \tag{6.18}$$

with $\Omega = \sum_a t^a \otimes t_a$ the Casimir element dual to the Killing form. This is *exact*: the rational (Yang) r-matrix has a simple pole and no higher-order terms in z^{-1}. The higher-order corrections appearing in the vertex R-matrix expansion (9.32) (the z^{-2}, z^{-3}, … terms) contribute to cochain-level operations but do not modify the classical r-matrix, which is determined by the $O(h_{\mathrm{KZ}})$ coefficient of $S(z)$ at the z^{-1} term.

The P_3 bracket on cohomology is the residue

$$\{X, Y\}_q = \mathrm{Res}_{z=0}\, r(z)(X \otimes Y)\, dz = \mathrm{Res}_{z=0}\, \frac{h_{\mathrm{KZ}}\,\Omega(X \otimes Y)}{z}\, dz = h_{\mathrm{KZ}}\,(X, Y), \tag{6.19}$$

where the last equality uses $\Omega(X \otimes Y) = (X, Y)$ (the Killing form pairing through the Casimir). Since $r(z)$ has *only* a simple pole (no z^{-k} terms for $k \geq 2$), this residue receives no corrections at any order in h_{KZ}. Hence $f(h_{\mathrm{KZ}}) = h_{\mathrm{KZ}}$.

The essential point is the distinction between rational, trigonometric, and elliptic R-matrices. The Yang R-matrix $R(z) = zI + h_{\mathrm{KZ}}P$ is the unique rational solution of the Yang–Baxter equation (up to gauge equivalence); it is polynomial of degree one in z. A trigonometric R-matrix (e.g., the XXZ solution $R(z) \sim \sinh(z + h_{\mathrm{KZ}}P)/\sinh(z)$, associated to $U_q(\widehat{\mathfrak{sl}}_2)$ rather than $Y_{h_{\mathrm{KZ}}}(\mathfrak{sl}_2)$) would have poles at $z \in 2\pi i\mathbb{Z}$, producing a Laurent expansion with infinitely many negative powers of z and potentially renormalising the P_3 bracket. For the EK quantum vertex algebra, which quantises affine Kac–Moody on the formal disk, the rational solution governs, and the P_3 bracket is protected by the polynomial structure of $R(z)$.

Step 4: GRT_1 *consistency check.* The preceding argument determines $f(h_{\mathrm{KZ}}) = h_{\mathrm{KZ}}$ from the explicit form of the Yang R-matrix; the GRT_1-action provides an independent consistency check. The group GRT_1 acts on the E_3-operad maps by conjugation via the Drinfeld associator Φ_{KZ}. On the one-dimensional deformation space $h_{\mathrm{KZ}}\,\mathbb{C}[[h_{\mathrm{KZ}}]]$ (since $H^3(\mathfrak{sl}_2) \cong \mathbb{C}$), this action is by rescaling. At the cohomological level, the conjugation shifts the P_3 bracket by Hochschild coboundaries, which vanish on $\mathrm{HH}^* = H^*(\overline{B}^{\Sigma}, \overline{B}^{\Sigma})$. The P_3 bracket is therefore GRT_1-invariant on cohomology, consistent with the scalar $f(h_{\mathrm{KZ}}) = h_{\mathrm{KZ}}$ being independent of the choice of associator. □

Remark 6.10 (Cochain-level corrections and bulk rigidity). At the *cochain* level, the P_3 bracket on $C^1(\overline{B}^{\Sigma}, \overline{B}^{\Sigma})$ acquires quantum corrections $\{D_X, D_Y\}^{\mathrm{ch}} = h_{\mathrm{KZ}}(X, Y)\,\mathrm{id} + h_{\mathrm{KZ}}^2\,\Gamma_2 + \cdots$, where each Γ_n is d_{Hoch}-exact. These corrections encode the braided monoidal structure on $\mathrm{Rep}(Y_\hbar(\mathfrak{sl}_2))$ (here $\hbar = k + h^\vee$ is the departure from critical level); they are invisible on cohomology but recover the Drinfeld–Jimbo quantum group at the cochain level. The cohomological

E_3 operations (cup product, Gerstenhaber bracket, P_3 bracket, BV operator) are identical for the classical $V_k(\mathfrak{sl}_2)$ and the EK quantum V_{EK}. The sole non-trivial cohomological E_3 invariant is $h_{\mathrm{KZ}} = 1/(k+2)$: the perturbative bulk observables of Chern–Simons are insensitive to the choice of associator.

Remark 6.11 (Why the quantum R-matrix does not renormalise the P_3 bracket). Four mechanisms, in decreasing order of directness:

(i) (Rational r-matrix exactness.) The Yang R-matrix $R(z) = zI + h_{\mathrm{KZ}}P$ is polynomial degree one in z, so the classical r-matrix $r(z) = h_{\mathrm{KZ}}\,\Omega/z$ has a simple pole and no higher-order z-corrections. The P_3 bracket is the residue of $r(z)$, hence $h_{\mathrm{KZ}}\,(X, Y)$ exactly.

(ii) (Equivariance.) $\mathfrak{sl}_2$-equivariance constrains $\{X, Y\}_q$ to be proportional to the Killing form (X, Y), reducing the problem to a single scalar $f(h_{\mathrm{KZ}})$.

(iii) (GRT_1-invariance.) $\dim H^3(\mathfrak{sl}_2) = 1$ makes the deformation space a formal line; the Drinfeld associator shifts the P_3 bracket by Hochschild coboundaries, which vanish on cohomology.

(iv) (Chern–Simons perturbative finiteness.) The P_3 bracket is the tree-level cubic vertex; pure Chern–Simons has no higher-loop corrections.

For simple $\mathfrak{g}$ with $\dim H^3 = 1$, mechanisms (i)–(iii) all apply and the rigidity persists. For non-simple $\mathfrak{g}$ with $\dim H^3 > 1$, mechanism (ii) still constrains the bracket to the space of invariant symmetric forms, but the higher-dimensional deformation space could allow parameter mixing under the GRT_1-action; mechanism (i) remains valid whenever the governing R-matrix is rational.

Both the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ and the CFG E_3-algebra $\mathcal{A}^\lambda$ deform $C^*(\mathfrak{g})$, but through different mechanisms. The chiral bar complex produces a filtered E_3-chiral algebra $\mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})$: a factorisation $\mathcal{D}$-module on $\mathrm{Ran}(X)$ whose global sections on the formal disk recover the CFG E_3-algebra $\mathcal{A}^\lambda$.

6.5. **The chiral P_3 bracket from the PVA λ-bracket.** The E_3 structure on the derived chiral centre is abstract; it comes from Higher Deligne applied to the E_2-coalgebra $\overline{B}^\Sigma(\mathcal{A})$. To make it explicit, one needs a concrete bracket. The Poisson vertex algebra (PVA) λ-bracket on $V_k(\mathfrak{g})$ provides exactly this: a degree-(-2) Lie bracket on the chiral Chevalley–Eilenberg complex, yielding a chiral P_3 structure in the category of $\mathcal{D}$-modules on X.

Definition 6.12 (The chiral Chevalley–Eilenberg complex)**.** Let $\mathfrak{g}$ be a simple Lie algebra with Killing form $(\cdot,\cdot)$ and dual Coxeter number $h^\vee$. Write $\hbar = k + h^\vee$ for the departure from critical level. The *chiral Chevalley–Eilenberg complex* is the curved chiral coalgebra

$$\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_\hbar) \;:=\; \big(\mathrm{Sym}^c(\mathfrak{g}^*[-1]) \otimes \omega_X,\; d_{\mathrm{CE}},\; m_0 = \tfrac{\hbar}{2h^\vee}\,\kappa\big), \tag{6.20}$$

where ω_X denotes the dualising sheaf of X, d_{CE} is the chiral Chevalley–Eilenberg differential encoding the Lie bracket of $\mathfrak{g}$, and the curvature m_0

is the image of the level under the bar construction (the notation κ here denotes the Koszul invariant, not the Killing form). At $\hbar = 0$ (critical level $k = -h^\vee$), the curvature vanishes and $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_0) = (\mathrm{Sym}^c(\mathfrak{g}^*[-1]) \otimes \omega_X,\, d_{\mathrm{CE}})$ is an uncurved commutative (E_∞) factorisation coalgebra on X.

Construction 6.13 (The chiral P_3 bracket)**.** The PVA λ-bracket on $V_k(\mathfrak{g})$ is (see [23], §5.11):

$$\{J^a{}_\lambda\, J^b\} \;=\; f^{ab}{}_c\, J^c \;+\; k\,(a,b)\,\lambda, \tag{6.21}$$

where $f^{ab}{}_c$ are the structure constants of $\mathfrak{g}$ and (a,b) is the Killing form.

At the level of the chiral Chevalley–Eilenberg complex, the λ-bracket induces a degree-(-2) bracket

$$\{-,-\}^{\mathrm{ch}} \colon \mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_\hbar) \otimes \mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_\hbar) \;\longrightarrow\; \mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_\hbar), \tag{6.22}$$

defined on generators $\phi_a = (J^a)^* \in \mathfrak{g}^*[-1]$ by the two components of the PVA bracket:

(i) (Lie bracket term.) The structure constant term $f^{ab}{}_c\, J^c$ contributes the Chevalley–Eilenberg differential d_{CE} at the algebra level. On the dual (coalgebra) side, it induces a coderivation. This is the classical P_3 bracket of the uncurved complex $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_0)$: the degree-$(-2)$ Lie bracket

$$\{\phi_a, \phi_b\}^{\mathrm{ch}}_{\mathrm{Lie}} \;=\; f^{ab}{}_c\, \phi_c. \tag{6.23}$$

This bracket is the coinvariant shadow of the ordered bar R-matrix at the critical level.

(ii) (Cocycle term.) The level-dependent term $k\,(a,b)\,\lambda$ contributes a degree-(-2) symmetric pairing valued in $\mathcal{O}_X$:

$$\{\phi_a, \phi_b\}^{\mathrm{ch}}_{\mathrm{cocycle}} \;=\; k\,(a,b)\cdot\partial, \tag{6.24}$$

where ∂ denotes the $\mathcal{D}$-module action (the spectral parameter λ of the PVA bracket becomes the derivation ∂_z in the $\mathcal{D}$-module structure). This term is the cocycle that controls the deformation away from the commutative (E_∞) factorisation structure.

The total chiral P_3 bracket is

$$\{\phi_a, \phi_b\}^{\mathrm{ch}} \;=\; f^{ab}{}_c\, \phi_c \;+\; k\,(a,b)\,\partial. \tag{6.25}$$

Proposition 6.14 (The chiral P_3 structure)**.** *The bracket* (6.25) *equips the chiral Chevalley–Eilenberg complex* $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_\hbar)$ *with the structure of a* P_3*-algebra object in* $\mathcal{D}$-$\mathrm{mod}(X)$*. Concretely:*

(i) *The bracket is a degree-*(-2) *Lie bracket satisfying graded antisymmetry and the Jacobi identity. The Jacobi identity follows from the Jacobi identity of the PVA* λ*-bracket, which in turn encodes the Jacobi identity of the Lie algebra* $\mathfrak{g}$ *together with the* $\mathfrak{g}$*-invariance of the Killing form.*

(ii) *The bracket is compatible with the commutative product on* $\mathrm{Sym}^c(\mathfrak{g}^*[-1])$ *via the Leibniz rule:*

$$\{a, b \cdot c\}^{\mathrm{ch}} = \{a, b\}^{\mathrm{ch}} \cdot c + (-1)^{(|a|-2)|b|}\, b \cdot \{a, c\}^{\mathrm{ch}}.$$

(iii) *The bracket is* $\mathcal{D}$*-linear in the sense that it respects the flat connection on* $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_\hbar)$ *inherited from the* $\mathcal{D}$*-module structure: the chiral* P_3 *bracket is a morphism of* $\mathcal{D}$*-modules, not merely of* $\mathcal{O}_X$*-modules.*

Proof. The PVA λ-bracket on $V_k(\mathfrak{g})$ satisfies the PVA axioms (sesquilinearity, Leibniz, skew-symmetry, Jacobi); these are the λ-bracket translations of the classical master equation for the Poisson structure. The identification of the λ-bracket with a degree-(-2) Lie bracket on the shifted dual follows the general correspondence between PVA brackets and shifted Poisson structures (Beilinson–Drinfeld [4], §3.8): a PVA λ-bracket of conformal weight 1 (where λ has weight 1) corresponds to a P_3 bracket (degree $-2 = -\dim_{\mathbb{C}} X - 1$, where $\dim_{\mathbb{C}} X = 1$) on the $\mathcal{D}$-module of chiral cochains.

The $\mathcal{D}$-linearity follows because the PVA λ-bracket is a $\mathcal{D}$-module morphism by construction: the derivation ∂ acts via $\lambda \mapsto \lambda$ in the λ-bracket formalism, and this action is compatible with the Leibniz rule by the sesquilinearity axiom of the PVA. □

6.6. **The filtered E_3-chiral algebra.** The chiral P_3 bracket and the chiral Chevalley–Eilenberg complex assemble into a single object: a filtered E_3-chiral algebra $\mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})$, a factorisation $\mathcal{D}$-module on $\mathrm{Ran}(X)$ whose global sections on the formal disk recover the CFG E_3-algebra.

Definition 6.15 (The filtered E_3-chiral algebra)**.** Let $\hbar = k + h^\vee$. The *chiral* E_3*-algebra*

$$\mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g}) \;:=\; \big(\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_\hbar),\ \{-,-\}^{\mathrm{ch}},\ \mathcal{D}\text{-module structure}\big) \tag{6.26}$$

is the chiral Chevalley–Eilenberg complex (Definition 6.12) equipped with the chiral P_3 bracket (Construction 6.13) and the $\mathcal{D}$-module structure from the factorisation algebra on X. It carries an $\hbar$-adic filtration $F^\bullet \mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})$ defined by:

$$F^p \mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g}) \;=\; \hbar^p \cdot \mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g}). \tag{6.27}$$

Theorem 6.16 (Structure of the chiral E_3-algebra)**.** *Let* $\mathfrak{g}$ *be a simple Lie algebra and* $\hbar = k + h^\vee$*. The chiral* E_3*-algebra* $\mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})$ *satisfies:*

(i) (Associated graded.) *The associated graded with respect to the* $\hbar$*-adic filtration is the uncurved chiral Chevalley–Eilenberg complex at the critical level:*

$$\mathrm{gr}_F\, \mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g}) \;\simeq\; \mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_0) \;=\; \big(\mathrm{Sym}^c(\mathfrak{g}^*[-1]) \otimes \omega_X,\ d_{\mathrm{CE}}\big), \tag{6.28}$$

which is an E_∞*-chiral algebra (commutative factorisation* $\mathcal{D}$*-module on* $\mathrm{Ran}(X)$*).*

(ii) (The E_3 structure.) *The chiral* P_3 *bracket (Proposition 6.14) deforms the* E_∞ *structure to a filtered* E_3*-chiral algebra:* E_2 *from the curve geometry (Warning 2.5)* $+1$ *from the chiral* P_3 *bracket* $= \mathsf{E}_3$*.*

(iii) (Curvature and the deformation parameter.) *The curvature* $m_0 = \frac{\hbar}{2h^\vee}\,\kappa$ *of the chiral Chevalley–Eilenberg complex is the obstruction to the* E_∞ *structure: at* $\hbar = 0$*, the curvature vanishes and the algebra is commutative; at* $\hbar \neq 0$*, the* P_3 *bracket and the curvature together deform the commutative structure to a filtered* E_3*-chiral algebra. The deformation parameter is* $\hbar \in H^3(\mathfrak{g})[[\hbar]]$*, matching the deformation space of Theorem 1.3*(ii)*.*

(iv) (Factorisation property.) *The* E_3*-chiral algebra* $\mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})$ *is a factorisation algebra on* X *in the sense that the restriction to disjoint opens* $U_1 \sqcup U_2 \hookrightarrow X$ *induces*

$$\mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})\big|_{U_1\sqcup U_2} \;\simeq\; \mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})\big|_{U_1} \otimes \mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})\big|_{U_2}, \tag{6.29}$$

where the tensor product is in the derived category of $\mathcal{D}$*-modules. The* E_3 *structure is compatible with the factorisation isomorphism.*

Proof. Part (i) is immediate from the filtration: the $\hbar$-adic filtration is exhaustive, and at $\hbar = 0$ the curvature and the cocycle term of the P_3 bracket both vanish, leaving the uncurved commutative chiral coalgebra $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_0)$.

Part (ii) requires three things: that the chiral P_3 bracket is a degree-(-2) Lie bracket (Jacobi), that it satisfies the Leibniz rule with respect to the commutative product (Poisson compatibility), and that it is compatible with the E_2 structure from the curve geometry, so that the combined data assembles into an E_3-algebra. We prove each in turn.

Step 1: the P_3 *Jacobi identity from the PVA Jacobi.* The PVA λ-bracket on $V_k(\mathfrak{g})$ satisfies the (-1)-shifted Jacobi identity (see [24], Appendix A, Proposition A.5):

$$\{a\,{}_\lambda\,\{b\,{}_\mu\,c\}\} - (-1)^{(|a|+1)(|b|+1)}\,\{b\,{}_\mu\,\{a\,{}_\lambda\,c\}\} = \{\{a\,{}_\lambda\,b\}\,{}_{\lambda+\mu}\,c\}. \tag{6.30}$$

The chiral P_3 bracket $\{-,-\}^{\mathrm{ch}}$ on $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_\hbar)$ is obtained from the λ-bracket by specialising $\lambda \mapsto \partial$ (the $\mathcal{D}$-module derivation) and passing to the chiral Chevalley–Eilenberg complex. On generators $\phi_a, \phi_b, \phi_c \in \mathfrak{g}^*[-1]$ (cohomological degree 1, shifted bracket-degree $|\phi_a| - 2 = -1$), the degree-(-2) Jacobi identity is:

$$\{\phi_a, \{\phi_b, \phi_c\}^{\mathrm{ch}}\}^{\mathrm{ch}} + \text{graded cyclic} = 0. \tag{6.31}$$

To derive this from (6.30), set $\lambda = \mu = 0$ (the constant-coefficient specialisation that extracts the P_3 bracket from the λ-bracket). The left side of (6.30) becomes $\{\phi_a, \{\phi_b, \phi_c\}^{\mathrm{ch}}\}^{\mathrm{ch}} - (-1)^{(|\phi_a|+1)(|\phi_b|+1)}\,\{\phi_b, \{\phi_a, \phi_c\}^{\mathrm{ch}}\}^{\mathrm{ch}}$, and the right side becomes $\{\{\phi_a, \phi_b\}^{\mathrm{ch}}, \phi_c\}^{\mathrm{ch}}$ (at $\lambda + \mu = 0$). The PVA λ-bracket has degree -1 (it is (-1)-shifted); the chiral P_3 bracket has degree -2, accounting for the additional degree shift from the $\mathcal{D}$-module derivation ∂ (conformal weight 1). The Koszul sign $(-1)^{(|\phi_a|+1)(|\phi_b|+1)}$ in the PVA identity translates to $(-1)^{(|\phi_a|-2)(|\phi_b|-2)}$ in the P_3 identity after incorporating the degree shift $|-| \mapsto |-| - 2$ appropriate to a degree-(-2) bracket. On $\mathfrak{g}^*[-1]$ where $|\phi_a| = 1$, both signs evaluate to $(-1)^{(-1)(-1)} = -1$. Rearranging gives (6.31).

For general elements of $\mathrm{Sym}^c(\mathfrak{g}^*[-1])$ (not just generators), the Jacobi identity extends by the Leibniz rule (Step 2 below): the bracket is a biderivation of the free commutative algebra, so a trilinear identity that holds on generators holds on all elements.

Step 2: the Leibniz rule from PVA sesquilinearity. The PVA right Leibniz rule (see [24], Appendix A, Proposition A.6) states:

$$\{a\,{}_\lambda\,(b\cdot c)\} = \{a\,{}_\lambda\,b\}\cdot c + (-1)^{(|a|+1)|b|}\,b\cdot\{a\,{}_\lambda\,c\} + \int_0^\lambda \{\{a\,{}_\mu\,b\}\,{}_\lambda\,c\}\,d\mu.$$

At $\lambda = 0$, the integral term vanishes (the integration range collapses to a point), leaving:

$$\{a, b\cdot c\}^{\mathrm{ch}} = \{a, b\}^{\mathrm{ch}}\cdot c + (-1)^{(|a|+1)|b|}\,b\cdot\{a, c\}^{\mathrm{ch}}.$$

Adjusting the Koszul sign from the (-1)-shifted convention to the degree-(-2) convention: $(|a|+1)|b|$ becomes $(|a|-2)|b|$ after the degree shift, recovering the P_3 Leibniz rule

$$\{a, b\cdot c\}^{\mathrm{ch}} = \{a, b\}^{\mathrm{ch}}\cdot c + (-1)^{(|a|-2)|b|}\,b\cdot\{a, c\}^{\mathrm{ch}},$$

as stated in Proposition 6.14(ii). This identifies the chiral P_3 bracket as a derivation of the commutative product μ on $\mathrm{Sym}^c(\mathfrak{g}^*[-1])\otimes\omega_X$: a commutative algebra equipped with a compatible degree-(-2) Lie bracket is a P_3-algebra.

Step 3: compatibility with E_2 *and the* E_3 *assembly.* The E_2-coalgebra structure on $\overline{B}^\Sigma$ comes from the curve geometry (Proposition 6.2(i), Warning 2.5): the configuration spaces $\mathrm{Conf}_n(X)$ provide operations parametrised by $\mathrm{Conf}_k(\mathbb{R}^2)$, the topological content of E_2. The framed E_2 structure (from genus-1 sewing) equips the derived centre with a BV operator Δ satisfying $[f, g] = \Delta(f\cdot g) - (\Delta f)\cdot g - (-1)^{|f|}\,f\cdot(\Delta g)$ (the BV relation (6.10)). The P_3 bracket of degree -2 supplements this E_2 structure with one additional degree of operadic freedom. The dimension count: E_2 operations are parametrised by $\mathrm{Conf}_k(\mathbb{R}^2)$, and the P_3 bracket (degree $-2 = -(3-1)$) corresponds to the fundamental class of $S^2 = \mathrm{Conf}_2(\mathbb{R}^3)/\mathbb{R}^+$, the extra generator needed for E_3 beyond E_2.

The compatibility condition is that the P_3 bracket must be a derivation of the Gerstenhaber bracket $[-,-]$ of degree -1 from the E_2 structure. Since the Gerstenhaber bracket is determined by Δ and μ via the BV relation, it suffices to show that $\{-,-\}^{\mathrm{ch}}$ commutes with Δ and is a derivation of μ (already proved in Step 2). The BV operator Δ arises from genus-1 sewing (the S^1-action on the framed E_2 structure), while the P_3 bracket arises from the arity-2 r-matrix residue at genus 0. These operations act on independent geometric data (genus-1 vs. genus-0, sewing vs. collision), so Δ commutes with $\{-,-\}^{\mathrm{ch}}$ as $\mathcal{D}$-module endomorphisms. Given this, for any

$a, b, c \in \mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_\hbar)$:

$$\begin{aligned}\{a, [b, c]\}^{\mathrm{ch}} &= \{a, \Delta(b \cdot c) - (\Delta b) \cdot c - (-1)^{|b|}\, b \cdot (\Delta c)\}^{\mathrm{ch}} \\ &= \Delta\{a, b \cdot c\}^{\mathrm{ch}} - \{a, (\Delta b) \cdot c\}^{\mathrm{ch}} - (-1)^{|b|}\, \{a, b \cdot (\Delta c)\}^{\mathrm{ch}},\end{aligned}$$

where the first term uses $[\Delta, \{a, -\}^{\mathrm{ch}}] = 0$. Expanding each term by the Leibniz rule of Step 2 and collecting via the BV relation yields

$$\{a, [b, c]\}^{\mathrm{ch}} = [\{a, b\}^{\mathrm{ch}}, c] + (-1)^{(|a|-2)|b|}\, [b, \{a, c\}^{\mathrm{ch}}] :$$

the P_3 bracket is a derivation of the Gerstenhaber bracket.

This is the mixed Leibniz rule that characterises $\mathsf{E}_3 = \mathsf{E}_2 + \mathsf{P}_3$: the P_3 bracket acts as a derivation of the E_2 Gerstenhaber bracket. By the recognition principle for E_n-algebras (Lurie [18], Theorem 5.1.2.2; Francis [15], Theorem 4.1): a commutative algebra with a compatible E_2 structure and a degree-(-2) Lie bracket satisfying the mutual Leibniz rules (Jacobi, Poisson compatibility with the product, and derivation of the Gerstenhaber bracket) is an E_3-algebra. The three verifications above establish all the required relations.

Part (iii) combines two observations. The curvature $m_0 = \frac{\hbar}{2h^\vee}\,\kappa$ is the bar-complex encoding of the level departure (the identification $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_\hbar) \simeq B^{\mathrm{ch}}(V_k(\mathfrak{g}))$ at $\hbar = k + h^\vee$). The cocycle term $k\,(a, b)\,\partial$ of the P_3 bracket is linear in k and hence linear in $\hbar$ (up to the shift $k = \hbar - h^\vee$). Both vanish at $\hbar = 0$, so the deformation is controlled by $\hbar$. The identification of the deformation space as $\hbar H^3(\mathfrak{g})[[\hbar]]$ then follows from the computation of Theorem 1.3(ii): $H^3(\mathfrak{g})$ is one-dimensional for simple $\mathfrak{g}$, and the Cartan 3-form $\omega_3(X, Y, Z) = (X, [Y, Z])$ is the generator.

Part (iv) is the factorisation property of the chiral bar complex. The chiral Chevalley–Eilenberg complex, being a chiral coalgebra, satisfies the factorisation axiom by construction (Beilinson–Drinfeld [4], Chapter 3). The P_3 bracket, being a $\mathcal{D}$-module morphism (Proposition 6.14(iii)), is compatible with the factorisation isomorphisms. □

6.7. **The $\mathcal{D}$-module structure and the KZ connection.** The chiral E_3-algebra $\mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})$ is not an abstract E_3-algebra: it is an E_3-algebra object in the category of $\mathcal{D}$-modules on X, or equivalently on $\mathrm{Ran}(X)$. This $\mathcal{D}$-module structure is the content that distinguishes the chiral construction from the topological CFG construction.

Proposition 6.17 (The $\mathcal{D}$-module structure)**.** *The chiral E_3-algebra $\mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})$ carries a flat connection*

$$\nabla^{\mathrm{ch}} \;=\; d \;+\; \sum_{i<j} \frac{k\,\Omega_{ij}}{z_i - z_j}\, dz_{ij} \tag{6.32}$$

on $\mathrm{Conf}_n(X)$, where Ω_{ij} is the Casimir element acting on the i-th and j-th tensor factors. This is the Knizhnik–Zamolodchikov connection.

(i) *The flatness* $(\nabla^{\mathrm{ch}})^2 = 0$ *follows from the classical Yang–Baxter equation for the* r*-matrix* $r(z) = k\,\Omega/z$ *(trace-form convention; at* $k = 0$*,* $r = 0$*): the curvature of* ∇^{ch} *is proportional to the LHS of the CYBE, which vanishes.*

(ii) *The* E_3*-algebra operations are horizontal with respect to* ∇^{ch}*: the cup product, the Gerstenhaber bracket, and the* P_3 *bracket are all flat sections of the corresponding* Hom*-bundles. This is the* $\mathcal{D}$*-linearity of Proposition* 6.14(iii) *in the multi-point form.*

(iii) *The monodromy representation of* ∇^{ch} *on* $\pi_1(\mathrm{Conf}_n(X))$ *gives the braid group action. For* $X = \mathbb{C}$*, the monodromy of the KZ connection is computed by the Drinfeld associator* Φ_{KZ} *and recovers the braided monoidal structure on* $\mathrm{Rep}(U_\hbar(\mathfrak{g}))$*.*

Proof. The KZ connection on the n-point conformal blocks of $V_k(\mathfrak{g})$ is classical (Knizhnik–Zamolodchikov, 1984). The presentation here identifies this connection as the $\mathcal{D}$-module structure on the chiral Chevalley–Eilenberg complex, viewed as a factorisation algebra on $\mathrm{Ran}(X)$.

For (i): the curvature of ∇^{ch} on $\mathrm{Conf}_n(X)$ is $[\nabla_i, \nabla_j] = k^2\,[\Omega_{ij}/(z_i - z_j),\, \Omega_{ik}/(z_i - z_k) + \Omega_{jk}/(z_j - z_k)]\, dz_{ij} \wedge dz_{ik}$ plus cyclic permutations. The vanishing of this expression is equivalent to the classical Yang–Baxter equation $[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0$ for $r_{ij} = k\,\Omega_{ij}/(z_i - z_j)$.

For (ii): the E_3 operations are defined in terms of the PVA λ-bracket, which is itself a $\mathcal{D}$-module morphism. The cup product is a $\mathcal{D}$-module map by the commutative algebra structure on the factorisation algebra. The P_3 bracket involves the derivation ∂ (from the λ in the PVA bracket), which is the $\mathcal{D}$-module action; its compatibility with the connection follows from the PVA sesquilinearity axiom.

For (iii): this is the Drinfeld–Kohno theorem (Drinfeld [12], Kohno, Kassel): the monodromy of the KZ connection on $\mathrm{Conf}_n(\mathbb{C})$ factors through the pure braid group P_n and gives, via the Drinfeld associator $\Phi_{\mathrm{KZ}} \in \exp(\hat{\mathfrak{t}}_3)$, a braided monoidal structure on $\mathrm{Rep}(U_\hbar(\mathfrak{g}))$. □

6.8. **Comparison with CFG: formal disk global sections.** The chiral E_3-algebra $\mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})$ lives on $\mathrm{Ran}(X)$ and carries the KZ connection; the CFG algebra $\mathcal{A}^\lambda$ lives on $\mathbb{R}^3$ and is purely topological. The formal disk $D = \mathrm{Spec}\,\mathbb{C}[[z]]$ is the bridge: restricting to D trivializes the $\mathcal{D}$-module structure and recovers the CFG algebra as the fiber.

Theorem 6.18 (Formal disk restriction recovers CFG)**.** *Let* $D = \mathrm{Spec}\,\mathbb{C}[[z]]$ *be the formal disk in* X*. There is a natural map of filtered* E_3*-algebras*

$$\Gamma(D,\, \mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})) \;\longrightarrow\; \mathcal{A}^\lambda \tag{6.33}$$

from the global sections of the chiral E_3*-algebra on the formal disk to the CFG* E_3*-algebra of Theorem* 6.4*, where* $\lambda = \hbar$ *(the Chern–Simons coupling equals the departure from critical level).*

(i) (Associated graded.) *At the level of associated graded algebras,*

$$\operatorname{gr}\Gamma(D,\,\mathrm{CE}^{\mathrm{ch}}_{\hbar}(\mathfrak{g}))\;\simeq\;C^*(\mathfrak{g})\;=\;\mathrm{Sym}(\mathfrak{g}^\vee[-1]),\tag{6.34}$$

the Chevalley–Eilenberg cochains of $\mathfrak{g}$*: both the chiral construction restricted to the formal disk and the CFG construction have the same classical limit.*

(ii) (Matching of P_3 brackets.) *The restriction of the chiral* P_3 *bracket* (6.25) *to the formal disk recovers the CFG* P_3 *bracket. On the formal disk, the derivation* $\partial=\partial_z$ *acts on* $\mathbb{C}[[z]]$*; the global sections* $\Gamma(D,\mathfrak{g}^*[-1]\otimes\omega_D)\cong\mathfrak{g}^*[-1][[z]]\,dz$ *produce, after extracting the residue at* $z=0$*, the finite-dimensional* P_3 *bracket*

$$\{\phi_a,\phi_b\}_{\mathrm{CFG}}\;=\;f^{ab}{}_c\,\phi_c\;+\;k\,(a,b)\cdot\delta_0,$$

where δ_0 *is the delta-function at the origin, the formal-disk avatar of* ∂*. Under the identification* $\Gamma(D,\omega_D)\cong\mathbb{C}\,dz$ *(taking the constant mode), this recovers the CFG bracket on* $C^*(\mathfrak{g})$*.*

(iii) (Matching of deformation spaces.) *Both* $\mathrm{CE}^{\mathrm{ch}}_{\hbar}(\mathfrak{g})\big|_D$ *and* $\mathcal{A}^\lambda$ *are deformations of* $C^*(\mathfrak{g})$ *over the base* $\hbar\,H^3(\mathfrak{g})[[\hbar]]$*, with the same one-dimensional fibre at each* $\hbar$*-order for simple* $\mathfrak{g}$*. The map* (6.33) *intertwines the* $\hbar$*-adic filtrations.*

Proof. The argument has two steps: construct the E_3-algebra structure on $\Gamma(D,\mathrm{CE}^{\mathrm{ch}}_{\hbar}(\mathfrak{g}))$ by symmetric monoidal transport, then identify it with the CFG algebra $\mathcal{A}^\lambda$.

Step 1: symmetric monoidal trivialization on D. The formal disk $D=\operatorname{Spec}\mathbb{C}[[z]]$ is contractible: every $\mathcal{D}$-module on D (that is, a $\mathbb{C}[[z]]$-module with flat connection) is isomorphic to its fibre at the origin, since every flat connection on D is trivializable ($\pi_1(D^n)=0$ for all n, so there is no monodromy obstruction). The global sections functor

$$\Gamma(D,-)\colon\mathcal{D}\text{-mod}(D)\;\xrightarrow{\sim}\;\mathrm{Vect}\tag{6.35}$$

is an equivalence of categories. This equivalence is symmetric monoidal: the tensor product of $\mathcal{D}$-modules on D (the !-tensor product, which on the contractible formal disk coincides with the $*$-tensor product and with the ordinary tensor product of the underlying vector spaces) corresponds under (6.35) to the tensor product in Vect.

An equivalence of symmetric monoidal categories preserves all algebraic structures definable in terms of the monoidal structure. In particular, it preserves E_3-algebra structures: if $\mathcal{B}$ is an E_3-algebra in $\mathcal{D}$-mod(D), then $\Gamma(D,\mathcal{B})$ is an E_3-algebra in Vect, with structure maps obtained by applying $\Gamma(D,-)$ to those of $\mathcal{B}$.

Applying this to $\mathcal{B}=\mathrm{CE}^{\mathrm{ch}}_{\hbar}(\mathfrak{g})\big|_D$: the restriction of the chiral E_3-algebra to D is an E_3-algebra in $\mathcal{D}$-mod(D) (by restriction of the E_3-structure on X), and the equivalence (6.35) endows $\Gamma(D,\mathrm{CE}^{\mathrm{ch}}_{\hbar}(\mathfrak{g}))$ with an E_3-algebra structure in Vect. This E_3-structure is functorial and carries the $\hbar$-adic filtration inherited from the chiral side.

Step 2: identification with $\mathcal{A}^\lambda$. It remains to identify $\Gamma(D, \mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g}))$ with the CFG algebra $\mathcal{A}^\lambda$ as filtered E_3-algebras, where $\lambda = \hbar$. Both are filtered E_3-algebras deforming the same classical limit over the same base, and the identification proceeds through three structural invariants.

Part (i): on D, the $\mathcal{D}$-module structure trivializes, so the chiral Chevalley–Eilenberg complex reduces to the ordinary one. The associated graded of the chiral E_3-algebra on D is

$$\operatorname{gr}\Gamma(D,\, \mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})) \;=\; \Gamma(D,\, \operatorname{gr}\mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})) \;=\; \Gamma(D,\, \mathrm{Sym}^c(\mathfrak{g}^*[-1]) \otimes \omega_D).$$

On the formal disk, $\omega_D \cong \mathcal{O}_D\, dz$; the equivalence (6.35) extracts the constant mode in the z-expansion, giving $\mathrm{Sym}(\mathfrak{g}^\vee[-1]) = C^*(\mathfrak{g})$, the Chevalley–Eilenberg cochains of $\mathfrak{g}$ as a commutative algebra. This is the associated graded of $\mathcal{A}^\lambda$ (Theorem 6.4(i)).

Part (ii): the chiral P_3 bracket on X (equation (6.25)) is defined via the PVA λ-bracket. On the formal disk, the spectral parameter λ becomes the formal variable of the completed enveloping algebra: the translation generator ∂_z acts on $\mathbb{C}[[z]]$, and taking global sections via (6.35) extracts the residue at $z = 0$. The resulting bracket on $\Gamma(D, \mathfrak{g}^*[-1] \otimes \omega_D) \cong \mathfrak{g}^*[-1][[z]]\, dz$ is

$$\{\phi_a, \phi_b\}_{\mathrm{CFG}} \;=\; f^{ab}{}_c\, \phi_c \;+\; k\,(a,b) \cdot \delta_0,$$

where $f^{ab}{}_c$ are the structure constants of $\mathfrak{g}$, (a, b) is the Killing form, and δ_0 is the delta-function at the origin (the formal-disk avatar of the derivation ∂). Under the identification $\Gamma(D, \omega_D) \cong \mathbb{C}\, dz$, this is the CFG P_3 bracket on $C^*(\mathfrak{g})$ (Theorem 6.4(ii)).

Part (iii): the deformation theory of E_3-algebras with fixed associated graded $C^*(\mathfrak{g})$ is controlled by the E_3-Hochschild cohomology, which for $C^*(\mathfrak{g})$ with $\mathfrak{g}$ simple is $\hbar\, H^3(\mathfrak{g})[[\hbar]]$, one-dimensional at each $\hbar$-order. Both $\Gamma(D, \mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g}))$ and $\mathcal{A}^\lambda$ are deformations of $C^*(\mathfrak{g})$ over this base (Theorem 1.3(ii) and Theorem 6.4(iv) respectively). Since the deformation space is one-dimensional at each order, and both deformations have matching associated graded (Part (i)) and matching P_3 bracket (Part (ii)), they are isomorphic as filtered E_3-algebras. The $\hbar$-adic filtration on the chiral side maps to the λ-adic filtration on the CFG side under $\lambda = \hbar = k + h^\vee$. □

Remark 6.19 (The chain of spaces). The restriction passes through $\mathrm{pt} \leftarrow D \to \mathbb{A}^1 \to \mathbb{P}^1 \to X$, with each step adding geometric content: the formal disk provides the derivation ∂ (cocycle term $k(a, b)\partial$); $\mathbb{A}^1$ adds Arnold relations and braid monodromy; $\mathbb{P}^1$ and higher-genus X add compactness and modular curvature. None of these restrictions is an identity; each step loses geometric information.

6.9. **The Khan–Zeng topological enhancement.** The chiral E_3-algebra is holomorphic: it depends on the complex structure of X through the KZ connection. At non-critical level $k \neq -h^\vee$, the Sugawara Virasoro element provides a homotopy between the holomorphic and topological directions,

upgrading the E_3 structure from holomorphic to topological. At the critical level, the Sugawara element is undefined and the enhancement fails.

Proposition 6.20 (Topological enhancement via Sugawara)**.** *At generic non-critical level $k \neq -h^\vee$, the affine Kac–Moody vertex algebra $V_k(\mathfrak{g})$ possesses a Sugawara Virasoro element*

$$T_{\mathrm{Sug}} \;=\; \frac{1}{2(k+h^\vee)} \sum_{a=1}^{\dim \mathfrak{g}} :J^a J_a:, \tag{6.36}$$

which generates an inner action of the Virasoro algebra at central charge $c = k\,\dim(\mathfrak{g})/(k+h^\vee)$.

By Khan–Zeng [20], the Virasoro element upgrades the holomorphic-topological symmetry of the $3d$ gauge theory on $\Sigma \times \mathbb{R}_t$ to fully topological symmetry on $\Sigma \times \mathbb{R}_t$. At the level of the chiral E_3-algebra, this means:

(i) (Holomorphic-topological.) *The chiral E_3-algebra $\mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})$ lives on $X \times \mathbb{R}$, where X carries holomorphic structure and $\mathbb{R}$ carries topological structure. The E_2 from the curve geometry is holomorphic; the $+1$ from the P_3 bracket is topological (from the $\mathbb{R}$ factor).*

(ii) (Full topological.) *The Sugawara element provides an identification of the holomorphic direction in X with a topological direction: the stress-energy tensor T_{Sug} generates translations along X, making them homotopically trivial. After this identification, the E_3-structure becomes* topological E_3*: an E_3-algebra in the category of locally constant sheaves on $\mathbb{R}^3 \cong X \times \mathbb{R}$ (identifying $X \cong \mathbb{C} \cong \mathbb{R}^2$).*

(iii) (Failure at critical level.) *At $k = -h^\vee$, the Sugawara element (6.36) is undefined (the denominator $2(k+h^\vee)$ vanishes). The critical-level chiral E_3-algebra $\mathrm{CE}^{\mathrm{ch}}_0(\mathfrak{g})$ is genuinely holomorphic, not topological: the E_2 from the curve geometry depends essentially on the complex structure of X. Perturbatively, at $k = -h^\vee + \hbar\lambda_1 + \cdots$, the topological enhancement holds order-by-order in $\hbar$ (since T_{Sug} exists at any nonzero $\hbar$).*

Proof. Part (i) is the HT structure of the 3d gauge theory whose boundary chiral algebra is $V_k(\mathfrak{g})$ (Costello–Gwilliam [10]). The factorisation algebra of the HT theory lives on $\Sigma \times \mathbb{R}$ with holomorphic dependence on Σ and topological dependence on $\mathbb{R}$; the chiral E_3-algebra is the algebraic encoding of this factorisation structure.

Part (ii). Khan–Zeng [20] prove the following at the classical level: an inner Virasoro element in a Poisson vertex algebra with non-degenerate central charge provides a homotopy between the holomorphic and topological directions, upgrading the HT theory to a fully topological theory. We extend this to the quantum vertex algebra $V_k(\mathfrak{g})$. The Sugawara construction (Frenkel–Ben-Zvi [14], Chapter 15) produces $T_{\mathrm{Sug}} \in V_k(\mathfrak{g})$ satisfying the

exact Virasoro OPE

$$T_{\mathrm{Sug}}(z)\,T_{\mathrm{Sug}}(w) \;\sim\; \frac{c/2}{(z-w)^4} + \frac{2\,T_{\mathrm{Sug}}(w)}{(z-w)^2} + \frac{\partial T_{\mathrm{Sug}}(w)}{z-w}$$

at central charge $c = k\,\dim(\mathfrak{g})/(k+h^\vee)$, and the Sugawara OPE with the current generators

$$T_{\mathrm{Sug}}(z)\,J^a(w) \;\sim\; \frac{J^a(w)}{(z-w)^2} + \frac{\partial J^a(w)}{z-w}.$$

The second relation gives $L_{-1}\cdot J^a = \partial J^a$: the holomorphic translation operator $L_{-1} = (T_{\mathrm{Sug}})_{(1)}$ acts as ∂ on all generators. These are identities in the quantum vertex algebra, not formal-deformation statements; the normal ordering in (6.36) absorbs all quantum corrections. The BRST operator Q of the 3d gauge theory satisfies $[Q,b] = L_{-1}$ where b is the antighost zero-mode (this is the standard BRST relation for the Sugawara Virasoro; see [14], §15.4), so L_{-1} is BRST-exact. The Khan–Zeng argument [20] then applies verbatim: BRST-exactness of holomorphic translation is the datum required to upgrade the HT symmetry to full topological symmetry. At generic k, the central charge $c = k\,\dim(\mathfrak{g})/(k+h^\vee) \neq 0$, so the hypothesis is satisfied.

Part (iii): at $k = -h^\vee$, the Feigin–Frenkel centre $Z(V_{-h^\vee}(\mathfrak{g})) = \mathrm{Fun}(\mathrm{Op}_{\mathfrak{g}^\vee}(D))$ (Feigin–Frenkel) is a commutative algebra with non-trivial dependence on the complex structure of the formal disk (through the notion of $\mathfrak{g}^\vee$-oper). The absence of the Sugawara element means no homotopy between holomorphic and topological directions exists, and the chiral E_3-algebra is genuinely holomorphic. The perturbative statement follows because the Sugawara element exists as a formal power series in $\hbar$ for $\hbar \neq 0$. □

6.10. **Example: $\mathfrak{sl}_2$ chiral E_3-algebra.** The general construction specialises cleanly to $\mathfrak{g} = \mathfrak{sl}_2$: the chiral P_3 bracket is determined by six pairings on the generators ϕ_e, ϕ_f, ϕ_h, each consisting of a Lie bracket term (from the structure constants) and a cocycle term (proportional to the level k, vanishing at $k = 0$). Restricting to the formal disk and extracting the zero mode recovers the CFG bracket of Proposition 6.7.

Example 6.21 (Explicit chiral P_3 bracket for $\mathfrak{sl}_2$)**.** For $\mathfrak{g} = \mathfrak{sl}_2$ with basis $\{e,f,h\}$, Killing form $(e,f) = (f,e) = 1$, $(h,h) = 2$, structure constants $[e,f] = h$, $[h,e] = 2e$, $[h,f] = -2f$, and $h^\vee = 2$, the chiral P_3 bracket (Construction 6.13) on the generators $\phi_e, \phi_f, \phi_h \in \mathfrak{sl}_2^*[-1]$ is:

$$\{\phi_e,\phi_f\}^{\mathrm{ch}} = \phi_h + k\,\partial, \tag{6.37}$$

$$\{\phi_h,\phi_e\}^{\mathrm{ch}} = 2\,\phi_e, \tag{6.38}$$

$$\{\phi_h,\phi_f\}^{\mathrm{ch}} = -2\,\phi_f, \tag{6.39}$$

$$\{\phi_h,\phi_h\}^{\mathrm{ch}} = 2k\,\partial, \tag{6.40}$$

$$\{\phi_e,\phi_e\}^{\mathrm{ch}} = 0, \tag{6.41}$$

$$\{\phi_f,\phi_f\}^{\mathrm{ch}} = 0. \tag{6.42}$$

The Lie bracket terms (ϕ_h in (6.37), $2\phi_e$ in (6.38), $-2\phi_f$ in (6.39)) persist at $k=0$; the cocycle terms ($k\,\partial$ in (6.37) and (6.40)) vanish at $k=0$ (AP126 check: at $k=0$, the level-dependent terms vanish, and the bracket reduces to the Chevalley–Eilenberg coderivation).

At the critical level $k=-2$ ($\hbar=0$), the cocycle terms become $-2\,\partial$, and the chiral E_3-algebra degenerates: the curvature $m_0=0$ (uncurved), and the commutative factorisation structure is undeformed.

Taking formal disk global sections and extracting the zero mode recovers the CFG P_3 bracket of Proposition 6.7: $\{\phi_a,\phi_b\}_{\mathrm{CFG}} = f^{ab}{}_c\,\phi_c + k\,(a,b)\,\delta_0$, which at the cohomological level gives $\{X,Y\} = h_{\mathrm{KZ}}\,(X,Y)$ for $X,Y\in \mathrm{HH}^1=\mathfrak{sl}_2$ (equation (6.6)), with $h_{\mathrm{KZ}}=1/(k+2)$.

7. Examples

The four shadow classes G, L, C, M produce qualitatively different ordered chiral homology theories. Class G (Heisenberg) has regular singularities, finite shadow tower, and trivial kernel ker(av). Class L (affine Kac–Moody) has regular singularities but a non-trivial R-matrix kernel. Class C ($\beta\gamma$ system) has irregular singularities of bounded Poincaré rank. Class M (Virasoro) has irregular singularities of unbounded rank and an infinite shadow tower.

Example 7.1 (Heisenberg: class G, $r=2$; explicit ordered chiral Hochschild cohomology)**.** The Heisenberg algebra $\mathcal{H}_k$ has a single generating field $J(z)$ of conformal weight 1, OPE $J(z)J(w)\sim k/(z-w)^2$, and r-matrix $r(z)=k/z$ (trace-form convention; at $k=0$, $r=0$). The Koszul invariant is $\kappa=k$. The algebra is E_∞-chiral (symmetric OPE; no simple pole) and class G (shadow depth $r=2$: $S_4=0$, so $\Delta=8\kappa S_4=0$). By the formality bridge (Theorem 5.4), ordered and symmetric chiral homologies agree. We compute the ordered chiral Hochschild cohomology $\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(\mathcal{H}_k,\mathcal{H}_k)$ on the formal disk D and on $\mathbb{P}^1$ explicitly, arity by arity.

Setup. The ordered chiral Hochschild cochain complex (Definition 6.1) is

$$\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(\mathcal{H}_k,\mathcal{H}_k) = \bigoplus_{n\geq 0}\Big(\Omega^*\big(\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})\big)\otimes \mathrm{Hom}(\mathcal{H}_k^{\otimes n},\mathcal{H}_k),\ d_{\mathrm{dR}}+d_{\mathrm{Hoch}}+\nabla_{\mathrm{KZ}}\Big),$$

where d_{Hoch} is the Hochschild differential (encoding OPE residues at boundary strata) and ∇_{KZ} is the KZ connection. The bar complex is $B(\mathcal{H}_k)=T^c(s^{-1}\bar{\mathcal{H}}_k)$ (desuspended augmentation ideal; $|s^{-1}v|=|v|-1$), with deconcatenation coproduct.

Arity 0: the vacuum. The chiral centre $Z(\mathcal{H}_k)=\{a : a_{(n)}b=0 \text{ for all } b,\ n\geq 0\}$ consists of the scalars only (since $J_{(1)}J=k\neq 0$ but $J_{(0)}J=0$).

$$\mathrm{HH}^0=\mathbb{C}. \tag{7.1}$$

Arity 1: outer derivations. A chiral derivation D satisfies $D(J)=c\cdot\mathbf{1}$ for $c\in\mathbb{C}$ (the Leibniz rule and the absence of fields with regular OPE against

J force this). Since $J_{(0)}J = 0$ (no simple pole), all inner derivations vanish: $\mathrm{Inn}(\mathcal{H}_k) = 0$.

$$\mathrm{HH}^1 = \mathbb{C}, \tag{7.2}$$

generated by the translation $D(J) = \mathbf{1}$. The classical Hochschild HH^1 of the Weyl algebra is 0 (every derivation is inner); the discrepancy arises because chiral inner derivations use $a_{(0)}$, not $[a, -]$.

Arity 2: level deformation and the KZ connection. Setting $w = z_1 - z_2$, the KZ connection is

$$\nabla_{\mathrm{KZ}} = d + k\, d\log w, \tag{7.3}$$

with solution $\Phi(w) = w^k$ and scalar monodromy

$$M = e^{2\pi i k}. \tag{7.4}$$

For generic $k \notin \mathbb{Z}$, the twisted de Rham cohomology of $\mathbb{C}^\times$ vanishes. The level deformation cocycle $\phi_k(J \otimes J) = k \cdot \mathbf{1}$ represents

$$[\phi_k] \in \mathrm{HH}^2 = \mathbb{C}, \tag{7.5}$$

the unique deformation $J(z)J(w) \sim (k+\epsilon)/(z-w)^2$.

Arity $n \geq 3$: vanishing. The Heisenberg is chirally Koszul with resolution of length 2, so the cochain complex at arity $n \geq 3$ is acyclic:

$$\mathrm{HH}^{n,\mathrm{ord}}_{\mathrm{ch}}(\mathcal{H}_k, \mathcal{H}_k) = 0 \quad \text{for } n \geq 3. \tag{7.6}$$

Summary on D. The ordered chiral Hochschild cohomology on the formal disk is

$$\mathrm{HH}^{n,\mathrm{ord}}_{\mathrm{ch}}(\mathcal{H}_k, \mathcal{H}_k) = \begin{cases} \mathbb{C} & n = 0 \text{ (vacuum: centre} = \mathbb{C}\cdot\mathbf{1}), \\ \mathbb{C} & n = 1 \text{ (outer derivation: } D(J) = \mathbf{1}), \\ \mathbb{C} & n = 2 \text{ (level deformation: } k \mapsto k+\epsilon), \\ 0 & n \geq 3 \text{ (Koszul resolution of length 2)}, \end{cases} \tag{7.7}$$

with Poincaré polynomial $P_{\mathcal{H}_k}(t) = 1 + t + t^2$. The total dimension is 3, and the cohomological amplitude is $[0, 2]$, consistent with Theorem H. The derived chiral centre is $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{H}_k) \cong \mathbb{C} \oplus \mathbb{C}[-1] \oplus \mathbb{C}[-2]$ as a graded vector space.

The computation on $\mathbb{P}^1$. The computation on $\mathbb{P}^1$ gives the same result:

$$\mathrm{HH}^{n,\mathrm{ord}}_{\mathrm{ch},\mathbb{P}^1}(\mathcal{H}_k, \mathcal{H}_k) \cong \mathrm{HH}^{n,\mathrm{ord}}_{\mathrm{ch},D}(\mathcal{H}_k, \mathcal{H}_k) \quad \text{for all } n. \tag{7.8}$$

The factorisation $\mathcal{D}$-module is insensitive to the compactification for chirally Koszul algebras: the Koszul resolution has length 2, so only arities 0, 1, 2 contribute, and at these arities the global geometry of $\mathbb{P}^1$ introduces no new cohomology beyond what the formal disk sees.

The E_3 structure. The derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{H}_k)$ is an E_3-algebra by Proposition 6.2. The obstruction space $\mathrm{HH}^3_{\mathsf{E}_2}(\overline{B}^\Sigma(\mathcal{H}_k), \overline{B}^\Sigma(\mathcal{H}_k)) \cong \Lambda^3(\mathbb{C}^1) = 0$ vanishes, so the E_3-algebra structure on $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{H}_k) \cong \mathbb{C} \oplus \mathbb{C}[-1] \oplus \mathbb{C}[-2]$ is the unique (trivial) one. The 3d theory is abelian Chern–Simons at level k;

the triviality reflects the absence of perturbative quantum corrections in the abelian theory.

Comparison with topological Hochschild. By the formality bridge (Theorem 5.4), $\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(\mathcal{H}_k, \mathcal{H}_k) \simeq \mathrm{HH}^{*,\mathrm{top}}(\mathcal{H}_k, \mathcal{H}_k)$: no information is lost in the passage from holomorphic (ordered, z-dependent) to topological (S^1, z-independent).

Example 7.2 (Affine Kac–Moody: class L, $r = 3$; explicit computation for $\widehat{\mathfrak{sl}}_2$)**.** We carry out the ordered chiral Hochschild computation for $V_k(\mathfrak{sl}_2)$ on the formal disk, the simplest non-abelian Kac–Moody algebra.

Data. $\mathfrak{g} = \mathfrak{sl}_2$, $h^\vee = 2$, $\dim(\mathfrak{g}) = 3$. Generators $\{e, f, h\}$ with brackets $[e, f] = h$, $[h, e] = 2e$, $[h, f] = -2f$. Normalised invariant form: $(e, f) = (f, e) = 1$, $(h, h) = 2$. Inverse Casimir: $\Omega = e \otimes f + f \otimes e + \frac{1}{2} h \otimes h$. The Koszul invariant is $\kappa = 3(k + 2)/4$. The r-matrix in trace-form convention is $r(z) = k\Omega/z$ (at $k = 0$, $r = 0$); in KZ normalisation, $r(z) = \Omega/((k + 2)z)$. Class L (shadow depth $r = 3$): ℓ_1, ℓ_2, ℓ_3 non-trivial, $\ell_k = 0$ for $k \geq 4$.

Arity 2: KZ conformal blocks. The ordered chiral Hochschild at arity 2 is the de Rham cohomology of $\mathrm{Conf}_2(\mathbb{C})$ with coefficients in the KZ local system. The KZ equation at arity 2, setting $z = z_1 - z_2$, is

$$\frac{\partial \Phi}{\partial z} = \frac{\Omega}{(k + 2)\, z}\, \Phi. \tag{7.9}$$

Since $V_k(\mathfrak{sl}_2)$ is E_∞-chiral, the coefficients live in $V_k(\mathfrak{sl}_2)^{\otimes 2}$, which decomposes under $\mathfrak{sl}_2$ by Clebsch–Gordan. For the adjoint representation acting on itself ($j = 1$): $V_1 \otimes V_1 = V_0 \oplus V_1 \oplus V_2$. The Casimir eigenvalue of Ω on the isospin-J component is

$$\lambda_J = \frac{C_2(J) - C_2(1) - C_2(1)}{2} = \frac{J(J + 1) - 4}{2}, \tag{7.10}$$

where $C_2(j) = j(j + 1)$ in the normalisation $(\theta|\theta) = 2$. The eigenvalues are:

$$\lambda_0 = -2, \quad \lambda_1 = -1, \quad \lambda_2 = 1.$$

On each isospin component, the KZ equation (7.9) has the explicit solution $\Phi_J(z) = z^{\lambda_J/(k+2)}$. The monodromy around $z = 0$ (the diagonal $\Delta_{12} \subset X^2$) is

$$M_J = \exp\bigl(2\pi i\, \lambda_J/(k + 2)\bigr).$$

The de Rham cohomology $H^*_{\mathrm{dR}}(\mathrm{Conf}_2(\mathbb{C}), \nabla_{\mathrm{KZ}})$ decomposes as:

- $H^0 = \ker(M - \mathrm{id})$: the subspace on which the monodromy is trivial. At generic k, this is zero (no isospin channel has $\lambda_J/(k + 2) \in \mathbb{Z}$).
- $H^1 = \mathrm{coker}(M - \mathrm{id})$: the space of KZ conformal blocks at arity 2. At generic k, $H^1 \cong V_0 \oplus V_1 \oplus V_2$ (three-dimensional over the isospin decomposition: one block per channel).

The averaging map $\mathrm{av}_2 \colon H^1_{\mathrm{ord}} \to H^1_\Sigma$ projects each conformal block to its Σ_2-coinvariant. Under the flip $z \mapsto -z$, the monodromy on V_J picks up a sign $(-1)^J$ (from the permutation of the two adjoint insertions); at generic

k, av_2 is surjective with $\ker(\mathrm{av}_2)$ carrying the odd-J blocks. The scalar $\kappa = \mathrm{av}(r(z)) + \dim(\mathfrak{g})/2 = 3k/(2\cdot 2) + 3/2 = 3(k+2)/4$ is the complete information surviving the averaging.

Arity 3: the Drinfeld associator. At arity 3, the KZ system on $\mathrm{Conf}_3(\mathbb{C})$ is

$$d\Phi = \frac{1}{k+2}\left(\frac{\Omega_{12}}{z_1-z_2}\,d(z_1-z_2) + \frac{\Omega_{13}}{z_1-z_3}\,d(z_1-z_3) + \frac{\Omega_{23}}{z_2-z_3}\,d(z_2-z_3)\right)\Phi. \tag{7.11}$$

The fundamental group $\pi_1(\mathrm{Conf}_3(\mathbb{C})) = PB_3$ (the pure braid group on three strands) has generators $\sigma_{12}^2, \sigma_{23}^2$ (Dehn twists around the diagonals). The KZ monodromy gives a representation $PB_3 \to \mathrm{GL}(V_k(\mathfrak{g})^{\otimes 3})$. The Drinfeld associator $\Phi_{\mathrm{KZ}} \in \exp(\widehat{\mathrm{Lie}}(t_{12}, t_{23}))$ is the regularised holonomy along the path from the boundary stratum $|z_1 - z_2| \ll |z_2 - z_3|$ to the stratum $|z_2 - z_3| \ll |z_1 - z_2|$ in the real locus of $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$.

The kernel of the averaging map av_3 is the space of ordered data invisible to $\overline{B}^{\Sigma}$. For $\mathfrak{g} = \mathfrak{sl}_2$, this kernel is one-dimensional: $\ker(\mathrm{av}_3) \cong \Lambda^3(\mathbb{C}^3) = \mathbb{C}$, carrying the Drinfeld associator. The associator Φ_{KZ} is the unique (up to GRT_1-action) non-trivial element of this kernel.

For general $\mathfrak{g}$, the ordered data at arity 3 is controlled by $\Lambda^3(\mathfrak{g}) = \Lambda^3(\mathbb{C}^{\dim \mathfrak{g}})$, with $\dim \Lambda^3(\mathfrak{g}) = \binom{\dim \mathfrak{g}}{3}$. At $\mathfrak{g} = \mathfrak{sl}_2$: $\binom{3}{3} = 1$.

E_3 obstruction. The symmetric bar $\overline{B}^{\Sigma}(V_k(\mathfrak{sl}_2))$ carries an E_2-coalgebra structure from the curve geometry (Proposition 6.2(i): the coshuffle coproduct on configuration spaces of the curve, not from bar-cobar). The obstruction to promoting $\overline{B}^{\Sigma}$ to an E_3-coalgebra lives in

$$\mathrm{HH}^3_{\mathsf{E}_2}(\overline{B}^{\Sigma}, \overline{B}^{\Sigma}) \;\cong\; H^3(\mathfrak{sl}_2) \;\cong\; \Lambda^3(\mathfrak{sl}_2^*)^{\mathfrak{sl}_2} \;=\; \mathbb{C}, \tag{7.12}$$

which is one-dimensional and carries the non-trivial obstruction class (Warning 6.3). The E_3-algebra is the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{sl}_2))$, not $\overline{B}^{\Sigma}$ itself.

Identification with CFG. The E_3-deformation space is $\hbar H^3(\mathfrak{sl}_2)[[\hbar]] \cong \hbar\mathbb{C}[[\hbar]]$: one-dimensional at each $\hbar$-order. For $\mathfrak{sl}_2$, $H^3(\mathfrak{sl}_2) = \mathbb{C}$ (generated by the three-form $\omega_3 = (e^* \wedge f^* \wedge h^*) \circ [\cdot,\cdot]$, the Cartan form). The deformation parameter is the departure from critical level: $k = -2 + \hbar\lambda_1 + \hbar^2\lambda_2 + \cdots$. By Theorem 6.18, the formal disk restriction $\Gamma(D, \mathrm{CE}^{\mathrm{ch}}_{\hbar}(\mathfrak{sl}_2))$ recovers the CFG filtered E_3-algebra $C^*_{\hbar}(\mathfrak{sl}_2)$ of perturbative Chern–Simons quantisations; the full identification of the deformation families is Conjecture 6.5. At the associative level, the Drinfeld–Kohno theorem identifies the KZ monodromy with the R-matrix of $U_q(\mathfrak{sl}_2)$ at $q = \exp(\pi i/(k+2))$: the chiral quantum group equivalence (Theorem 9.5) lifts this from monodromy representations to the algebra.

Summary of ordered chiral Hochschild for $\widehat{\mathfrak{sl}}_2$.

Arity n	Ordered datum	Symmetric shadow
1	$V_k(\mathfrak{sl}_2)$	$V_k(\mathfrak{sl}_2)$
2	KZ local system; conformal blocks $V_0 \oplus V_1 \oplus V_2$	scalar $\kappa = 3(k+2)/4$
3	KZ associator Φ_{KZ}; $\ker(\mathrm{av}_3) = \mathbb{C}$	trivial (associator lost)
≥ 4	higher KZ holonomy	trivial

The KZ connection (3.6) has simple poles: regular singularities. The 3d theory is Chern–Simons gauge theory with gauge group G.

Example 7.3 ($\beta\gamma$ system: class C, $r = 4$; explicit ordered chiral Hochschild cohomology)**.** The $\beta\gamma$ system at conformal weight λ has two generating fields $\beta(z)$ of weight λ and $\gamma(z)$ of weight $1-\lambda$, with OPE

$$\beta(z)\gamma(w) \sim \frac{1}{z-w}, \qquad \gamma(z)\beta(w) \sim \frac{1}{z-w}, \qquad \beta(z)\beta(w) \sim 0, \qquad \gamma(z)\gamma(w) \sim 0.$$

The central charge is $c_{\beta\gamma}(\lambda) = 2(6\lambda^2 - 6\lambda + 1)$ and the Koszul invariant is $\kappa = c_{\beta\gamma}/2 = 6\lambda^2 - 6\lambda + 1$. (Verification: at $\lambda = 1/2$, $c_{\beta\gamma} = -1$; at $\lambda = 2$, $c_{\beta\gamma} = 26$.) The algebra is E_∞-chiral (the OPE is symmetric after clearing the simple pole by locality) and class C (shadow depth $r = 4$: the quartic contact shadow is nontrivial, but the quintic obstruction vanishes by rank-one abelian rigidity).

By the formality bridge (Theorem 5.4), ordered and symmetric chiral homologies agree. The distinctive feature of this example is that the KZ connection has *unipotent* monodromy (rather than scalar or semisimple), and the nilpotency $\Theta^2 = 0$ makes the parallel transport computable in closed form at every arity. The $\beta\gamma$ system is the *hydrogen atom* for ordered chiral homology with unipotent local systems: all KZ solutions are explicit polynomials in logarithms, with no hypergeometric functions and no Drinfeld associator.

The collision residue and R-matrix. The generating space is $V = \mathbb{C}\beta \oplus \mathbb{C}\gamma$ (two-dimensional). The OPE has a simple pole between β and γ. After d-log absorption (pole order $1 - 1 = 0$), the collision residue is a *constant* nilpotent endomorphism

$$\Theta \in \mathrm{End}(V \otimes V), \qquad \Theta(\beta \otimes \gamma) = \mathbf{1}, \quad \Theta(\gamma \otimes \beta) = \mathbf{1}, \quad \Theta(\beta \otimes \beta) = 0, \quad \Theta(\gamma \otimes \gamma) = 0, \tag{7.13}$$

where $\mathbf{1}$ denotes the vacuum in the full algebra A (not in $V \otimes V$). The map Θ sends $V \otimes V \to A$, with image in the one-dimensional vacuum subspace $\mathbb{C} \cdot \mathbf{1} \subset A$. The key algebraic fact: $\Theta^2 = 0$ (nilpotent of order 2), because the image of Θ is the vacuum subspace, and Θ annihilates the vacuum. Concretely: Θ has rank 1 on the four-dimensional space $V^{\otimes 2}$ (both $\beta \otimes \gamma$ and $\gamma \otimes \beta$ map to $\mathbf{1}$, hence the image is one-dimensional), with $\ker(\Theta) \supset \mathbb{C}(\beta \otimes \beta) \oplus \mathbb{C}(\gamma \otimes \gamma)$ (the "same-type" tensors, on which the OPE is regular) and $\mathrm{im}(\Theta) = \mathbb{C} \cdot \mathbf{1}$.

The R-matrix is the monodromy $R = \exp(2\pi i\,\Theta)$. Since $\Theta^2 = 0$, the exponential terminates:

$$R_{\beta\gamma} = I + 2\pi i\,\Theta. \tag{7.14}$$

This termination at the linear term is the defining feature of class C. For affine Kac–Moody (class L), the R-matrix is an infinite series in the spectral parameter; for $\beta\gamma$, nilpotency truncates the series.

The KZ connection. At arity n, the KZ connection on $\mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{C})$ is

$$\nabla_n^{\beta\gamma} = d + \sum_{1\le i<j\le n} \Theta_{ij}\,\omega_{ij}, \qquad \omega_{ij} = d\log(z_i - z_j), \tag{7.15}$$

where $\Theta_{ij} \in \mathrm{End}(V^{\otimes n})$ acts as Θ on the i-th and j-th tensor factors and as the identity elsewhere. The connection is logarithmic: regular singularities along every diagonal Δ_{ij}. This is the analogue of the KZ connection for affine KM, with the simplification that $\Theta_{ij}^2 = 0$.

Flatness: the condition $(\nabla_n^{\beta\gamma})^2 = 0$ reduces to the classical Yang–Baxter equation for Θ:

$$[\Theta_{12}, \Theta_{13}] + [\Theta_{12}, \Theta_{23}] + [\Theta_{13}, \Theta_{23}] = 0 \quad \text{in } \mathrm{End}(V^{\otimes 3}). \tag{7.16}$$

This holds because Θ is the collision residue of a vertex algebra OPE. Each commutator $[\Theta_{ij}, \Theta_{jk}]$ produces a "triple collision" term (contraction at all three sites via the shared index j), and the three such terms cancel by the cyclic symmetry of the Arnold relation.

Explicit KZ solutions via nilpotency. Since $\Theta^2 = 0$, the parallel transport is polynomial in logarithms. At arity 2:

$$\Phi(w) = \exp(\Theta_{12}\log w) = I + \Theta_{12}\,\log w, \tag{7.17}$$

with monodromy $R = I + 2\pi i\,\Theta_{12}$. At arity 3, the solution terminates at degree two in logarithms (triple contractions vanish on the two-dimensional V), and the Drinfeld associator is trivial. Since $\Theta^2 = 0$, the iterated-integral corrections from non-commutativity of the connection vanish: the Picard series $\Phi = I + \sum_{i<j}\Theta_{ij}\log(z_i - z_j) + \sum_{(i<j),(k<l)}\Theta_{ij}\Theta_{kl}\log(z_i - z_j)\log(z_k - z_l) + \cdots$ reduces to ordinary products of logarithms (rather than Chen iterated integrals), because the ordering ambiguity in the iterated integral $\int \Theta_{ij}\omega_{ij} \circ \Theta_{kl}\omega_{kl}$ versus $\int \Theta_{kl}\omega_{kl} \circ \Theta_{ij}\omega_{ij}$ differs by the commutator $[\Theta_{ij}, \Theta_{kl}]$, which vanishes when $\Theta^2 = 0$ (the image of each Θ_{ij} is the vacuum, annihilated by every Θ_{kl}).

The chiral Hochschild computation. The $\beta\gamma$ system is chirally Koszul with Koszul resolution of length 2 (generating space $V = \mathbb{C}\beta \oplus \mathbb{C}\gamma$, relation space $R = \mathbb{C}\cdot(\beta_{(0)}\gamma - \mathbf{1})$), so only arities 0, 1, 2 contribute.

Arity 0: the vacuum. Since $\beta_{(0)}\gamma = \mathbf{1} \neq 0$, neither β nor γ is central; only the scalars survive.

$$\mathrm{HH}^0 = \mathbb{C}. \tag{7.18}$$

Arity 1: outer derivations. The simple-pole OPE $\beta_{(0)}\gamma = \mathbf{1}$ makes $a_{(0)}$ surjective on the generating space, so every chiral derivation is inner ($\beta \mapsto \beta + \epsilon$ is generated by $(\epsilon\gamma)_{(0)}$).

$$\mathrm{HH}^1 = 0. \tag{7.19}$$

This inverts the Heisenberg pattern: a double pole makes inner derivations trivial ($\mathrm{HH}^1 = \mathbb{C}$); a simple pole makes all derivations inner ($\mathrm{HH}^1 = 0$). The dichotomy is a shadow of the Koszul duality $(\beta\gamma)^! \simeq bc$.

Arity 2: the unipotent KZ connection. The KZ connection (7.15) at arity 2 is

$$\nabla^{\beta\gamma}_{\mathrm{KZ}} = d + \Theta_{12}\,\omega_{12}, \qquad \omega_{12} = d\log(z_1 - z_2). \tag{7.20}$$

The solution is $\Phi(w) = I + \Theta_{12}\log w$ (equation (7.17)), with *unipotent* monodromy

$$M = R_{\beta\gamma} = I + 2\pi i\,\Theta_{12}, \qquad (M - I)^2 = 0. \tag{7.21}$$

For a unipotent local system ($N = M - I$, $N^2 = 0$), the de Rham cohomology is $H^0 = \ker(N)$, $H^1 = \mathrm{coker}(N)$. Since Θ_{12} has rank 2 on $V^{\otimes 2} \cong \mathbb{C}^4$ (killing $\beta \otimes \beta$ and $\gamma \otimes \gamma$), $\dim\ker = \dim\mathrm{coker} = 2$. The monodromy is scalar for the Heisenberg ($e^{2\pi i k}$, generic), semisimple for affine KM ($e^{2\pi i \lambda_J/(k+2)}$, channel-dependent), and unipotent for $\beta\gamma$ (always nontrivial flat sections).

The unique deformation cocycle $\phi(\beta \otimes \gamma) = \epsilon \cdot \mathbf{1}$ represents

$$[\phi] \in \mathrm{HH}^2 = \mathbb{C}, \tag{7.22}$$

corresponding to $\beta(z)\gamma(w) \sim (1+\epsilon)/(z-w)$.

Arity $n \geq 3$: vanishing. By the Koszul resolution of length 2:

$$\mathrm{HH}^{n,\mathrm{ord}}_{\mathrm{ch}}(\beta\gamma, \beta\gamma) = 0 \quad \text{for } n \geq 3. \tag{7.23}$$

Summary on D. The ordered chiral Hochschild cohomology on the formal disk is

$$\mathrm{HH}^{n,\mathrm{ord}}_{\mathrm{ch}}(\beta\gamma, \beta\gamma) = \begin{cases} \mathbb{C} & n = 0 \text{ (vacuum: centre} = \mathbb{C}\cdot\mathbf{1}), \\ 0 & n = 1 \text{ (all derivations inner: simple-pole OPE)}, \\ \mathbb{C} & n = 2 \text{ (OPE residue deformation: } 1/(z-w) \mapsto (1+\epsilon)/(z-w)), \\ 0 & n \geq 3 \text{ (Koszul resolution of length 2)}, \end{cases} \tag{7.24}$$

with Poincaré polynomial $P_{\beta\gamma}(t) = 1 + t^2$. The total dimension is 2, and the cohomological amplitude is $[0, 2]$, consistent with Theorem H. The derived chiral centre is $Z^{\mathrm{der}}_{\mathrm{ch}}(\beta\gamma) \cong \mathbb{C} \oplus \mathbb{C}[-2]$ as a graded vector space.

Remark 7.4 (Poincaré polynomials and Koszul duality). $P_{\mathcal{H}_k}(t) = 1 + t + t^2$ versus $P_{\beta\gamma}(t) = 1 + t^2$: the missing t-term reflects $\mathrm{HH}^1 = 0$. By Koszul duality $(\beta\gamma)^! \simeq bc$, the bc ghost system has $P_{bc}(t) = 1 + t^2$ as well ($\mathrm{HH}^n(\beta\gamma) \cong \mathrm{HH}^{2-n}(bc)^*$).

The computation on $\mathbb{P}^1$. The computation on $\mathbb{P}^1$ gives the same result (as for the Heisenberg, the factorisation $\mathcal{D}$-module is insensitive to the compactification for chirally Koszul algebras):

$$\mathrm{HH}^{n,\mathrm{ord}}_{\mathrm{ch},\mathbb{P}^1}(\beta\gamma,\beta\gamma) \cong \mathrm{HH}^{n,\mathrm{ord}}_{\mathrm{ch},D}(\beta\gamma,\beta\gamma) \quad \text{for all } n. \tag{7.25}$$

Koszul duality and complementarity. The Koszul dual is the bc ghost system at weight λ, with $c_{bc}(\lambda) = 1 - 3(2\lambda-1)^2$, satisfying the complementarity

$$\kappa(\beta\gamma) + \kappa(bc) = 0. \tag{7.26}$$

(Verification: $c_{\beta\gamma}(2) = 26$, $c_{bc}(2) = -26$, sum $= 0$.) The complementarity $K = 0$ places $\beta\gamma$ in the same family as the Heisenberg and lattice algebras: the Koszul conductor vanishes.

The E_3 structure. As for the Heisenberg, the E_3 obstruction space $\mathrm{HH}^3_{\mathsf{E}_2}(\overline{B}^{\Sigma}(\beta\gamma), \overline{B}^{\Sigma}(\beta\gamma)) \cong \Lambda^3(\mathbb{C}^2) = 0$ vanishes for dimensional reasons, and the E_3-algebra structure on $Z^{\mathrm{der}}_{\mathrm{ch}}(\beta\gamma) \cong \mathbb{C} \oplus \mathbb{C}[-2]$ is the unique (trivial) one.

Comparison with topological Hochschild. By the formality bridge (Theorem 5.4), $\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(\beta\gamma,\beta\gamma) \simeq \mathrm{HH}^{*,\mathrm{top}}(\beta\gamma,\beta\gamma)$. The transferred A_∞-structure carries nontrivial operations $m_3^{\mathrm{tr}} \neq 0$ and $m_4^{\mathrm{tr}} \neq 0$ (shadow depth $r = 4$): the formality bridge preserves the chain complex but does not kill the higher operations. The information lost in passing from ordered to topological is the explicit z-dependence of the flat sections: $\Phi(w) = I + \Theta \log w$ is replaced by the abstract homotopy transfer data.

Example 7.5 (Yangian $Y_\hbar(\mathfrak{sl}_2)$ as a chiral quantum group: three structures and the ordered chiral center)**.** The Yangian $Y_\hbar(\mathfrak{sl}_2)$ is the first example in which the ordered chiral homology differs from the Beilinson–Drinfeld (symmetric) chiral homology. It is a genuinely E_1-chiral algebra: the R-matrix is independent input, not derived from any E_∞-chiral OPE. The formality bridge (Theorem 5.4) does not apply, because the factorisation $\mathcal{D}$-module is not Σ_n-equivariant. We exhibit all three structures of the chiral quantum group equivalence (Theorem 9.5) concretely, then compute the ordered chiral center at low arities and identify $\ker(\mathrm{av}) \neq 0$.

Data. Let $\mathfrak{g} = \mathfrak{sl}_2$ with standard basis $\{e, f, h\}$, brackets $[e,f] = h$, $[h,e] = 2e$, $[h,f] = -2f$, and normalised Killing form $(e,f) = 1$, $(h,h) = 2$. The dual Coxeter number is $h^\vee = 2$. The deformation parameter is $\hbar = 1/(k+h^\vee) = 1/(k+2)$ ($= h_{\mathrm{KZ}}$ of Remark 6.6), where k is the level of the affine input $V_k(\mathfrak{sl}_2)$. The inverse Casimir is

$$\Omega = e \otimes f + f \otimes e + \tfrac{1}{2}\, h \otimes h \ \in\ \mathfrak{g} \otimes \mathfrak{g}. \tag{7.27}$$

In the fundamental representation $V = \mathbb{C}^2$: $(\rho \otimes \rho)(\Omega) = P - \frac{1}{2} I$, where P is the permutation operator on $V \otimes V$.

Structure (I): the R-matrix. The Yang R-matrix on $V \otimes V$ is

$$R(u) = u\,I + \hbar\,P \;\in\; \mathrm{End}(V \otimes V)[u], \tag{7.28}$$

where u is the spectral parameter (the difference of evaluation points on $\mathbb{C}$). This satisfies the quantum Yang–Baxter equation

$$R_{12}(u-v)\,R_{13}(u)\,R_{23}(v) = R_{23}(v)\,R_{13}(u)\,R_{12}(u-v) \tag{7.29}$$

in $\mathrm{End}(V^{\otimes 3})[u,v]$, and the unitarity condition $R_{12}(u)\,R_{21}(-u) = (u^2-\hbar^2)\,I$ (scalar: invertible for $u \neq \pm\hbar$).

At the classical level, $R(u) = u\,I + \hbar\,P = u(I + \hbar\,\Omega/u + O(\hbar^2))$ after restoring the Casimir from $P = \Omega + \frac{1}{2}I$ on $V \otimes V$. The classical r-matrix $r(z) = \hbar\,\Omega/z$ is in KZ normalisation; at $\hbar = 0$ (i.e. $k \to \infty$), $r = 0$; at the critical level $k = -2$, $\hbar$ diverges.

The eigenvalues of $R(u)$ on $V \otimes V$ are $u + \hbar$ on $\mathrm{Sym}^2 V$ (dimension 3) and $u - \hbar$ on $\bigwedge^2 V$ (dimension 1); the R-matrix has a zero on $\bigwedge^2 V$ at $u = \hbar$, selecting the antisymmetric channel.

Structure (II): the A_∞ structure. The chiral A_∞-structure maps $m_k^{\mathrm{ch}}\colon Y_\hbar^{\otimes k} \to Y_\hbar$ are the operations extracted from the ordered bar complex via the bar-cobar correspondence. The transfer matrix

$$T(u) = \begin{pmatrix} A(u) & B(u) \\ C(u) & D(u) \end{pmatrix} \;\in\; \mathrm{End}(V) \otimes Y_\hbar(\mathfrak{sl}_2)[\![u^{-1}]\!] \tag{7.30}$$

encodes the generating functions of the algebra.

The binary operation m_2^{ch} is the RTT relation: for any two entries $T_{ij}(u)$ and $T_{kl}(v)$ of the transfer matrix,

$$R_{12}(u-v)\,T_1(u)\,T_2(v) = T_2(v)\,T_1(u)\,R_{12}(u-v), \tag{7.31}$$

where subscripts indicate the auxiliary space factor. Expanding this 4×4 matrix equation produces the Yangian relations in RTT form. The ten independent component relations include:

- diagonal commutativity: $[A(u), A(v)] = 0$, $[D(u), D(v)] = 0$;
- off-diagonal commutativity: $[B(u), B(v)] = 0$, $[C(u), C(v)] = 0$;
- the cross-relations, e.g. $(u-v)[B(u), C(v)] = \hbar\big(D(u)A(v) - D(v)A(u)\big)$.

The higher operations m_k^{ch} for $k \geq 3$ involve the Drinfeld associator Φ_{KZ}. The binary product m_2^{ch} is strictly associative only up to the holonomy of the KZ connection on $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$: the correction is $m_3^{\mathrm{ch}}(a,b,c) = \Phi_{\mathrm{KZ}} \cdot m_2(m_2(a,b),c) - m_2(a, m_2(b,c))$, where Φ_{KZ} is the regularised holonomy between the two maximal boundary strata of $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$.

The Gauss decomposition $T(u) = F(u)\,K(u)\,E(u)$ (lower-triangular, diagonal, upper-triangular) translates the RTT relations into the Drinfeld

presentation: generators $\{E_r, F_r, H_r\}_{r\geq 0}$ with relations

$$[H_r, H_s] = 0, \tag{7.32}$$

$$[H_0, E_s] = 2E_s, \quad [H_0, F_s] = -2F_s, \tag{7.33}$$

$$[E_r, F_s] = H_{r+s}, \tag{7.34}$$

$$[H_{r+1}, E_s] - [H_r, E_{s+1}] = \hbar(H_r E_s + E_s H_r), \tag{7.35}$$

and the Serre relation $[E_{r+1}, [E_r, E_s]] - [E_r, [E_{r+1}, E_s]] = 0$ (and similarly for F). The leading modes $E_0 = e$, $F_0 = f$, $H_0 = h$ generate a copy of $U(\mathfrak{sl}_2)$ inside $Y_\hbar(\mathfrak{sl}_2)$; the PBW filtration has $\operatorname{gr} Y_\hbar(\mathfrak{sl}_2) \cong U(\mathfrak{sl}_2[t])$.

Structure (III): the chiral coproduct. The Drinfeld coproduct on $Y_\hbar(\mathfrak{sl}_2)$ is

$$\Delta_z(T(u)) = T(u) \dot{\otimes} T(u - z), \tag{7.36}$$

where $\dot{\otimes}$ denotes matrix multiplication in the auxiliary space $\mathrm{End}(V)$ and ordinary (completed) tensor product in the $Y_\hbar$ factor. The spectral parameter z is the difference coordinate $z_1 - z_2$ on $\mathrm{Conf}_2(\mathbb{C})$. On Drinfeld generators:

$$\Delta_z(E_0) = E_0 \otimes 1 + 1 \otimes E_0 + \hbar\, H_0 \otimes E_0 \cdot z^{-1} + O(z^{-2}), \tag{7.37}$$

$$\Delta_z(H_0) = H_0 \otimes 1 + 1 \otimes H_0 + 2\hbar\,(E_0 \otimes F_0 - F_0 \otimes E_0) \cdot z^{-1} + O(z^{-2}). \tag{7.38}$$

The coproduct is coassociative up to the Drinfeld associator Φ:

$$(\Delta_{z_1} \otimes \mathrm{id}) \circ \Delta_{z_2} = \Phi_{123}(z_1, z_2) \cdot (\mathrm{id} \otimes \Delta_{z_2}) \circ \Delta_{z_1} \cdot \Phi_{123}(z_1, z_2)^{-1},$$

where $\Phi_{123}(z_1, z_2) \in \exp(\widehat{\mathrm{Lie}}(\Omega_{12}, \Omega_{23}))$ is the KZ associator with spectral parameters.

The R-matrix is recovered from the coproduct: $R(u) = \sigma \circ \Delta_z \circ (\Delta_z^{\mathrm{op}})^{-1}$ evaluated on $V \otimes V$, where σ is the flip. This is the content of the equivalence (I)$\Leftrightarrow$(III) in Theorem 9.5.

The three structures are equivalent. The ordered bar complex $\overline{B}^{\mathrm{ord}}(Y_\hbar(\mathfrak{sl}_2)) = T^c(s^{-1}\overline{Y}_\hbar)$ (desuspended augmentation ideal, deconcatenation coproduct) is the universal datum from which all three structures are recovered:

(a) The bar differential encodes the A_∞ maps (coderivations of T^c biject with multilinear operations): the RTT relation (7.31) is the degree-2 bar differential, and the Serre relations are the degree-3 component.
(b) The deconcatenation coproduct on T^c induces the Drinfeld coproduct (7.36) via cobar dualisation.
(c) The R-matrix (7.28) is the arity-2 KZ monodromy: the parallel transport of the connection $\nabla = d + \hbar\,\Omega/(z_1 - z_2) \cdot d(z_1 - z_2)$ around the diagonal.

The passage $r(z) = \hbar\,\Omega/z \xrightarrow{\text{exponentiation}} R(u) = u\,I + P \xrightarrow{\text{RTT}}$ Yangian relations $\xrightarrow{\text{Gauss}}$ Drinfeld presentation is the ordered bar construction applied to $V_k(\mathfrak{sl}_2)$, producing $Y_\hbar(\mathfrak{sl}_2)$ as the E_1-chiral Koszul dual.

Part B: the ordered chiral center. We compute $\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(Y_\hbar(\mathfrak{sl}_2),\, Y_\hbar(\mathfrak{sl}_2))$ at low arities on the formal disk D, and identify the kernel of the averaging map av.

Arity 0*: the center.* The arity-0 cohomology is the chiral center $Z(Y_\hbar(\mathfrak{sl}_2))$: those elements a with $a_{(n)}b = 0$ for all b and $n \geq 0$. The quantum determinant $\mathrm{qdet}\, T(u) = A(u)\, D(u-\hbar) - B(u)\, C(u-\hbar)$ is central in $Y_\hbar(\mathfrak{sl}_2)$ [25]. The center is the polynomial algebra generated by the coefficients of $\mathrm{qdet}\, T(u)$:

$$\mathrm{HH}^{0,\mathrm{ord}}_{\mathrm{ch}} = Z(Y_\hbar(\mathfrak{sl}_2)) = \mathbb{C}[\mathrm{qdet}\, T(u)]. \tag{7.39}$$

The averaging map av_0 is an isomorphism at arity 0 (there is nothing to symmetrise), so $\ker(\mathrm{av}_0) = 0$.

Arity 2*: the KZ local system and* $\ker(\mathrm{av}_2) \neq 0$*.* This is the arity where the ordered theory diverges from the symmetric. The ordered chiral Hochschild complex at arity 2 involves the de Rham cohomology of $\mathrm{Conf}_2^{\mathrm{ord}}(\mathbb{C})$ with coefficients in the KZ local system. At arity 2, the leading-order KZ connection $\nabla = d + \hbar\,\Omega\, d\log(z_1 - z_2)$ captures the classical r-matrix data $r(z) = \hbar\,\Omega/z$. The full quantum R-matrix $R(u) = u\, I + \hbar\, P$ contributes higher-order corrections visible at arity ≥ 3 through the Drinfeld associator; at arity 2 the connection is:

$$\nabla_{\mathrm{KZ}} = d + \frac{\hbar\,\Omega}{z_1 - z_2}\, d(z_1 - z_2) = d + \hbar\,\Omega\, d\log(z_1 - z_2), \tag{7.40}$$

with coefficients in $\mathrm{End}(Y_\hbar^{\otimes 2})$. Setting $w = z_1 - z_2$, the flat sections on $\mathbb{C}^\times$ satisfy $\partial\Phi/\partial w = (\hbar\,\Omega/w)\,\Phi$.

For the Yangian on the fundamental representation $V \otimes V$, the Casimir decomposes along the irreducible $\mathfrak{sl}_2$-components:

$$V \otimes V = \mathrm{Sym}^2 V \oplus \bigwedge^2 V, \qquad \Omega\big|_{\mathrm{Sym}^2 V} = +\tfrac{1}{2}, \quad \Omega\big|_{\bigwedge^2 V} = -\tfrac{3}{2}.$$

The monodromy of ∇_{KZ} around $w = 0$ on each isospin component is

$$M_{\mathrm{sym}} = e^{2\pi i\hbar/2} = e^{\pi i/(k+2)}, \qquad M_{\mathrm{alt}} = e^{-3\pi i/(k+2)}. \tag{7.41}$$

The averaging map at arity 2 is $\mathrm{av}_2 \colon H^*_{\mathrm{dR}}(\mathrm{Conf}_2^{\mathrm{ord}}, \nabla_{\mathrm{KZ}}) \to H^*_{\mathrm{dR}}(\mathrm{Conf}_2, \nabla_{\mathrm{KZ}})^{\Sigma_2}$, where Σ_2 acts by the flip $w \mapsto -w$ composed with the R-matrix.

For an E_∞-chiral algebra, $R = \mathrm{id}$ and the flip acts by the Koszul sign; av_2 is an isomorphism and $\ker(\mathrm{av}_2) = 0$. For the Yangian, $R(u) \neq \mathrm{id}$: the R-matrix mixes the $\mathrm{Sym}^2 V$ and $\bigwedge^2 V$ channels non-trivially, and the Σ_2-action on the ordered de Rham cohomology is the composition of the geometric flip with R.

The kernel is spanned by those flat sections that are antisymmetric under this twisted Σ_2-action:

$$\ker(\mathrm{av}_2) \cong \bigwedge^2 V = \mathbb{C}. \tag{7.42}$$

The generator is the flat section $\Phi_{\mathrm{alt}}(w) = w^{-3\hbar/2}$ on the antisymmetric channel of $V \otimes V$. The full R-matrix $R(u) = u\, I + \hbar\, P$ evaluates on

the antisymmetric component as $(u-\hbar)\,\mathrm{id}$; the Σ_2-flip sends $\Phi_{\mathrm{alt}}(w)$ to $R_{21}(-w)\,\Phi_{\mathrm{alt}}(-w) = -\Phi_{\mathrm{alt}}(w)$ (the R-matrix eigenvalue -1 on $\bigwedge^2 V$ at the leading order cancels the geometric sign). This class is in the ordered cohomology but maps to zero under av_2: it is invisible to the symmetric chiral homology.

The scalar shadow surviving av_2 is $\kappa = \mathrm{av}(r(z)) + \dim(\mathfrak{g})/2 = 3k/(2\cdot 2) + 3/2 = 3(k+2)/4$.

Arity 3*: the Drinfeld associator in* $\ker(\mathrm{av}_3)$*.* At arity 3, the KZ system on $\mathrm{Conf}_3^{\mathrm{ord}}(\mathbb{C})$ is

$$d\Phi = \hbar\left(\frac{\Omega_{12}}{z_1-z_2}\,d(z_1-z_2) + \frac{\Omega_{13}}{z_1-z_3}\,d(z_1-z_3) + \frac{\Omega_{23}}{z_2-z_3}\,d(z_2-z_3)\right)\Phi.$$

The fundamental group $\pi_1(\mathrm{Conf}_3^{\mathrm{ord}}(\mathbb{C})) = PB_3$ (pure braid group on three strands). The KZ monodromy gives a representation $\rho_{\mathrm{KZ}}\colon PB_3 \to \mathrm{GL}(V^{\otimes 3})$.

The Drinfeld associator $\Phi_{\mathrm{KZ}} \in \exp(\widehat{\mathrm{Lie}}(t_{12}, t_{23}))$ is the regularised holonomy of this connection along the real path from the boundary stratum $|z_1-z_2| \ll |z_2-z_3|$ to the stratum $|z_2-z_3| \ll |z_1-z_2|$ in $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$.

The averaging map av_3 sends ordered data to Σ_3-coinvariants. At arity 3, the relevant representation is the adjoint $\mathfrak{g} = \mathfrak{sl}_2$ (dimension 3), not the fundamental $V = \mathbb{C}^2$ (dimension 2) used at arity 2: the convolution algebra acts on $\mathfrak{g}^{\otimes n}$ via the adjoint action (the Casimir operators Ω_{ij} act through the adjoint representation on each tensor factor), and the Drinfeld associator lives in $\Lambda^3(\mathfrak{g})$. The kernel of av_3 contains the associator:

$$\ker(\mathrm{av}_3) \cong \Lambda^3(\mathfrak{g}) = \mathbb{C}, \tag{7.43}$$

one-dimensional, carrying the associator class. This computation follows from the structure of the cohomology of $\mathrm{Conf}_3^{\mathrm{ord}}(\mathbb{C})$: the top-degree Arnold class $\omega_{12}\wedge\omega_{23} \in H^2(\mathrm{Conf}_3^{\mathrm{ord}})$ maps to the alternating representation of Σ_3, and the KZ-twisted version (with coefficients in $\mathfrak{g}^{\otimes 3}$) gives $\ker(\mathrm{av}_3) \cong \Lambda^3(\mathfrak{g})$. For $\mathfrak{g} = \mathfrak{sl}_2$: $\dim \Lambda^3(\mathfrak{g}) = \binom{3}{3} = 1$.

For E_∞-chiral algebras, the formality bridge forces $\ker(\mathrm{av}_3) = 0$: the associator is in the image of a Σ_3-equivariant quasi-isomorphism and carries no independent ordered data. For the Yangian, the $\mathcal{D}$-module is not Σ_3-equivariant (because $R \neq \mathrm{id}$), and the associator survives as a non-trivial element of the ordered chiral center invisible to any symmetric theory.

Arity n*: braid group monodromy.* At arity n, the ordered chiral center carries the full pure braid group monodromy of the KZ system with the quantum R-matrix:

$$\rho_{\mathrm{KZ}}^{(n)}\colon PB_n \;\to\; \mathrm{GL}\big(V^{\otimes n}\big).$$

The symmetric shadow at arity n retains only the scalar invariants: the traces of the monodromy matrices and the Markov-trace data. The kernel $\ker(\mathrm{av}_n)$ carries the full matrix-valued monodromy, from which knot invariants (the Jones polynomial and its coloured generalisations) are extracted.

The Poincaré series of the ordered chiral center has the form

$$P^{\mathrm{ord}}_{Y_\hbar}(t) = P^{\Sigma}_{Y_\hbar}(t) + \Delta P(t),$$

where $\Delta P(t) = t^2 + t^3 + \cdots$ records the extra data in ker(av) at each arity. At arity 2: $\Delta P_2 = 1$ (the antisymmetric KZ block). At arity 3: $\Delta P_3 = 1$ (the Drinfeld associator class).

Why this is new. This is the first instance where the formality bridge (Theorem 5.4) fails: the $\mathcal{D}$-module is not Σ_n-equivariant because $R(u) \neq \mathrm{id}$, so the ordered chiral homology carries strictly more data than the symmetric theory.

Arity n	Ordered datum	$\ker(\mathrm{av}_n)$	Symmetric shadow
0	$Z(Y_\hbar) = \mathbb{C}[\mathrm{qdet}]$	0	$Z(Y_\hbar)$
2	full R-matrix $R(u) = uI + \hbar P$	$\bigwedge^2 V = \mathbb{C}$	scalar $\kappa = 3(k{+}2)/4$
3	KZ associator Φ_{KZ}	$\Lambda^3(\mathfrak{g}) = \mathbb{C}$	trivial
n	PB_n-monodromy of KZ	matrix-valued	scalar traces only

Example 7.6 (Virasoro: class M, $r = \infty$; the irregular $\mathcal{D}$-module and Stokes phenomenon)**.** The Virasoro algebra Vir_c has a single generating field $T(z)$ of conformal weight 2, OPE

$$T(z)\,T(w) \;\sim\; \frac{c/2}{(z-w)^4} + \frac{2\,T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w},$$

and r-matrix $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$ (quartic OPE pole, d-log absorption gives cubic r-matrix pole). The Koszul invariant is $\kappa = c/2$. The algebra is E_∞-chiral (symmetric OPE: the commutator $[T(z), T(w)]$ is commutative after clearing poles) and class M (shadow depth $r = \infty$: all higher shadow operations $\ell_k \neq 0$ in the symmetric convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$).

Formality bridge and E_∞ descent. By the formality bridge (Theorem 5.4), ordered and symmetric chiral homologies agree; ker(av) = 0 at each arity. The infinite shadow tower of class M is a *symmetric* phenomenon. The content genuinely new to class M is the *singularity type* of the factorisation $\mathcal{D}$-module: irregular, with Stokes data that has no analogue in class $G/L/C$.

The irregular KZ connection at arity 2. At arity 2, setting $z = z_1 - z_2$, the KZ equation (Proposition 3.6(ii)) reads

$$\frac{\partial \Phi}{\partial z} = \left(\frac{c/2}{z^3} + \frac{2T}{z}\right)\Phi. \tag{7.44}$$

This has a pole of order 3 in z: the singular point $z = 0$ is *irregular* of Poincaré rank 2 (pole order minus one). The classical Riemann–Hilbert correspondence of Deligne does not apply; one needs the irregular Riemann–Hilbert correspondence of D'Agnolo–Kashiwara [11].

For comparison, the affine Kac–Moody KZ equation at arity 2 (Example 7.2) is $\partial\Phi/\partial z = (\Omega/((k+h^\vee)z))\,\Phi$, a *simple* pole. Its solution is the power function $\Phi(z) = z^{\Omega/(k+h^\vee)}$: purely algebraic monodromy, no exponential corrections, no Stokes data. The difference between class L and class M is the difference between regular and irregular singularities.

Formal solution and the exponential factor. On the primary sector, T acts by the eigenvalue h (conformal weight of the field being transported). Equation (7.44) restricts to the scalar ODE

$$\frac{d\varphi}{dz} = \left(\frac{c/2}{z^3} + \frac{2h}{z}\right)\varphi, \tag{7.45}$$

which integrates exactly:

$$\varphi(z) = z^{2h}\,\exp\!\left(-\frac{c}{4z^2}\right). \tag{7.46}$$

The factor z^{2h} is the monodromy piece: it contributes $\exp(4\pi i h)$ upon circling $z = 0$, matching the conformal spin. The factor $\exp(-c/(4z^2))$ is the *irregular* piece: it has an essential singularity at $z = 0$, invisible to monodromy, carrying exponentially small corrections in different angular sectors of the z-plane.

In the Hukuhara–Turrittin–Levelt normal form [32, 28], the formal fundamental matrix is $\Phi^{\mathrm{form}}(z) = \widehat{F}(z)\,z^L\,\exp(-c/(4z^2)\,\mathrm{Id})$, where $\widehat{F}(z) \in \mathrm{GL}(V)[[z^{-1}]]$ is a Gevrey-1/2 formal gauge transformation.

Stokes phenomenon. The exponential $\exp(-c/(4z^2))$ produces four Stokes rays at $\theta = 0, \pi/2, \pi, 3\pi/2$ (matching Poincaré rank 2: $2 \times 2 = 4$ rays). Across each ray, the true solution jumps by a Stokes multiplier $\mathfrak{S}$:

$$\Phi_+^{\mathrm{true}}(z) = \Phi_-^{\mathrm{true}}(z) \cdot (\mathrm{Id} + \mathfrak{S}). \tag{7.47}$$

The flat sections are classified by a *Stokes-filtered local system*, not a monodromy representation. This contrasts with class $G/L/C$, where regular singularities produce no Stokes data.

Wall-crossing at $c = -22/5$. The Stokes multipliers depend on the central charge c. At $c = -22/5$ (the $(2,5)$ minimal model, the Yang–Lee edge singularity), two Stokes rays coalesce in a wall-crossing: the connection resonates, and the Stokes multipliers undergo a discontinuous change. In the language of the shadow obstruction tower [23], the shadow invariant $S_4 = 10/(c(5c+22))$ has a pole at $c = -22/5$, signalling the coalescence.

De Rham cohomology of the irregular $\mathcal{D}$-module. The factorisation $\mathcal{D}$-module $\mathcal{F}_n^{\mathrm{ord}}(\mathrm{Vir}_c)$ is holonomic on X^n for any smooth proper curve X (Proposition 3.6(i)). By Mebkhout's theorem [27], holonomic $\mathcal{D}$-modules on proper varieties have finite-dimensional de Rham cohomology regardless of regularity. The finite-dimensionality of $H^*_{\mathrm{dR}}(X^n, \mathcal{F}_n^{\mathrm{ord}})$ is therefore unconditional: the irregular singularities of class M do not obstruct finiteness.

The de Rham cohomology computes the flat sections of the connection with coefficients in the *Stokes-filtered local system*, not the naive local system (which is the wrong coefficient system for irregular connections). For $X = \mathbb{P}^1$ at arity $n = 2$: the irregular de Rham space $H^*_{\mathrm{dR}}(\mathrm{Conf}_2(\mathbb{P}^1), \nabla^{\mathrm{Vir}}_{\mathrm{KZ}})$ is finite-dimensional, with the dimension depending on c through the Stokes multipliers.

Shadow tower and the gravitational partition function. The class M shadow obstruction tower is infinite:

$$\begin{aligned}
S_2 &= c/2 = \kappa,\\
S_3 &= 2 \quad (c\text{-independent}),\\
S_4 &= 10/(c(5c+22)),\\
S_5 &= -48/(c^2(5c+22)),\\
S_6 &= 80(45c+193)/(3\,c^3(5c+22)^2),
\end{aligned}$$

with all S_r belonging to $\mathbb{Q}(c)$ and having poles only at $c = 0$ and $c = -22/5$. The spectral discriminant is $\Delta = 8\kappa S_4 = 40/(5c+22) \neq 0$: the tower does not terminate, confirming class M.

The scalar gravitational partition function on the scalar lane is

$$Z^{\mathrm{scal}}_{\mathrm{grav}}(\mathrm{Vir}_c;\,\hbar) = \frac{c}{2}\cdot\frac{\sqrt{\hbar}/2}{\sin(\sqrt{\hbar}/2)} \qquad \text{(UNIFORM-WEIGHT)}, \tag{7.48}$$

converging for $|\hbar| < 4\pi^2$. This is $\kappa\cdot(\sqrt{\hbar}/2)/\sin(\sqrt{\hbar}/2)$ evaluated at $\kappa = c/2$. At $c = 26$ (the bosonic string), $\kappa_{\mathrm{eff}} = \kappa(\mathrm{matter}) + \kappa(\mathrm{ghost}) = 13 + (-13) = 0$: the scalar gravitational partition function vanishes, and the non-scalar corrections from the shadow tower become the leading contribution.

The full gravitational partition function receives corrections from all arities of the shadow tower. For class M, infinitely many arities contribute at each genus $g \geq 2$. The leading non-scalar correction involves the quartic contact invariant $S_4 = 10/(c(5c+22))$, which encodes the quartic gravitational counterterm in the 3d theory.

Koszul duality and complementarity. The Koszul dual is Vir_{26-c}, with $\kappa + \kappa' = c/2 + (26-c)/2 = 13$ (self-dual at $c = 13$; Virasoro complementarity $K = 13$, distinct from the vanishing $K = 0$ of KM/Heis/lattice). Under $c \mapsto 26 - c$:

$$\Delta(c) + \Delta(26-c) = \frac{40}{5c+22} + \frac{40}{152-5c} = \frac{6960}{(5c+22)(152-5c)}. \tag{7.49}$$

The shadow growth rates are complementary: $\rho(c)\cdot\rho(26-c)$ is *not* constant, but both vanish in the limit $c \to \infty$.

The 3d gravitational theory. The 3d theory dual to Vir_c is holomorphic-topological gravity in the Beltrami parametrisation. The irregular singularity of the factorisation $\mathcal{D}$-module is the algebraic counterpart of the unbounded

spin tower in the gravitational theory: the $T(z)$ OPE generates fields of arbitrarily high spin through repeated collisions, producing the infinite shadow tower $S_2, S_3, S_4, \ldots$. In the 3d language, each S_r is a perturbative gravitational counterterm at r-point contact. The essential singularity $\exp(-c/(4z^2))$ in the flat sections of the KZ connection is the chiral-algebraic incarnation of the non-perturbative gravitational instanton corrections.

Summary.

Feature	**Class L (KM)**	**Class M (Vir)**
r-matrix pole	1 (simple)	3 (cubic)
Singularity type	regular (Fuchsian)	irregular (Poincaré rk. 2)
Flat sections	$z^{\Omega/(k+h^\vee)}$	$z^{2h}\exp(-c/(4z^2))$
Monodromy data	representation of π_1	Stokes-filtered local system
RH correspondence	Deligne (classical)	D'Agnolo–Kashiwara
Stokes rays	0	4
Shadow tower	terminates ($r = 3$)	infinite ($r = \infty$)
3d theory	CS gauge theory	HT gravity

8. Comparison with related work

Two existing programmes address the integration of braided algebraic structures over surfaces: Ben-Zvi–Brochier–Jordan integrate braided monoidal *categories*, while Latyntsev constructs chiral homology for factorisation *coalgebras*. Ordered chiral homology occupies a different position: it integrates chiral *algebras* at the chain level, applies uniformly to all four shadow classes, and does not require a finite ribbon category.

Remark 8.1 (Ben-Zvi–Brochier–Jordan). Ben-Zvi, Brochier, and Jordan [2, 3] construct a theory of factorisation homology for braided monoidal categories: for a ribbon category $\mathcal{C}$ and a surface Σ, they define $\int_\Sigma \mathcal{C}$ as a categorical invariant. Their construction and the present one operate at different levels:

- BBJ integrate braided monoidal *categories* over surfaces (categorical level).
- Ordered chiral homology integrates chiral *algebras* over curves (chain level).

For affine Kac–Moody: BBJ's $\int_\Sigma \mathrm{Rep}_q(\mathfrak{g})$ is a category, while $\int_X^{\mathrm{ord}} V_k(\mathfrak{g})$ is a chain complex. The expected comparison is:

$$\mathrm{HH}_*\Big(\int_\Sigma \mathrm{Rep}_q(\mathfrak{g})\Big) \;\simeq\; \int_X^{\mathrm{ord}} V_k(\mathfrak{g}), \tag{8.1}$$

where the left side is the Hochschild homology of the categorical invariant and the right side is the ordered chiral homology of the algebraic one.

For class M (Virasoro at generic c): the BBJ framework requires a finite ribbon category, which is not available for generic central charge. Ordered chiral homology applies without restriction to class M, providing

an integration theory where the categorical approach does not (currently) reach.

Remark 8.2 (Latyntsev). Latyntsev [22] constructs chiral homology for factorisation coalgebras on curves, providing an algebraic counterpart to the analytic construction of this paper. The relationship between the two approaches at the chain level, particularly for irregular singularities (class M), deserves further investigation.

9. Chiral quantum groups: R-matrices, A_∞-structures, and chiral coproducts

The classical theory of quantum groups rests on an equivalence between three descriptions of the same structure: a quasi-triangular Hopf algebra (H, Δ, R) can be specified by its R-matrix, by its coproduct, or by an operadic A_∞-algebra map. The goal of this section is to establish the chiral analogue: an equivalence between three descriptions of an E_1-chiral algebra equipped with quantum-group-type structure. The primitive datum is the E_1 ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with its deconcatenation coproduct (the topological color of the Swiss-cheese operad); the R-matrix and the chiral coproduct are both recoverable from this datum.

9.1. **Three structures on an E_1-chiral algebra.** An E_1-chiral algebra admits three equivalent descriptions: by a vertex R-matrix $S(z)$ (braiding data), by chiral A_∞ structure maps m_k^{ch} (higher associativity data), and by a chiral coproduct Δ^{ch} (coalgebra data). Each description captures a different projection of the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$.

Definition 9.1 (E_1-chiral algebra with vertex R-matrix)**.** An E_1-chiral algebra with vertex R-matrix is a pair $(\mathcal{A}, S(z))$ where $\mathcal{A}$ is a graded vertex algebra and $S(z) \in \mathrm{End}(\mathcal{A} \otimes \mathcal{A})[[z, z^{-1}]]$ is a vertex R-matrix satisfying:

(i) (S-locality.) For all $a, b \in \mathcal{A}$ and sufficiently large N:

$$(z-w)^N\, Y(a,z)\, Y(b,w) \;=\; (z-w)^N\, Y(b,w)\, Y(a,z) \cdot S(z-w). \tag{9.1}$$

(ii) (Quantum Yang–Baxter equation.) In $\mathrm{End}(\mathcal{A}^{\otimes 3})$:

$$S_{12}(z_1-z_2)\, S_{13}(z_1-z_3)\, S_{23}(z_2-z_3) \;=\; S_{23}(z_2-z_3)\, S_{13}(z_1-z_3)\, S_{12}(z_1-z_2). \tag{9.2}$$

(iii) (Unitarity.) $S_{12}(z)\, S_{21}(-z) = \mathrm{id}$.

(iv) (Shift condition.) The R-matrix intertwines the translation operator T: $[T \otimes \mathrm{id} + \mathrm{id} \otimes T,\, S(z)] = \partial_z S(z)$.

(v) (Hexagon axiom.) On $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$, the braiding and associativity satisfy the coherence: for the two standard paths in $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$ connecting the same boundary strata,

$$S_{12}(z_{12})\, \Phi_{123}^{-1}\, S_{13}(z_{13})\, \Phi_{132} \;=\; \Phi_{213}^{-1}\, S_{23}(z_{23})\, \Phi_{123} \tag{9.3}$$

where Φ_{ijk} is the Drinfeld associator evaluated on the three-point KZ connection.

These are the axioms of Etingof–Kazhdan [13] for a quantum vertex operator algebra.

Definition 9.2 (A_∞-algebra in the chiral endomorphism operad)**.** An A_∞-algebra structure on $\mathcal{A}$ in the chiral endomorphism operad $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$ is an operadic map

$$A_\infty \;\longrightarrow\; \mathrm{End}^{\mathrm{ch}}_{\mathcal{A}} \tag{9.4}$$

where $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}(n) = \mathrm{Hom}(\mathcal{A}^{\otimes n},\ \mathcal{A}((\lambda_1))\cdots((\lambda_{n-1})))$ is the spectral substitution operad ([23, §3]). Concretely, the data consist of chiral structure maps

$$m_k^{\mathrm{ch}}\colon \mathcal{A}^{\otimes k} \longrightarrow \mathcal{A}((\lambda_1))\cdots((\lambda_{k-1})), \qquad k \geq 1, \tag{9.5}$$

satisfying the Stasheff A_∞-relations in the chiral endomorphism operad: for each $n \geq 1$,

$$\sum_{\substack{i+j=n+1\\ 1\leq p\leq i}} (-1)^{\epsilon}\, m_i^{\mathrm{ch}} \circ_p m_j^{\mathrm{ch}} \;=\; 0, \tag{9.6}$$

where $\circ_p$ is the partial composition via spectral substitution (λ-variable nesting), and ϵ carries the standard Koszul sign from the suspended degree $\|m_k\| = k-2$. The structure maps m_k^{ch} are meromorphic in the spectral parameters $\lambda_1, \ldots, \lambda_{k-1}$. At $k=2$, the map m_2^{ch} is the OPE vertex operation $Y(-,z)$. At $k=1$, $m_1^{\mathrm{ch}} = d$ is the differential. The operations m_k^{ch} for $k \geq 3$ record the obstruction to strict associativity of the chiral product.

Definition 9.3 (E_1-chiral algebra with compatible chiral coproduct)**.** An E_1-chiral algebra with compatible chiral coproduct is a triple $(\mathcal{A}, Y, \Delta^{\mathrm{ch}})$ where $(\mathcal{A}, Y)$ is a chiral algebra and

$$\Delta^{\mathrm{ch}}\colon \mathcal{A} \longrightarrow (\mathcal{A} \mathbin{\hat{\otimes}} \mathcal{A})((z)) \tag{9.7}$$

is a chiral coproduct (a map to the completed tensor product with spectral parameter) satisfying:

(i) (Coassociativity up to associator.) The two iterated coproducts agree up to conjugation by the Drinfeld associator Φ:

$$(\Delta^{\mathrm{ch}} \otimes \mathrm{id}) \circ \Delta^{\mathrm{ch}} \;=\; \Phi \cdot (\mathrm{id} \otimes \Delta^{\mathrm{ch}}) \circ \Delta^{\mathrm{ch}} \cdot \Phi^{-1}. \tag{9.8}$$

(ii) (OPE compatibility.) The coproduct intertwines the vertex operation: for $a, b \in \mathcal{A}$,

$$\Delta^{\mathrm{ch}}(Y(a,z)\,b) \;=\; Y^{(2)}(\Delta^{\mathrm{ch}}(a), z) \cdot \Delta^{\mathrm{ch}}(b) \tag{9.9}$$

where $Y^{(2)}$ denotes the tensorwise vertex operation on $\mathcal{A} \mathbin{\hat{\otimes}} \mathcal{A}$.

(iii) (Counit.) There exists a counit $\varepsilon\colon \mathcal{A} \to \mathbb{C}$ with $(\varepsilon \otimes \mathrm{id}) \circ \Delta^{\mathrm{ch}} = \mathrm{id} = (\mathrm{id} \otimes \varepsilon) \circ \Delta^{\mathrm{ch}}$, recovering the augmentation of $\mathcal{A}$.

The chiral coproduct determines a vertex R-matrix by

$$S(z) \;:=\; \sigma \circ \Delta^{\mathrm{ch}} \circ (\Delta^{\mathrm{ch,op}})^{-1} \tag{9.10}$$

where σ is the graded flip and $\Delta^{\mathrm{ch,op}} = \sigma \circ \Delta^{\mathrm{ch}}$ is the opposite coproduct. This is the chiral analogue of the classical formula $R = \Delta^{\mathrm{op}}(\Delta^{-1}(-))$ for a quasi-triangular Hopf algebra.

Remark 9.4 (The role of the Drinfeld associator)*.* The Drinfeld associator $\Phi \in \exp(\widehat{\mathrm{Lie}}(t_{12}, t_{23}))$ controls the failure of strict coassociativity: the two parenthesizations traverse different paths in $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$, and Φ is the KZ parallel transport between them. It also appears in $\ker(\mathrm{av}_3)$ of Proposition 4.3, invisible to the symmetric shadow. For E_∞-chiral algebras, Φ acts trivially on coinvariants; for genuinely E_1-chiral algebras, it is essential structural data.

9.2. **The equivalence theorem.** The three descriptions of the preceding subsection are not independent: a vertex R-matrix determines the chiral A_∞ operations via the bar complex, the A_∞ operations determine the chiral coproduct via cobar dualization, and the coproduct recovers the R-matrix. The equivalences close into a triangle, each arrow depending on the choice of a Drinfeld associator Φ.

Theorem 9.5 (Chiral quantum group equivalence)**.** *The following three structures on an E_1-chiral algebra $\mathcal{A}$ determine each other, up to the choice of a Drinfeld associator Φ:*

(I) E_1*-chiral algebra with vertex R-matrix $S(z)$* (*Definition* 9.1)

(II) A_∞*-algebra in the chiral endomorphism operad* $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$ (*Definition* 9.2)

(III) E_1*-chiral algebra with compatible chiral coproduct* Δ^{ch} (*Definition* 9.3)

The equivalences form a triangle:

$$\begin{tikzcd}
\text{(I)} \arrow[rr, shift left, "\alpha"] \arrow[dr, shift left, "\gamma"] & & \text{(II)} \arrow[ll, shift left, "\alpha^{-1}"] \arrow[dl, shift left, "\beta"] \\
& \text{(III)} \arrow[ul, shift left, "\gamma^{-1}"] \arrow[ur, shift left, "\beta^{-1}"] &
\end{tikzcd} \tag{9.11}$$

where each arrow is an equivalence of categories.

Proof. We prove each direction of the triangle (9.11).

(I) → (II): R-matrix gives A_∞ structure. The Stasheff associahedron K_n is isomorphic to the real Fulton–MacPherson compactification $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{R})$ (Sinha [29]); the inclusion $\mathbb{R} \hookrightarrow \mathbb{C}$ gives the embedding $K_n \hookrightarrow \overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$

of (2.2). The chiral A_∞-maps m_k^{ch} are defined by restricting the OPE to this real locus:

$$m_k^{\mathrm{ch}}(a_1,\dots,a_k) \;=\; \int_{K_k} Y(a_1,z_1)\cdots Y(a_k,z_k)\,\omega_k, \tag{9.12}$$

where ω_k is the canonical volume form on K_k induced by the bar propagator.

The Stasheff identity at arity n (equation (9.6)) follows from Stokes' theorem on $K_{n+1} = \overline{\mathrm{FM}}^{\mathrm{ord}}_{n+1}(\mathbb{R})$. The codimension-1 boundary ∂K_{n+1} has exactly n facets, one for each consecutive collision $z_i \to z_{i+1}$ $(1 \le i \le n)$. Each facet is isomorphic to $K_{r+1} \times K_{s+1}$ with $r+s=n$, recording the cluster and the residual configuration. The integral of $d\omega_{n+1}$ over K_{n+1} vanishes, and the boundary contributions give $\sum(-1)^\epsilon\, m_i^{\mathrm{ch}} \circ_p m_j^{\mathrm{ch}} = 0$: the ordered AOS reduction ([23, §5]).

At codimension 2, where two simultaneous consecutive collisions $(z_i \to z_{i+1}, z_j \to z_{j+1})$ with $|i-j| \ge 2$ occur, the compatibility condition is the hexagon axiom (9.3): the two codimension-2 strata correspond to the two standard paths in $\overline{\mathrm{FM}}^{\mathrm{ord}}_3(\mathbb{C})$, and the hexagon ensures that the iterated boundary contributions match. At codimension 3 (triple simultaneous collision in $\overline{\mathrm{FM}}^{\mathrm{ord}}_4(\mathbb{C})$), the compatibility is the pentagon relation for Φ, which is equivalent to the QYBE (9.2) for $S(z)$.

S-locality (9.1) ensures that the integrals (9.12) converge: the factor $(z-w)^N$ in S-locality bounds the pole orders of the OPE along each boundary divisor.

This construction defines the functor α. Its inverse α^{-1} recovers the R-matrix from the A_∞-structure as the arity-2 holonomy: $S(z) = \mathcal{P}\exp\bigl(\int_{\gamma(z)} m_2^{\mathrm{ch}}\bigr)$, where $\gamma(z)$ is the KZ parallel-transport path in $\mathrm{Conf}_2^{\mathrm{ord}}(\mathbb{C})$.

(II) → (III): A_∞ structure gives chiral coproduct. The ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ (where $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal) carries the deconcatenation coproduct

$$\Delta[a_1|\cdots|a_n] = \sum_{i=0}^{n} [a_1|\cdots|a_i] \otimes [a_{i+1}|\cdots|a_n]. \tag{9.13}$$

The A_∞-maps m_k^{ch} determine the bar differential $d_{\overline{B}^{\mathrm{ord}}}$ by the standard formula $d_{\overline{B}^{\mathrm{ord}}} = \sum_{k\ge1} \hat{m}_k^{\mathrm{ch}}$, where $\hat{m}_k^{\mathrm{ch}}$ is the coderivation extension of m_k^{ch} to $T^c(s^{-1}\bar{\mathcal{A}})$. The relation $d^2_{\overline{B}^{\mathrm{ord}}} = 0$ is equivalent to the Stasheff identities (9.6), so the A_∞ data make $(\overline{B}^{\mathrm{ord}}(\mathcal{A}), d_{\overline{B}^{\mathrm{ord}}}, \Delta)$ into a dg coalgebra in the $\mathsf{SC}^{\mathrm{ch,top}}$-coalgebra sense ([23, Theorem 1.2]): the differential $d_{\overline{B}^{\mathrm{ord}}}$ is the closed (holomorphic) color and the deconcatenation Δ is the open (topological) color.

The chiral coproduct on $\mathcal{A}$ is constructed as follows. On the Koszul locus (where the bar-cobar adjunction is an equivalence), the cobar functor Ω applied to $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ recovers $\mathcal{A}$: $\mathcal{A} \simeq \Omega(\overline{B}^{\mathrm{ord}}(\mathcal{A}))$ ([23, Theorem 1.1(i)]). The

deconcatenation Δ on $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ is a morphism of dg coalgebras $\Delta\colon \overline{B}^{\mathrm{ord}}(\mathcal{A}) \to \overline{B}^{\mathrm{ord}}(\mathcal{A}) \otimes \overline{B}^{\mathrm{ord}}(\mathcal{A})$; applying Ω produces

$$\Delta^{\mathrm{ch}}\colon \mathcal{A} \simeq \Omega(\overline{B}^{\mathrm{ord}}(\mathcal{A})) \xrightarrow{\Omega(\Delta)} \Omega(\overline{B}^{\mathrm{ord}}(\mathcal{A}) \otimes \overline{B}^{\mathrm{ord}}(\mathcal{A})) \to (\mathcal{A} \hat{\otimes} \mathcal{A})((z)), \tag{9.14}$$

where the last map uses the Künneth quasi-isomorphism $\Omega(C \otimes D) \xrightarrow{\sim} \Omega(C) \hat{\otimes} \Omega(D)$ for conilpotent coalgebras.

The deconcatenation coproduct is strictly coassociative on $T^c(s^{-1}\bar{\mathcal{A}})$: the two iterated coproducts $(\Delta \otimes \mathrm{id}) \circ \Delta = (\mathrm{id} \otimes \Delta) \circ \Delta$ by direct computation (the $(n+1)(n+2)/2$ terms on each side match by the unique decomposition of a word into three consecutive subwords). The associator Φ enters when passing through the cobar functor: the formality quasi-isomorphism underlying Ω depends on Φ, so Δ^{ch} is coassociative only up to conjugation by Φ, as required by Definition 9.3(i).

The OPE compatibility (9.9) follows from the coderivation property of $d_{\overline{B}^{\mathrm{ord}}}$ with respect to Δ: the bar differential commutes with the coproduct, which dualizes under Ω to the intertwining of Δ^{ch} with the vertex operation Y. The counit $\varepsilon\colon \mathcal{A} \to \mathbb{C}$ is the augmentation, and the counit axiom (Definition 9.3(iii)) holds because the bar complex is built on $\bar{\mathcal{A}} = \ker(\varepsilon)$.

This construction defines the functor β. Its inverse β^{-1} recovers the A_∞-maps from a chiral coproduct by reading off the bar differential: the k-th structure map m_k^{ch} is the component of $d_{\overline{B}^{\mathrm{ord}}}$ at tensor degree k.

(III) → (I): chiral coproduct gives R-matrix. Given a chiral coproduct Δ^{ch} satisfying Definition 9.3, the vertex R-matrix is defined by (9.10):

$$S(z) \;:=\; \sigma \circ \Delta^{\mathrm{ch}} \circ (\Delta^{\mathrm{ch,op}})^{-1}, \tag{9.15}$$

the chiral analogue of the classical Drinfeld formula for quasi-triangular quasi-Hopf algebras [8]. We verify the five axioms of Definition 9.1.

QYBE. Expand the left side of (9.2) using the definition (9.15). Each factor $S_{ij}(z_{ij})$ introduces one coproduct and one inverse opposite coproduct; composing the three factors yields an expression involving three applications of the coassociativity relation (9.8). In each application, the two iterated coproducts differ by conjugation by Φ. The pentagon identity for Φ (Drinfeld [8]; see also Etingof–Kazhdan [12], §3) ensures that the three associator insertions cancel pairwise, and the remaining coproduct factors rearrange to give the right side of (9.2).

Unitarity. By definition, $S_{21}(-z) = \sigma \circ S_{12}(-z) \circ \sigma$, and $\Delta^{\mathrm{ch,op,op}} = \Delta^{\mathrm{ch}}$ (applying the flip twice is the identity). Composing:

$$S_{12}(z)\, S_{21}(-z) = \sigma \circ \Delta^{\mathrm{ch}} \circ (\Delta^{\mathrm{ch,op}})^{-1} \cdot \sigma \circ \Delta^{\mathrm{ch,op}} \circ (\Delta^{\mathrm{ch}})^{-1} = \mathrm{id}.$$

Hexagon. The hexagon (9.3) encodes the compatibility of $S(z)$ with Φ on three-point configurations. Substituting (9.15) into the left side, each S_{ij} contributes a coproduct and an inverse opposite coproduct, and each $\Phi_{ijk}^{\pm 1}$ reparenthesizes. The OPE compatibility (9.9) (the chiral Leibniz rule: Δ^{ch} intertwines Y) ensures that the product-coproduct rearrangement through Φ

matches the right side. This computation is the vertex-algebraic analogue of Drinfeld's proof that a quasi-Hopf coproduct produces a quasi-triangular structure [8]; the adaptation to the chiral setting (spectral-parameter dependence, meromorphic completions) follows Etingof–Kazhdan [13], §4.

Shift condition. The translation operator $T = L_{-1}$ satisfies $[T, Y(a,z)] = \partial_z Y(a,z)$. The OPE compatibility (9.9) applied to the vacuum gives $\Delta^{\mathrm{ch}}(T \cdot a) = (T \otimes \mathrm{id} + \mathrm{id} \otimes T) \cdot \Delta^{\mathrm{ch}}(a)$ (the counit axiom ensures the two tensor factors share T additively). Differentiating (9.15) in z and using this intertwining yields $[T \otimes \mathrm{id} + \mathrm{id} \otimes T,\, S(z)] = \partial_z S(z)$.

S-locality. The OPE compatibility (9.9) maps singular OPE terms to tensor products of singular terms. For $a, b \in \mathcal{A}$ and N sufficiently large, $(z-w)^N Y(a,z)Y(b,w)$ is regular. Applying Δ^{ch} to both sides of the OPE and using the definition of $S(z-w)$ via (9.15) gives $(z-w)^N Y(a,z)Y(b,w) = (z-w)^N Y(b,w)Y(a,z) \cdot S(z-w)$, which is (9.1).

This construction defines γ. Its inverse γ^{-1} recovers the coproduct from $(\mathcal{A}, S(z))$ by composing α and β: $\gamma^{-1} = \beta \circ \alpha$.

Triangle coherence. The three functors satisfy $\gamma = \alpha^{-1} \circ \beta^{-1}$. Each functor is an equivalence with explicit inverse as constructed above, and the triangle (9.11) commutes: $\alpha \circ \gamma \circ \beta = \mathrm{id}$. □

9.3. **The classical limit.** The chiral quantum group equivalence (Theorem 9.5) is a vertex-algebraic lift of Drinfeld's classical equivalence between quasi-triangular Hopf algebras, R-matrices, and coproducts. The lift replaces finite-dimensional vector spaces with modules over $\mathbb{C}((z))$, the symmetric group with the braid group, and strict coassociativity with coassociativity up to the Drinfeld associator Φ.

Remark 9.6 (Comparison with the classical Hopf algebra story). The chiral quantum group equivalence (Theorem 9.5) is the vertex-algebraic analogue of the following classical equivalence, due to Drinfeld:

- $(\mathrm{I})_{\mathrm{cl}}$ Quasi-triangular quasi-Hopf algebra (H, Δ, R, Φ)
- $(\mathrm{II})_{\mathrm{cl}}$ A_∞-algebra in the endomorphism operad End_H
- $(\mathrm{III})_{\mathrm{cl}}$ Bialgebra with R-matrix: $R \cdot \Delta(x) = \Delta^{\mathrm{op}}(x) \cdot R$ for all $x \in H$

The passage from classical to chiral introduces two features: the coproduct acquires a spectral parameter ($\Delta^{\mathrm{ch}} \colon \mathcal{A} \to (\mathcal{A} \hat{\otimes} \mathcal{A})((z))$), and strict coassociativity is replaced by coassociativity up to the Drinfeld associator Φ, which appears universally in the chiral setting because the KZ connection on $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$ has non-trivial holonomy.

9.4. **Examples.** The three structures are most transparent for affine Kac–Moody, where the R-matrix is the KZ monodromy, the A_∞ operations are the Sugawara OPE restricted to the associahedron, and the chiral coproduct is the Drinfeld coproduct. For the Yangian $Y_\hbar(\mathfrak{g})$, the R-matrix is the starting point rather than derived from an E_∞-chiral OPE. For E_∞-chiral algebras, all three structures are trivial.

Example 9.7 (Affine Kac–Moody: Sugawara coproduct)**.** For $V_k(\mathfrak{g})$ at generic level $k \neq -h^\vee$, the three structures are:

(i) (R-matrix.) The vertex R-matrix is the KZ R-matrix: the monodromy of the KZ connection $\nabla = d + k\Omega/(z_1 - z_2) \cdot d(z_1 - z_2)$ around Δ_{12}. At the classical level ($\hbar \to 0$, i.e. $k \to \infty$), this reduces to $R = 1 + \hbar\, r(z) + O(\hbar^2)$ where $r(z) = k\Omega/z$ is the classical r-matrix (trace-form; at $k = 0$, $r = 0$).

(ii) (A_∞ structure.) The chiral A_∞ structure maps are the Sugawara OPE operations restricted to $K_n \hookrightarrow \overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$. At $k = 2$: $m_2^{\mathrm{ch}}(J^a, J^b; \lambda) = [J^a, J^b] + k\,\delta^{ab}/\lambda$ is the affine OPE. The transferred higher operations m_k^{ch} for $k \geq 3$ encode the associator corrections from the KZ holonomy.

(iii) (Chiral coproduct.) The Drinfeld coproduct on $Y_\hbar(\mathfrak{g})$ is the paradigmatic example. On generators: $\Delta^{\mathrm{ch}}(J^a) = J^a \otimes 1 + 1 \otimes J^a + \hbar\, f^a_{bc}\, J^b \otimes J^c/z + \cdots$, where the spectral parameter z comes from the difference coordinate $z_1 - z_2$ on $\mathrm{Conf}_2(\mathbb{C})$. The deconcatenation coproduct on the ordered bar $\overline{B}^{\mathrm{ord}}(V_k(\mathfrak{g}))$ induces, via the cobar functor, this spectral Drinfeld coproduct.

The Drinfeld–Kohno theorem identifies the KZ monodromy representation (the R-matrix of (i)) with the quantum group R-matrix on $U_q(\mathfrak{g})$-modules, where $q = \exp(\pi i/(k + h^\vee))$. The chiral quantum group equivalence (Theorem 9.5) lifts this identification from monodromy representations to the level of the algebra itself.

Example 9.8 (Yangian: the Drinfeld coproduct as chiral primitive)**.** For the Yangian $Y_\hbar(\mathfrak{g})$, structure (I) is the starting point: the R-matrix $R(z) = 1 + \hbar\, r/z + O(\hbar^2)$ is independent input (not derived from any E_∞-chiral OPE). The Drinfeld coproduct

$$\Delta_z(T^{(n)}) \;=\; \sum_{k=0}^{n} T^{(k)} \otimes T^{(n-k)} \cdot z^{-k} \tag{9.16}$$

(on the generating series $T(u) = 1 + \sum_{n\geq 1} T^{(n)} u^{-n}$) provides structure (III). The ordered bar complex $\overline{B}^{\mathrm{ord}}(Y_\hbar(\mathfrak{g}))$ is the bridge: its deconcatenation coproduct is strictly coassociative and induces, via cobar dualization, the Drinfeld coproduct (9.16) on $Y_\hbar(\mathfrak{g})$. The associator Φ appears when passing from the strictly coassociative bar level to the algebra level.

The A_∞ structure maps m_k^{ch} for $k \geq 3$ are the RTT relations of the Yangian: they encode the obstruction to strict associativity of the spectral OPE, governed by the fusion procedure for the R-matrix. The bar-cobar inversion $\Omega(\overline{B}^{\mathrm{ord}}(Y_\hbar(\mathfrak{g}))) \xrightarrow{\sim} Y_\hbar(\mathfrak{g})$ identifies these operations with the standard Yangian relations.

Example 9.9 (Etingof–Kazhdan quantum vertex algebra for $\mathfrak{sl}_2$: full braided VOA structure)**.** The Yangian $Y_\hbar(\mathfrak{sl}_2)$ of Example 7.5 is a quasi-triangular

Hopf algebra in the category of topological algebras with a spectral parameter. The Etingof–Kazhdan construction [13] lifts it to a *quantum vertex operator algebra*: an algebra satisfying the full suite of braided VOA axioms (S-locality, vertex R-matrix, hexagon) of Definition 9.1. The purpose of this example is to exhibit the construction for $\mathfrak{g} = \mathfrak{sl}_2$ concretely, verify the axioms, compute the ordered chiral center with the full quantum R-matrix $S(z)$ (not merely its classical leading order), and connect S-locality to the chiral A_∞ structure.

Part A: the EK construction. The input is the Yang R-matrix $R(u) = u\,I + \hbar\,P \in \mathrm{End}(V \otimes V)[u]$ from (7.28), where $V = \mathbb{C}^2$ is the fundamental representation, $P = (\rho \otimes \rho)(\Omega) + \frac{1}{2}\,I$ is the permutation operator, and $\hbar = 1/(k+2)$. The EK quantum VOA is constructed in four steps: quantum function algebra, dual QUE algebra (state space), vertex operator (creation–annihilation decomposition), and vertex R-matrix.

Step 1: the quantum function algebra $F_0(R)$. The algebra $F_0(R)$ is generated by the matrix entries of a formal power series

$$T(u) = \sum_{n\geq 0} T^{(n)}\,u^n \;\in\; \mathrm{End}(V) \otimes F_0(R)[\![u]\!], \tag{9.17}$$

subject to the RTT relation

$$R_{12}(u-v)\,T_{13}(u)\,T_{23}(v) = T_{23}(v)\,T_{13}(u)\,R_{12}(u-v) \tag{9.18}$$

in $\mathrm{End}(V^{\otimes 2}) \otimes F_0(R)[\![u,v]\!]$. For $R(u) = u\,I + \hbar\,P$, writing $T^{(n)} = \left(\begin{smallmatrix} a_n & b_n \\ c_n & d_n \end{smallmatrix}\right)$, equation (9.18) expands at order $u^m v^n$ to give:

$$[a_m, a_n] = 0, \quad [d_m, d_n] = 0, \quad [b_m, b_n] = 0, \quad [c_m, c_n] = 0, \tag{9.19}$$

$$a_m\,b_n - b_n\,a_m = \hbar\,(b_m\,a_n - b_{m-1}\,a_{n+1}), \tag{9.20}$$

$$[a_m, d_n] = \hbar\,(c_m\,b_n - c_{m-1}\,b_{n+1} - b_m\,c_n + b_{m-1}\,c_{n+1}). \tag{9.21}$$

The algebra $F_0(R)$ carries a coalgebra structure $\Delta_F(T(u)) = T(u) \dot{\otimes} T(u)$ (matrix multiplication in $\mathrm{End}(V)$, tensor product in $F_0(R)$), a counit $\varepsilon_F(T(u)) = I$, and an antipode $S_F(T(u)) = T(u)^{-1}$.

Step 2: the state space $V_{\mathrm{EK}} = U_0(R)$. The quantum universal enveloping algebra $U_0(R)$ is the continuous dual of $F_0(R)$:

$$V_{\mathrm{EK}} \;:=\; U_0(R) \;=\; F_0(R)^*_{\mathrm{cont}}, \tag{9.22}$$

a topological Hopf algebra with multiplication dual to the comultiplication on $F_0(R)$. The vacuum vector is the counit:

$$\Omega = \varepsilon_F \in U_0(R), \qquad \langle \Omega, f\rangle = \varepsilon_F(f) \quad \text{for all } f \in F_0(R). \tag{9.23}$$

The translation operator $D\colon U_0(R) \to U_0(R)$ is defined by the shift of spectral parameter:

$$e^{zD}\,T(u_1)\cdots T(u_n)\,\Omega \;=\; T(u_1+z)\cdots T(u_n+z)\,\Omega. \tag{9.24}$$

On the generating modes: $D\,T^{(n)} = n\,T^{(n-1)}$ (with $T^{(-1)} := 0$), reproducing the conformal translation of mode operators.

For $\mathfrak{g} = \mathfrak{sl}_2$: the graded components of $U_0(R)$ are finite-dimensional, with

$$\dim U_0(R)_n \;=\; \dim \mathrm{Sym}^n(\mathfrak{sl}_2[t]) \;=\; \begin{cases} 1 & n=0, \\ 3 & n=1, \\ 9 & n=2, \\ 22 & n=3, \end{cases} \tag{9.25}$$

the Hilbert series of the PBW basis $\{E_r, F_r, H_r : r \geq 0\}$ of $Y_\hbar(\mathfrak{sl}_2)$, confirming the identification $U_0(R) \cong Y_\hbar(\mathfrak{sl}_2)$ as vector spaces.

Step 3: the vertex operator Y. The vertex operator $Y\colon V_{\mathrm{EK}} \to \mathrm{End}(V_{\mathrm{EK}})[\![z, z^{-1}]\!]$ is constructed from the creation–annihilation decomposition of the quantum current. Define:

- the *creation operator* $\mathbf{T}^+(u,z) = T(u+z)$, acting on V_{EK} by left multiplication;
- the *annihilation operator* $\mathbf{T}^-(u,z) = (T^*(u+z-K\hbar/2))^{-1}$, where T^* is the adjoint transfer matrix (the transpose of T with respect to the Killing form) and K is the level.

The quantum current $\mathbf{T}(u,z)$ is the product

$$\mathbf{T}(u,z) \;=\; \mathbf{T}^+(u,z)\cdot\big(\mathbf{T}^-(u,z)\big)^{-1} \;\in\; \mathrm{End}(V)\otimes\mathrm{End}(V_{\mathrm{EK}})[\![u^{-1}, z, z^{-1}]\!]. \tag{9.26}$$

The vertex operator on generators is

$$Y(T^{(n)}\,\Omega,\, z) \;=\; \mathrm{Res}_{u=0}\; u^n\,\mathbf{T}(u,z)\,du, \tag{9.27}$$

and extends to products by the normal-ordering prescription: $Y(a\,b,\,z) = :Y(a,z)\,Y(b,z):$, where the normal ordering places annihilation modes to the right.

On the leading generators ($n=0$): identifying $T^{(0)} = I + \hbar(e\otimes e_{12} + f\otimes e_{21} + \frac{1}{2}h\otimes(e_{11}-e_{22}))$ at leading order, the vertex operators reduce to the standard affine Kac–Moody currents

$$Y(J^a\,\Omega,\, z) \;=\; J^a(z) = \sum_{n\in\mathbb{Z}} J^a_n\, z^{-n-1} \tag{9.28}$$

for $a \in \{e, f, h\}$, matching the OPE of $V_k(\mathfrak{sl}_2)$ at leading order in $\hbar$.

Step 4: the vertex R-matrix $S(z)$. The vertex R-matrix is the braiding operator that governs the exchange of vertex operators. It is constructed from the R-matrix data via the universal formula of [13]:

$$S(z) \;=\; \mathcal{R}(z)\cdot\sigma \;\in\; \mathrm{End}(V_{\mathrm{EK}}\otimes V_{\mathrm{EK}})[\![z, z^{-1}]\!], \tag{9.29}$$

where σ is the graded flip and $\mathcal{R}(z) = \sum_{n\geq 0}\mathcal{R}_n\, z^{-n}$ is the *spectral universal R-matrix* of $Y_\hbar(\mathfrak{sl}_2)$, satisfying

$$\mathcal{R}(z)\cdot\Delta_z(x) \;=\; \Delta_z^{\mathrm{op}}(x)\cdot\mathcal{R}(z) \qquad \text{for all } x \in Y_\hbar(\mathfrak{sl}_2). \tag{9.30}$$

On the fundamental representation $V\otimes V$, the spectral universal R-matrix evaluates to the Yang R-matrix:

$$(\rho\otimes\rho)\big(\mathcal{R}(z)\big) \;=\; R(z) \;=\; z\,I + \hbar\,P \;\in\; \mathrm{End}(V\otimes V)[z]. \tag{9.31}$$

The leading-order expansion of $S(z)$ on $V_{\mathrm{EK}} \otimes V_{\mathrm{EK}}$ is

$$S(z) = \mathrm{id} + \frac{\hbar\,\Omega}{z} + \frac{\hbar^2}{z^2}\Big(\Omega^{(2)} + \tfrac{1}{2}\,\Omega^2\Big) + O(z^{-3}), \tag{9.32}$$

where $\Omega^{(2)} \in U(\mathfrak{g})^{\otimes 2}$ is the second-order Casimir correction (determined by the R-matrix unitarity condition) and $\Omega^2 = \Omega \cdot \Omega \in U(\mathfrak{g})^{\otimes 2}$. At leading order ($z^{-1}$ term), the classical r-matrix $r(z) = \hbar\,\Omega/z$ is recovered.

Part A (continued): verification of the braided VOA axioms. The five axioms of Definition 9.1 are verified for $(V_{\mathrm{EK}},\, Y,\, S(z))$ as follows.

S-locality. For the generating currents, the S-locality

$$(z-w)^N\, Y(a,z)\, Y(b,w) \;=\; (z-w)^N\, Y(b,w)\, Y(a,z) \cdot S(z-w) \tag{9.33}$$

holds at $N = 2$: the $\hbar\,\Omega/(z-w)$ correction from $S(z-w)$ accounts for the structure-constant term in the OPE, and the higher-order terms account for the double pole. For descendants, S-locality follows by induction on conformal weight via the shift condition.

QYBE, unitarity, shift, hexagon. The quantum Yang–Baxter equation on $V_{\mathrm{EK}}^{\otimes 3}$ reduces on $V^{\otimes 3}$ to (7.29), verified in Example 7.5; the extension to V_{EK} follows from the universal property of $\mathcal{R}(z)$. Unitarity $S_{12}(z) \cdot (\sigma \circ S_{12}(-z) \circ \sigma) = \mathrm{id}$ follows from Drinfeld–Jimbo unitarity. The shift condition

$$[D \otimes \mathrm{id} + \mathrm{id} \otimes D,\; S(z)] \;=\; \partial_z\, S(z) \tag{9.34}$$

follows from the translation-covariance of the Drinfeld coproduct. The hexagon follows from the pentagon relation for Φ (from the Stasheff boundary of $\overline{\mathrm{FM}}_4^{\mathrm{ord}}(\mathbb{C}) = K_4$) combined with the QYBE for S, by the standard argument of Drinfeld [12] lifted to the vertex-algebraic setting.

Part B: the ordered chiral center of the EK quantum VOA. We compute the ordered chiral Hochschild cohomology $\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(V_{\mathrm{EK}},\, V_{\mathrm{EK}})$ at low arities on the formal disk D, paralleling the computation of Example 7.5 (Part B) but now using the *full* vertex R-matrix $S(z)$ rather than the Yang R-matrix on $V \otimes V$ alone.

Arity 2: the KZ connection with the full vertex R-matrix. At arity 2, the KZ connection on $\mathrm{Conf}_2^{\mathrm{ord}}(\mathbb{C})$ with coefficients in $\mathrm{End}(V_{\mathrm{EK}}^{\otimes 2})$ is

$$\nabla^S_{\mathrm{KZ}} = d + \frac{\hbar\,\Omega}{z}\,dz + \frac{\hbar^2}{z^2}\,\Big(\Omega^{(2)} + \tfrac{1}{2}\,\Omega^2\Big)\,dz + O(z^{-3}), \tag{9.35}$$

where $z = z_1 - z_2$. The connection (9.35) is obtained from $S(z)$ via the general formula $\nabla = d + (\partial_z \log S(z))\,dz$ (logarithmic derivative of the vertex R-matrix).

For the classical $V_k(\mathfrak{sl}_2)$ (trivial braiding $S(z) = \mathrm{id}$), the flat sections are $\Phi(z) = z^{\Omega/(k+2)}$ with algebraic monodromy. For the EK quantum VOA ($S(z) \neq \mathrm{id}$), the flat sections acquire quantum corrections:

$$\Phi^S(z) = z^{\hbar\,\Omega}\,\Big(I + \frac{\hbar^2\,\Omega^{(2)}}{z} + O(z^{-2})\Big). \tag{9.36}$$

The kernel ker(av). The ordered chiral centre computation parallels Example 7.5; the result is identical on cohomology. On $V \otimes V$: $\ker(\mathrm{av}_2)\big|_{V\otimes V} \cong \bigwedge^2 V = \mathbb{C}$, the same one-dimensional space as equation (7.42). The quantum corrections from $S(z)$ shift the subleading terms of the flat sections but do not alter the $\ker(\mathrm{av}_2)$ dimension. The scalar shadow surviving av_2 is $\kappa = 3(k+2)/4$, identical to the classical computation. At arity 3, the kernel $\ker(\mathrm{av}_3) \cong \Lambda^3(\mathfrak{g}) = \mathbb{C}$ carries the quantum-corrected Drinfeld associator

$$\Phi^S_{\mathrm{KZ}} - 1 \;=\; \hbar^2\,\zeta(2)\,[\Omega_{12}, \Omega_{23}] + O(\hbar^3), \tag{9.37}$$

whose leading term agrees with the classical KZ associator.

Part C: S-locality as an A_∞ relation. The braided VOA axioms of the EK quantum VOA are equivalent to the chiral A_∞ relations for V_{EK} through the chiral quantum group equivalence (Theorem 9.5, (I)⇔(II)). In the ordered bar complex $\overline{B}^{\mathrm{ord}}(V_{\mathrm{EK}}) = T^c(s^{-1}\overline{V}_{\mathrm{EK}})$, the S-locality axiom (9.33) governs the reordering of two insertions: the braid group $\pi_1(\mathrm{Conf}_2^{\mathrm{ord}}(\mathbb{C})) = \mathbb{Z}$ acts on degree-2 bar chains by $\sigma_1 \cdot [a|b] = S(z) \cdot [b|a]$. The S-locality axiom is the $n = 2$ Stasheff relation, the hexagon is the $n = 3$ relation, and the QYBE is the $n = 4$ relation (from the pentagon for Φ on $K_4 = \overline{\mathrm{FM}}_4^{\mathrm{ord}}(\mathbb{R})$).

Summary. The EK quantum VOA (V_{EK}, Y, $S(z)$) provides the first concrete instance of all three structures of the chiral quantum group equivalence (Theorem 9.5) exhibited simultaneously on a single algebra with all axioms verified. The ordered chiral center carries the full quantum R-matrix $S(z)$ (not just the classical r-matrix $r(z) = \hbar\,\Omega/z$), and the kernel $\ker(\mathrm{av}_2) \neq 0$ demonstrates that the ordered theory is strictly richer than the symmetric Beilinson–Drinfeld chiral homology even at the quantum level.

Example 9.10 (Heisenberg: trivial case)**.** For the Heisenberg algebra $\mathcal{H}_k$ (class G), all three structures are trivial: $S(z) = \mathrm{id}$ (the braiding is symmetric); $m_k^{\mathrm{ch}} = 0$ for $k \geq 3$ (the algebra is strictly associative in the chiral sense); $\Delta^{\mathrm{ch}}(J) = J \otimes 1 + 1 \otimes J$ is cocommutative and strictly coassociative (Φ acts trivially). The ordered and symmetric data coincide.

Example 9.11 ($\mathcal{W}_{1+\infty}$ as a chiral quantum group via the cohomological Hall algebra)**.** The algebra $\mathcal{W}_{1+\infty}[\Psi]$ admits a chiral quantum group structure originating from the cohomological Hall algebra (CoHA) of the Jordan quiver Q_{Jor}. The CoHA $\mathcal{H}_{\mathrm{Jor}} = \bigoplus_{n\geq 0} H^{GL_n}_{\mathrm{BM},*}(\mathcal{M}_n^{\mathrm{nil}})$ (Kontsevich–Soibelman [21]) carries a Hecke product and a comultiplication from short exact sequences; Yang–Zhao [33] proved this is a bialgebra with Drinfeld double $D(\mathcal{H}_{\mathrm{Jor}}) \cong Y(\mathfrak{gl}_1)$. Schiffmann–Mellit–Minets–Vasserot [30] identified $\mathcal{H}_{\mathrm{Jor}} \cong U(\mathcal{W}_{1+\infty}[\Psi])^+$.

The three structures of Theorem 9.5 on $\mathcal{W}_{1+\infty}$ are:

(i) (R-matrix.) $R_{\mathrm{CoHA}} = q^{\alpha\cdot\beta}$ ($q = e^{\pi i \Psi}$), the universal R-matrix of quantum toroidal $\mathfrak{gl}_1$ restricted to the CoHA; classical limit $r(z) = \Psi/z$.

(ii) (A_∞ structure.) Higher operations m_k^{ch} for $k \geq 3$ encode the obstruction to strict commutativity at successive spins; these lie in $\ker(\mathrm{av}_k)$.

(iii) (Chiral coproduct.) The CoHA comultiplication gives $\Delta^{\mathrm{ch}}(J) = J \otimes 1 + 1 \otimes J$ (cocommutative at spin 1) and $\Delta^{\mathrm{ch}}(T) = T \otimes 1 + 1 \otimes T + \Psi \cdot (J \otimes J)/z^2 + \cdots$ (spectral corrections from the Virasoro OPE at spin 2).

Descent to $\mathcal{W}_N$. Drinfeld–Sokolov reduction gives $\pi_N \colon \mathcal{W}_{1+\infty}[\Psi] \twoheadrightarrow \mathcal{W}_N$ for each $N \geq 2$, compatible with the coproduct. The tower $\cdots \twoheadrightarrow \mathcal{W}_{N+1} \twoheadrightarrow \mathcal{W}_N \twoheadrightarrow \cdots \twoheadrightarrow \mathcal{W}_2 = \mathrm{Vir}_c$ is a projective system of chiral quantum groups. At $N = 2$: class M, $\kappa = c/2$, irregular KZ connection. At $N = 3$: $\kappa(\mathcal{W}_3) = 5c/6$ (where $H_3 - 1 = 5/6$).

Y-algebras and holomorphic gravity. The Gaiotto–Rapčák Y-algebras $Y_{N_1,N_2,N_3}[\Psi]$ [16] are truncations of $\mathcal{W}_{1+\infty}$; the S_3-triality is a symmetry of the chiral coproduct, corresponding to three Hecke correspondences on spiked-instanton moduli (Rapčák–Soibelman–Yang–Zhao [26]). The $\mathcal{D}$-module $\mathcal{F}_n^{\mathrm{ord}}(\mathcal{W}_{1+\infty})$ has irregular singularities of unbounded Poincaré rank (spin-s contributes pole order $2s-1$ after d-log absorption), providing the boundary algebra of higher-spin holomorphic gravity.

9.5. **Structural consequences.** The three equivalent descriptions of an E_1-chiral algebra all project from a single datum: the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with its deconcatenation coproduct. The bar differential encodes the A_∞ operations, the deconcatenation induces the chiral coproduct, and the KZ monodromy recovers the vertex R-matrix. This is the structural content of E_1 primacy.

Corollary 9.12 (The ordered bar complex encodes all three structures)**.** *The ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, equipped with its bar differential $d_{\overline{B}^{\mathrm{ord}}}$ and deconcatenation coproduct Δ, is the universal object from which all three structures in Theorem 9.5 are recovered:*

(i) *The bar differential $d_{\overline{B}^{\mathrm{ord}}}$ encodes the A_∞ structure maps m_k^{ch} (by the standard bar-cobar correspondence: coderivations of $T^c(s^{-1}\bar{\mathcal{A}})$ biject with elements of $\prod_k \mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}(k)$).*

(ii) *The deconcatenation coproduct Δ induces, via cobar dualization, the chiral coproduct Δ^{ch}.*

(iii) *The vertex R-matrix $S(z)$ is recovered as the monodromy of the KZ connection, which is itself the arity-2 projection of the Maurer–Cartan element $\Theta_{\mathcal{A}}$ in $\mathfrak{g}_{\mathcal{A}}^{\mathsf{E}_1}$.*

This is the structural content of the E_1 primacy principle: the ordered bar complex with deconcatenation is the primitive object; the R-matrix, the chiral coproduct, and the A_∞ operations are projections of this single datum.

Remark 9.13 (Relation to factorization quantum groups). When the hexagon holds strictly, the chiral quantum group $(\mathcal{A}, S(z), \Delta^{\mathrm{ch}})$ is a *factorization*

quantum group (Latyntsev [22]): its ordered bar complex defines a factorization coalgebra on the curve. When the hexagon holds only up to Φ, one obtains a *quasi-factorization quantum group* with regularized holonomy data (R, Φ).

10. Conclusion and open problems

This paper defines ordered chiral homology $\int_X^{\mathrm{ord}} \mathcal{A}$ for E_1-chiral algebras on algebraic curves, constructs the chiral E_3-algebra $\mathrm{CE}_\hbar^{\mathrm{ch}}(\mathfrak{g})$ as the holomorphic refinement of the Costello–Francis–Gwilliam perturbative Chern–Simons algebra, and establishes the chiral quantum group equivalence (Theorem 9.5) relating vertex R-matrices, A_∞-structures, and chiral coproducts as three faces of a single ordered bar complex. Five explicit computations across the shadow classification (Heisenberg: class G; affine Kac–Moody: class L; $\beta\gamma$: class C; Virasoro: class M; Yangian: genuinely E_1) verify the formality bridge, the regular/irregular dichotomy, and the Stokes phenomena that distinguish gauge theory from gravity at the level of the KZ $\mathcal{D}$-module.

The following problems remain open.

(a) (The E_3 identification for affine Kac–Moody.) Conjecture 6.5 asserts that the derived chiral centre $Z_{\mathrm{ch}}^{\mathrm{der}}(V_k(\mathfrak{g}))$ and the CFG algebra $\mathcal{A}^\lambda$ are isomorphic as E_3-algebras over the formal deformation space $\hbar H^3(\mathfrak{g})[[\hbar]]$. This paper proves that the deformation spaces agree (both one-dimensional at each order in $\hbar$ for simple $\mathfrak{g}$) and that the formal disk global sections match at the level of P_3-brackets (§6.3). What is missing is a chain-level comparison that identifies the E_3-structures as families, not merely fiber by fiber: is there a natural map between the two E_3-algebras that is a quasi-isomorphism at each non-critical level $k \neq -h^\vee$?

(b) (Holomorphic Chern–Simons invariants of curves.) The chiral E_3-algebra $\mathrm{CE}_\hbar^{\mathrm{ch}}(\mathfrak{g})$ on a smooth proper curve X of genus g should produce, via the higher-genus ordered chiral homology and factorisation on $\overline{\mathcal{M}}_{g,n}$, invariants of $(X, \mathfrak{g}, k)$ that refine Reshetikhin–Turaev invariants in the same way that perturbative Chern–Simons refines Witten's partition function. Concretely: does the genus-g ordered chiral homology $\int_X^{\mathrm{ord}} \mathrm{CE}_\hbar^{\mathrm{ch}}(\mathfrak{g})$ recover the perturbative Chern–Simons partition function $Z_{\mathrm{CS}}(X, \mathfrak{g}, k)$ as its Σ_n-coinvariant shadow, and what invariants live in the kernel of the averaging map av?

(c) (The $\mathcal{W}_{1+\infty}$ chiral quantum group.) Example 9.11 identifies $\mathcal{W}_{1+\infty}[\Psi]$ as a chiral quantum group whose CoHA origin provides the R-matrix, coproduct, and A_∞-structure. Two questions are unresolved. First, the CoHA construction of Schiffmann–Mellit–Minets–Vasserot [30] gives the positive half $U(\mathcal{W}_{1+\infty})^+$; does the full chiral quantum group structure (including the antipode and the Drinfeld double) admit a CoHA interpretation? Second, the KZ connection for $\mathcal{W}_N$ at finite N

has irregular singularities of Poincaré rank $N-1$; what are its Stokes multipliers, and do they encode the higher-spin BPS spectrum?

(d) (Higher-genus ordered chiral homology.) Section 4 defines ordered chiral homology at genus 0 and describes the curvature obstruction $d^2_{\mathrm{fib}} = \kappa \cdot \omega_g$ at genus $g \geq 1$, where $\omega_g = c_1(\lambda)$ is the Hodge class on $\overline{\mathcal{M}}_g$. The full genus-g theory requires a period-corrected differential whose coefficients involve the $(g \times g)$-matrix $(\operatorname{Im}\Omega)^{-1}$ of the Siegel period matrix, not the scalar $1/\operatorname{Im}(\tau)$ of the genus-1 degeneration. Does the ordered chiral chain complex at genus g define a flat connection over $\overline{\mathcal{M}}_g$ compatible with the Mumford–Morita class κ_1, and does the resulting system of conformal blocks satisfy factorisation at the boundary divisors of $\overline{\mathcal{M}}_g$?

(e) (Trigonometric and elliptic R-matrices.) This paper treats the rational Yang R-matrix $R(u) = 1 + \hbar\, r(u)/u + O(\hbar^2)$ with $r(z) = k\Omega/z$ (trace-form convention; at $k = 0$, $r = 0$). The trigonometric R-matrix $R^{\mathrm{trig}}(u) = q^{\Omega/u}$ (quantum affine algebras) and the elliptic Belavin R-matrix $R^{\mathrm{ell}}(u,\tau)$ produce richer ordered chiral homology with qualitatively different analytic behaviour: higher-order Stokes phenomena, theta-function monodromy, and flat sections valued in elliptic hypergeometric integrals. What is the ordered factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}_n(\mathcal{A})$ for an E_1-chiral algebra whose vertex R-matrix is trigonometric or elliptic, and does the regular/irregular dichotomy of Theorem 1.1 extend to a trichotomy (regular, irregular-rational, irregular-elliptic) correlated with the three types of quantum groups?

(f) (Categorification.) Ben-Zvi–Brochier–Jordan [2, 3] integrate braided monoidal categories over surfaces; ordered chiral homology integrates chiral algebras over curves. The expected bridge (Remark 8.1, equation (8.1)) identifies the Hochschild homology of the categorical invariant $\int_\Sigma \mathrm{Rep}_q(\mathfrak{g})$ with the ordered chiral homology $\int^{\mathrm{ord}}_X V_k(\mathfrak{g})$. Is there a direct functor from the BBJ categorical factorisation homology to ordered chiral homology that intertwines the E_3-structures, and does the kernel of this functor detect the higher categorical data (objects of $\int_\Sigma \mathrm{Rep}_q(\mathfrak{g})$ that are invisible at the chain level)?

(g) (The Khan–Zeng quantum extension.) Khan–Zeng [20] prove the topological enhancement of the holomorphic-topological 3d gauge theory for the classical Poisson vertex algebra (Proposition 6.20). The extension to the full quantum vertex algebra $V_k(\mathfrak{g})$ at non-critical level is used in Theorem 1.3(iv) but remains conjectural. Does the Sugawara Virasoro element at non-critical level provide a topological E_3-structure on the quantum chiral E_3-algebra $\mathrm{CE}^{\mathrm{ch}}_\hbar(\mathfrak{g})$, or does the quantum correction to the Sugawara element obstruct the enhancement at finite $\hbar$?

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